MASTER THESIS

Agent-Based Simulation of Emergency Evacuation in University Lecture Halls

submitted by

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Abstract

Emergency evacuations from crowded spaces such as university lecture halls present significant challenges for ensuring human safety. This thesis introduces a sophisticated agent-based model developed in NetLogo, to simulate and analyze the complex dynamics of emergency evacuations in these specific environments. The model investigates the interplay between architectural design, human behavior and evacuation efficiency. The simulation incorporates a heterogeneous population of agents (students and a professor) with diverse behavioral strategies including panic-driven rushing, calm and orderly movement, herding and rational decision-making based on the nearest exit. Key psychological factors like panic contagion, familiarity with the environment and variable reaction times are modeled to create a realistic representation of human responses during a crisis. The model also accounts for physical variables such as agent mobility impairments and the impact of architectural elements like stair aisles and tiered seating, which realistically affect movement speed. A comprehensive set of experiments was conducted to evaluate evacuation performance across various scenarios. These scenarios systematically vary parameters such as lecture hall layout (e.g., traditional, tiered, wide-auditorium), exit configurations, agent population and the presence of exit blockages. The primary metrics for evaluation include total evacuation time, agent flow rate, bottleneck formation and average panic levels. The results demonstrate a significant correlation between lecture hall architecture and evacuation outcomes, with certain layouts consistently outperforming others in reducing congestion and total egress time. The findings also underscore the critical role of human behavioral patterns, showing how phenomena like herding can lead to suboptimal exit choices and increased congestion, even when clear paths are available. This research offers valuable insights for university administrators and safety planners. It provides a powerful tool for assessing the safety of existing lecture halls and designing safer educational spaces.

Zusammenfassung

Notfall-Evakuierungen aus überfüllten Räumen wie Hörsälen von Universitäten stellen eine große Herausforderung für die Gewährleistung der Sicherheit von Menschen dar. Diese Arbeit stellt ein in NetLogo entwickeltes, hochentwickeltes agentenbasiertes Modell vor, mit dem die komplexe Dynamik von Notfall-Evakuierungen in diesen spezifischen Umgebungen simuliert und analysiert werden kann. Das Modell untersucht das Zusammenspiel zwischen architektonischer Gestaltung, menschlichem Verhalten und Evakuierungseffizienz. Die Simulation umfasst eine heterogene Population von Agenten (Studierende und ein Professor) mit unterschiedlichen Verhaltensstrategien, darunter panikgetriebenes Gedränge, ruhige und geordnete Bewegung, Herdenverhalten und rationale Entscheidungsfindung auf der Grundlage des nächstgelegenen Ausgangs. Wichtige psychologische Faktoren wie Panikansteckung, Vertrautheit mit der Umgebung und variable Reaktionszeiten werden modelliert, um ein realistisches Bild der menschlichen Reaktionen in einer Krisensituation zu erstellen. Das Modell berücksichtigt auch physikalische Variablen wie Mobilitätseinschränkungen der Agenten und die Auswirkungen architektonischer Elemente wie Treppenhäuser und Sitzreihen, die sich realistisch auf die Bewegungsgeschwindigkeit auswirken. Es wurde eine umfassende Reihe von Experimenten durchgeführt, um die Evakuierungsleistung in verschiedenen Szenarien zu bewerten. Diese Szenarien variieren systematisch Parameter wie die Anordnung des Hörsaals (z. B. traditionell, gestuft, breiter Saal), die Konfiguration der Ausgänge, die Anzahl der Personen und das Vorhandensein von Blockaden an den Ausgängen. Die wichtigsten Bewertungskriterien sind die Gesamtdauer der Evakuierung, die Durchflussrate der Personen, die Bildung von Engpässen und das durchschnittliche Panikniveau. Die Ergebnisse zeigen eine signifikante Korrelation zwischen der Architektur des Hörsaals und den Evakuierungsergebnissen, wobei bestimmte Layouts andere hinsichtlich der Verringerung von Staus und der Gesamtfluchtzeit durchweg übertreffen. Die Ergebnisse unterstreichen auch die entscheidende Rolle menschlicher Verhaltensmuster und zeigen, wie Phänomene wie Herdenverhalten zu suboptimalen Ausgangsentscheidungen und erhöhten Staus führen können, selbst wenn klare Wege verfügbar sind. Diese Forschung bietet wertvolle Erkenntnisse für Universitätsverwaltungen und Sicherheitsplaner. Sie stellt ein leistungsfähiges Instrument zur Bewertung der Sicherheit bestehender Hörsäle und zur Gestaltung sicherer Bildungsräume dar.

Declaration

I hereby declare that I have composed the thesis independently, have not used any sources or aids other than those indicated, and that all parts of the thesis that have been quoted or summarized from other sources are marked as such. I also confirm that the thesis has neither been submitted in this form nor in a similar form to any other examination authority.

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Chapter 1

Introduction

Human behavior during emergency evacuations is influenced by a complex interplay of psychological, physical, and environmental factors. In scenarios such as fire alarms or hazardous events within university lecture halls, response patterns may range from calm and orderly movement to disorganized panic. This variability is shaped by personal attributes like mobility, sensory impairments, or stress resilience, as well as by situational pressures such as rising congestion, visibility constraints, and perceived proximity to danger [1]. These human responses often deviate from the assumptions of purely rational pathfinding models, which makes it crucial to integrate behavioral realism into evacuation scenario analysis.

From a simulation perspective, understanding the dynamics of evacuation requires accounting for multiple sequential stages: detection of the event, decision-making time before initiating movement, travel on the floor of origin, and navigation through vertical evacuation routes such as staircases [2]. Each stage introduces potential delays, particularly if evacuees hesitate due to uncertainty or adhere to habitual exit routes rather than optimal ones. For example, individuals frequently prefer familiar paths even when alternatives might be faster, reflecting an inclination towards the known over the unknown [3]. This tendency can persist even in open spaces with clear sightlines to alternative exits.

Psychological states play an enormous role in shaping patterns of movement. Panic can manifest in behaviors like pushing toward a single congested door despite other viable escape options. Social dynamics compound these effects; for instance, herding behavior emerges when people lack sufficient information about what to do and instead follow visible groups [4]. In more nuanced social structures, small pre-formed groups, whether friends or classmates, may coordinate their movements implicitly or explicitly through a leader-follower dynamic. The resulting flow is a composite outcome of individual preferences and group-level decision processes.

At the same time, architectural layouts exert strong structural influences on evacuation efficiency. Seemingly minor factors like desk arrangements or door positioning can accelerate or hinder movement flow. Studies using symmetrical configurations have shown marked improvements in clearing times compared to certain asymmetric door placements [5]. Similarly, disparate utilization rates among multiple exits suggest that pedestrians are not making choices solely based on geometric shortest paths but also taking into account perceived crowding and spatial redirection costs [6]. These patterns indicate that both environmental affordances and real-time evaluations contribute to evacuation performance.

Given these behavioral and architectural complexities, modeling approaches that treat ev-

ery occupant as an independent actor without interaction effects risk oversimplification. Agent-based models address this shortcoming by simulating micro-level decision-making for each agent while still allowing emergent macro-patterns to be observed. Such models make it possible to incorporate frictional forces between agents during panic states, vision-based limitations in exit perception, and variations in risk tolerance across personality profiles [7]. Moreover, integrating autonomous decision-making agents into simulated environments enables exploration of cooperative versus competitive strategies under different stress intensities. Agent-based modeling in platforms like NetLogo [8] is particularly apt because it provides fine-grained control over individual agent attributes while enabling batch runs through tools like BehaviorSpace for examining varied parameter sets without statistical post-processing complexities. NetLogo's graphical interface further allows tracking how agents' interactions evolve spatially over time [9].

By embedding realistic perceptions, such as whether an agent sees an exit or trusts signage, as well as social attachments into these simulations, modelers can capture a richer spectrum of plausible behavior. A consideration worth noting is that emergency preparedness levels significantly influence outcomes. If individuals have prior familiarity with evacuation routes thanks to drills or repeated exposure to the space layout, their reaction speed improves. Conversely, untrained individuals using specialized evacuation devices (for instance in cases involving mobility-impaired evacuees) may move slower or hesitate more at critical junctures [2]. The gap between trained and untrained response underscores why models must discriminate between heterogeneous agent types.

This thesis builds upon frameworks that highlight the synergy between building characteristics and occupant traits during emergencies. Researchers have argued that effective evacuation planning must consider not just physical layouts but also human tendencies influenced by anxiety levels, environment familiarity, signage effectiveness, and even prior social commitments [1]. Such integration allows for more credible scenario testing, from worst-case panic surges clogging the main exit observed in some studies [10], to calmer dispersals where crowd self-organization alleviates bottlenecks. Ultimately, incorporating both psychological diversity and spatial variation into simulations grants insight into how bottlenecks form and how they might be alleviated without relying on overly idealized assumptions about human conduct during crises. By experimentally adjusting configurations, such as altering exit sizes or positions, and replicating various behavioral modes (panic rushes versus rational calculations), one can explore alternative strategies for risk reduction before actual emergencies occur. These insights set the foundation for systematically comparing evacuation approaches applied within university lecture hall contexts while retaining fidelity to both physical constraints and human unpredictability.

1.1 Research Gap and Questions

Research on agent-based models in evacuation scenarios is growing, but there aren't many studies specifically about using them in lecture halls. Existing models largely focus on generic public spaces such as shopping malls, transit hubs, or open arenas, with comparatively less attention devoted to the distinctive architectural and occupancy characteristics of educational buildings. Lecture halls, however, introduce unique challenges: fixed seating arrangements, constrained aisle widths, teaching areas, and high occupant densities during peak times. These factors shape evacuation dynamics in ways that differ significantly from more generic built environments. This thesis seeks to address these gaps by investigating how agent-based modeling can be adapted to capture the architectural, behavioral, and social-psychological complexities of lecture hall evac-

uations. The research is guided by the following questions:

- How can agent-based models be tailored to simulate emergency evacuations in lecture hall settings?
- What are the critical factors influencing evacuation efficiency in these environments?
- How do different architectural designs and occupancy patterns impact the effectiveness of evacuation strategies?
- How can an agent-based model be adapted to reflect the unique architectural and human behavioral characteristics found in lecture hall evacuations?
- What behavioral rules and decision-making processes should be incorporated to simulate realistic evacuation scenarios?
- How do variations in lecture hall design (e.g., exits, seating arrangements) affect evacuation efficiency and safety?

1.2 Research Objectives

This thesis aims to develop and analyze a comprehensive agent-based simulation model for emergency evacuation in university lecture halls. The primary objectives are to create a realistic agent-based model in NetLogo that accurately represents multiple lecture hall layouts with varying architectural features, incorporates heterogeneous agent behaviors grounded in established evacuation psychology, and accounts for physical constraints such as movement speeds, personal space, and stair navigation. Additionally, the model integrates dynamic factors including panic contagion and exit choice adaptation.

A central focus of the research is to investigate the impact of different evacuation strategies on overall evacuation performance. This includes quantifying the effectiveness of panic-driven versus orderly evacuation, analyzing the influence of social dynamics and herding behavior, and evaluating the benefits of exit familiarity and rational decision-making. The thesis further seeks to compare evacuation performance across a range of lecture hall designs, such as traditional single-exit layouts, center-aisle configurations, tiered seating arrangements, and dual-aisle designs.

Another key objective is the systematic identification and characterization of bottleneck patterns during evacuation. This involves mapping common bottleneck locations, quantifying the severity and duration of congestion, and analyzing the relationship between specific layout features and the formation of congestion. Based on these analyses, the thesis aims to provide practical, evidence-based recommendations for optimal lecture hall design to improve evacuation efficiency, inform emergency procedure planning and training, enhance exit signage and visibility, and support capacity planning in light of evacuation constraints.

In addition to these primary goals, secondary objectives include validating the simulation model against established evacuation time calculations and empirical observations, conducting sensitivity analyses to identify critical parameters affecting evacuation outcomes, developing visualization tools for effectively communicating evacuation dynamics to stakeholders, and creating a flexible modeling framework that can be extended to other educational environments.

1.3 Thesis Structure

Before going into detail, the following paragraphs give a short overview of this thesis and its structure:

Chapter 2 (Background and Context) provides a comprehensive review of the evolution of emergency evacuation research in public buildings, emphasizing the transition from observational studies to advanced agent-based modeling. This shift in research methodology enables the capture of nuanced human decision-making processes, group dynamics, and architectural constraints. The article emphasizes the significance of integrating behavioral science and spatial analytics in safety design, particularly in complex environments such as lecture halls. It demonstrates the superiority of agent-based simulation over traditional models in predicting realistic evacuation outcomes and informing proactive building safety strategies.

Chapter 3 (Literature review) explores how crowd dynamics and human behavior, especially under panic and stress, shape emergency evacuation outcomes, emphasizing the influence of social factors like herding and leadership. It contrasts microscopic (agent-based) and macroscopic modeling approaches, highlighting the advantages of agent-based models especially those built on platforms like NetLogo for capturing behavioral diversity and architectural constraints in educational environments such as lecture halls. Lessons from past research underline the need for holistic safety design integrating spatial, behavioral, and procedural factors, validated through both empirical data and simulation.

Chapter 4 (Methodology) details the use of agent-based modeling in NetLogo to simulate emergency evacuations in lecture halls, comparing NetLogo's flexibility and behavioral realism to other platforms. It describes the conceptual design of the model, including diverse agent behaviors and layouts, outlines a rigorous experimental protocol varying key parameters such as population size and panic levels, and specifies comprehensive metrics and validation measures for analyzing evacuation effectiveness under realistic conditions.

Chapter 5 (Results) analyzes over 9000 evacuation simulations, showing that evacuation strategy, architectural layout, population size, and occupant preparedness are decisive factors in lecture hall safety. Modern layouts with multiple aisles and exits consistently outperform traditional designs, especially under high-stress and high-density conditions, while calm and rational behaviors, environmental familiarity, and early alarm responses lead to the safest and fastest evacuations. The findings highlight strong interaction effects — optimal safety is achieved only through integrating effective design, training, and management.

Chapter 6 (Discussion) interprets simulation results, showing that both architectural design and human behavior critically shape evacuation safety and efficiency in lecture halls. Symmetrical exit placement, wide aisles, and clear sightlines foster balanced flow and lower crowding risk, but their benefits depend on occupant familiarity and composure; behavioral tendencies like herding or panic can undermine even optimal layouts. The chapter recommends integrated design and policy approaches combining spatial improvements, training, and dynamic guidance systems while acknowledging model limitations related to agent diversity, hazard realism, and architectural complexity, stressing the need for site-specific empirical validation before real-world application.

Chapter 7 (Conclusion) demonstrates that safe and efficient evacuation from lecture halls depends on the interplay of architectural design, behavioral dynamics, and environmental conditions, with balanced sightlines, wide aisles, and rational agent behavior yielding the best outcomes. Future directions suggest leveraging AI and machine learning for enhanced realism and optimization, integrating BIM for precise spatial modeling, and scaling methods to entire academic complexes ensuring that evacuation planning evolves alongside technology and empirical evidence for greater safety and adaptability.

Chapter 2

Background and Context

2.1 Emergency Evacuation in Public Buildings

Early research into evacuation behavior was largely observational, relying on recordings from fire drills, field notes, and photographic studies of pedestrian movement. Much of this work was motivated by events where crowd disasters occurred and the existing assumptions about human motion under stress proved inadequate. Studies from the social sciences emphasized that individual decision-making in emergencies is not purely instinctive panic but often follows recognizable patterns shaped by social interactions, environmental awareness, and prior training. This conceptual shift moved analyses away from treating evacuees as homogeneous particles and toward recognizing them as agents, people or groups with beliefs, goals, and adaptive strategies. [11]

2.1.1 Historical Perspective on Evacuation Studies

In the mid to late twentieth century, empirical investigations shed light on "normal pedestrian behavior" under non-crisis conditions. These formed a baseline for understanding deviations during emergencies [11]. Early categorization of crowd motion phenomena, such as string formation, stop-and-go waves, turbulence zones, and the so-called "faster-is-slower" effect, provided a vocabulary for researchers to link micro-interactions with macro traffic flows [1]. The observation that organized patterns can emerge spontaneously even without centralized control inspired later theoretical models focused on self-organization. By the 1990s, mathematical modeling efforts like the social force model began formalizing these behavioral processes. In its original form, this model treated pedestrians as particles subject to attractive and repulsive forces while moving toward their goals; yet it also incorporated nonlinear interaction terms that could naturally reproduce empirically observed lane formation or oscillations at narrow bottlenecks [12]. Significantly, these simulated patterns arose without assuming deliberate strategic planning, suggesting that much human motion in crowds may result from local reactions rather than top-down reasoning. Parallel to these theoretical developments, practical investigations concentrated on data from fire drills in both educational and commercial buildings. Researchers weighed the trade-offs between realism and convenience: drills provided structured opportunities for measurement but lacked the authentic stress of an actual hazard. Still, in situations without direct exposure to danger cues like smoke or heat, drill data has been considered a reasonable proxy for true emergency responses [13]. This assumption allowed accumulation of large

datasets under controlled circumstances, which helped refine timing benchmarks for building codes and safety guidelines. From there, experimental designs involving controlled modifications to furniture layout, exit visibility, or door configurations emerged. For example, some early classroom evacuation experiments explored how delays, such as keeping doors closed briefly before opening, might give individuals time to orient themselves better. Yet such delays could also extend total evacuation times when confusion increased [14]. These findings highlighted how relatively small architectural or procedural choices might significantly alter outcomes. Spatial choice has remained another recurring theme in historical studies: participants often select exits based on visibility or habitual use rather than optimal dispersal strategies. Evidence from lecture hall evacuations showed higher preference for doors in clear view or ones serving as main entrances [9]. Such tendencies reinforced arguments that visual fields and learned familiarity must be factored into design evaluations alongside nominal capacity calculations. Crowd disasters in high-density venues pushed historical research further towards integrating group psychology into modeling. Large social units frequently move cohesively even when conditions would suggest splitting up is faster. This cohesion can aggravate congestion if multiple such groups target the same exit [6]. Before computational tools matured enough to simulate these patterns at scale, such behaviors were documented qualitatively through post-incident analysis. It is within this context that the significance of group behavior became fully acknowledged. Another line in the historical trajectory is the investigation of herding tendencies when information is incomplete or conflicting.

Earlier theories often portrayed such collective movement negatively, as mere panic, but subsequent studies showed it could sometimes aid dispersion if it led people toward multiple exit options [10]. However, when coupled with limited visibility, herding could also generate dangerous pressure buildups at single points of evacuation. With accumulating empirical observations came increasingly complex simulation approaches. Agent-based modeling frameworks allowed incorporation of decision rules akin to cognitive processing models like Beliefs–Desires–Intentions (BDI), where occupants' goals are mediated by their perception of hazards and environmental changes [11]. The integration of such cognitive representations marked a departure from earlier flow-based methods by acknowledging intentionality and variability among individuals. Early applications of these advanced simulations validated many field observations while providing a toolset to explore hypothetical architectural configurations without physical reconstruction. What distinguished them historically was their ability to reproduce emergent social behaviors based on localized rules, a direct intellectual descendant of earlier self-organization work, which meant that long-recognized crowd phenomena could now be tested virtually against design alternatives before implementation. Viewed over time, evacuation research progressed from manual observation of drills and post-event reconstructions toward deeply parameterized computational experiments grounded in decades' worth of behavioral insight. Historical milestones include the codification of motion patterns through self-organization concepts [1], formal dynamical models capturing these patterns mathematically [12], critical reevaluations of drill representativeness [13], and layered agent-based decision systems reflecting psychological nuances documented across varied public building contexts [9]. This trajectory set the stage for current studies that increasingly merge spatial analytics with realistic depictions of human conduct under duress, a methodological synthesis already apparent in recent scenario-driven simulations deployed within educational environments.

2.1.2 Importance of Evacuation Modeling in Safety Design

The role of evacuation modeling in safety-oriented building design has become increasingly apparent as architectural complexity and occupancy densities have grown. It is not solely a matter of predicting how quickly a space can be cleared, but of identifying the interactions between human decision-making, group behaviors, and spatial constraints that will ultimately shape evacuation outcomes. Robust modeling enables architects, engineers, and safety planners to address critical questions before construction or retrofitting begins, questions about bottleneck locations, exit placement efficacy, or the adequacy of signage under impaired visibility conditions [6]. Evaluations based only on static capacity indicators, such as the nominal flow rate through a doorway, have proven insufficient. In high-density settings like lecture halls, aisles with varying slopes can alter evacuees' preferred routes even when absolute travel distances favor alternative exits. People may opt for a longer but downward-sloping aisle over climbing toward an exit above them, introducing route-choice asymmetries that would be overlooked without dynamic simulation [3]. Behavioral models uncover such subtleties by coupling environmental geometry with probable decision heuristics employed by occupants during crises. This capacity to anticipate non-intuitive patterns is one of the primary reasons simulation tools are integrated into modern safety design workflows.

Agent-based simulations add another dimension by making it possible to integrate diverse agent profiles within the same environment. Individuals unfamiliar with a building's layout exhibit slower progress due to navigational hesitation and obstacle encounters, particularly under reduced visibility from smoke [15]. If those individuals are also in poor health or mobility-impaired, their clearance times can increase dramatically. By accounting for these parameters in pre-construction analyses, designers can identify where additional guidance systems or physical modifications might mitigate excessive delays for vulnerable subpopulations. Group dynamics are another essential component captured effectively by behavioral modeling. Smaller social units tend to coordinate their actions through ongoing communication; this cohesion can either facilitate error avoidance in route choice or introduce delays while consensus forms [6]. Larger groups often display more complex patterns, splitting up in some cases or stubbornly adhering to mutual proximity in others, which must be accounted for to avoid overly optimistic clearance time estimates. Simulating these scenarios reveals how distributed decision-making interacts with physical constraints like doorway width or aisle narrowness to produce unexpected congestion points.

The historical pattern of treating layouts and occupant behavior independently has given way to integrated approaches where changes to either domain can be evaluated for their combined effect. For example, modifying classroom capacities from 30 to 25 students has been shown to significantly shorten calculated emergency evacuation times in simulations [1]. Such data has immediate policy relevance: regulatory bodies might consider maximum occupancy adjustments as an inexpensive intervention compared with costly structural renovations. Another important feature of simulation-based design is scenario flexibility. The same model can assess routine alarm evacuations under calm compliance as well as higher-stress situations involving panic surges toward visible exits. Visibility itself plays a major role: doors placed centrally on walls tend to attract more uniform use across the population than corner doors of equivalent width because they remain within more occupants' lines of sight across varied seating configurations [10]. Testing both arrangements computationally gives empirical backing for choices that confer real safety benefits yet would be expensive, or impossible, to trial physically. Certain models in-

corporate predictive spatial accessibility measures alongside human behavioral rules to evaluate how minor geometric changes ripple through overall efficiency. These approaches underscore that strategic placement of fire safety facilities and clear visual access routes matters as much as raw evacuation capacity. For instance, rerouting congested flows by repositioning exits in hall designs with dual evacuation points has been demonstrated to improve performance markedly compared with single-exit baselines [14]. Moreover, virtual modeling supports cost-benefit reasoning that extends beyond obvious construction expenses. The price for adding additional exits and signaling systems may appear high at first glance, but when weighed against potential legal liabilities from injury compensation after poorly managed evacuations, investment in optimized evacuation features becomes far more justifiable [1]. Such economic framing often proves persuasive for stakeholders whose priorities center on budgetary constraints rather than life-safety benchmarks alone. Empirical datasets collected from field drills feed directly into refining predictive accuracy. Data from drill evacuations in buildings with diverse stair widths, floor counts, and occupant densities has been used both to validate agent-based models and test their capability against real-world timings [2]. By adjusting model parameters until simulated results align with observed values, such as movement speeds under crowding, designers gain confidence that prospective alterations will perform as intended under actual emergency loads.

Modeling also allows designers to probe "what-if" cases relating to rare but catastrophic hazards without risking actual lives. This includes extreme scenarios involving fire spread progression coinciding with social clustering behaviors or partial exit blockages due to structural collapse. By mapping how congestion waves form and dissipate when an exit is removed mid-evacuation, planners understand tolerances within the current design and identify backup pathways or behavioral advisories that could avert fatal slowdowns. Ultimately, building safety planning benefits from recognizing evacuation modeling as more than a compliance exercise; it is an iterative process linking behavioral science with architectural pragmatism. Whether focusing on minimizing panic-driven crowd compression at choke points [10], anticipating asymmetric density build-ups due to exit placement [6], or safeguarding slower-moving individuals who cannot traverse obstacles optimally [15], these models give tangible foresight into design decisions whose impact cannot be easily reversed once construction is complete. When embedded early in the design cycle, evacuation simulations transform safety considerations from reactive risk mitigation into proactive hazard prevention grounded in realistic human behavior projections rather than abstract flow formulas alone.

2.2 Lecture Hall Design and Constraints

The spatial configuration of a lecture hall shapes both the physical possibilities and psychological preferences guiding occupants during evacuation. Features such as exit positioning, aisle width, door dimensions, and seating arrangements interact to influence not only the maximum theoretical throughput, but also the distribution of occupants across available routes. Even seemingly minor design variations can create significant disparities in evacuation efficiency by altering how congestion forms and dissipates. The concept of bottlenecks is especially relevant here: doors or aisles narrower than upstream spaces act as constraint points where flow becomes choked [3].

2.2.1 Architectural Features Affecting Evacuation

Once pedestrian density reaches a critical threshold around these points, queuing effects propagate backward into the crowd, lengthening overall clearance times. Within lecture halls, exits may be placed symmetrically, either along a vertical or horizontal axis, or asymmetrically, with respect to the seating layout. Symmetrical placement often promotes a more balanced distribution of evacuees across doors, whereas asymmetrical designs tend to bias flow toward the nearest or most visible exit, sometimes leading to underutilization of secondary routes. Multiple studies exploring symmetrical versus non-symmetrical door configurations have shown measurable impacts on total evacuation time when occupant numbers are high [5]. This stems partly from proximity effects and partly from visual accessibility; an exit positioned along a primary line-of-sight attracts disproportionate attention compared to one obscured by furniture or positioned at an oblique angle to most seats. Visibility plays a crucial role in exit choice. High visibility reduces uncertainty and promotes earlier initiation of movement toward less familiar doors. Conversely, if an exit remains hidden behind architectural features, such as tall stage elements or complex seating arrangements, it is less likely to draw evacuees even when it offers equal or shorter travel distances [9]. Haghani and Sarvi's observations indicated that distance alone does not determine choice; rather, it combines with perceived safety anchored in line-of-sight clarity. In situations where smoke or other visibility-reducing conditions are present, this effect is magnified. Seating arrangement dictates mobility pathways within the evacuation space. Rows with narrow intervening gaps impede swift movement toward aisles; staggered seat layouts can require longer detours before reaching a main passageway.

When experimenting with various desk placements combined with multiple door configurations, simulation models have demonstrated that certain pairings can reduce congestion waves substantially compared to others [5]. For instance, aligning wider aisles directly toward multiple exits rather than funneling all rows toward a single central corridor results in greater parallelization of flow. The structural incline of lecture halls further complicates navigation decisions and influences travel speed. An inclined floor sloping downward toward the stage may facilitate faster descent for those heading toward exits located at the front but will disincentivize movement against the slope when rear exits are viable alternatives. The presence of multiple vertical levels introduces differential effort costs that affect route selection under time pressure. Doorway size sets an upper limit on flow rates. In spaces where all exit doors share equal widths, such as three exits each at 0.71 m, the idealized capacity might seem evenly distributed; however, human tendencies disrupt this equilibrium in practice [16]. Aisle width interacts with these door constraints: narrow side aisles (e.g., 0.595 m) can prove more constraining than the door itself if they serve as feeders for large seat blocks. Simulation and empirical observation alike underscore that increasing aisle width often produces more pronounced gains in clearance times than marginal increases in door width alone, especially when room geometry forces converging flows at acute angles before doorway entry [3]. Occupant load density strongly modulates how these architectural factors are expressed in real evacuations. At lower densities, asymmetries in exit use may not produce excessive delays because queues do not reach saturation levels. At high densities approaching seating capacity, 140 students in some case studies, the interaction between narrow feeder aisles and concentrated demand at preferred exits causes rapid queue formation with cascading delays into downstream spaces [16]. Bottlenecks routinely appear in similar locations across configurations when other factors remain constant, highlighting predictable weak points for intervention planning [1]. Another dimension influencing evacuation performance is obsta-

cle placement relative to primary movement flows. Desks bolted to floors force detours around fixed objects; random spacing gaps between clusters exacerbate path inefficiencies when aligning toward an exit not located along the central axis [4]. Agent-based modeling within such environments incorporates these obstructions explicitly, revealing that partial blockage by obstacles can reduce achievable flow rates far below those predicted by simple capacity formulas. Importantly, many architectural parameters cannot be examined effectively without simultaneously accounting for behavior-driven variables such as group cohesion or herding tendency. Herding behaviors may either concentrate demand on certain exits or serendipitously balance it if influential leaders direct their followers toward underused routes [9]. However, reliance on this stochastic balancing effect is risky since it depends on contingent social dynamics rather than fixed design properties. While physical alterations like widening aisles or repositioning doors incur material costs, their potential benefits must be weighed against these enduring behavioral patterns shaped by layout choices at design time. Case simulations offer compelling evidence that changing spatial geometry can yield improvements even if behavioral tendencies remain constant, highlighting that while human decision patterns may be resistant to change without training or signage upgrades, well-designed layouts can mitigate their negative consequences without requiring such interventions upfront [5]. Thus, integrating knowledge of architectural influences, from macro-scale alignments like symmetrical door placement down to micro-scale details like desk spacing, not only improves predictive fidelity in simulations but also informs actionable recommendations for built environment modifications aimed at improving emergency evacuation outcomes under varied loading conditions and behavioral modes.

2.2.2 Impact of Exit Placement and Number

The configuration of exits in lecture halls, both their spatial placement and their total number, exerts a decisive influence on evacuation dynamics. Experimental work in controlled settings with multiple exits has demonstrated that the distribution of evacuees across these points rarely achieves the even balance assumed by capacity-based calculations. When exits are symmetrically placed with comparable visibility, usage tends to be more equitable, which in turn lowers the likelihood of severe congestion at any single point [5]. In contrast, asymmetrical arrangements often concentrate flows toward one or two preferred doors, generally those situated along a direct line of sight from the bulk of seating. This crowding effect becomes particularly pronounced in higher occupancy scenarios, where demand for evacuation can quickly exceed localized capacity. The number of exits available is not just a question of meeting regulatory minimums but also impacts decision-making under stress. Providing additional exit options, when well-positioned and clearly visible, can reduce average travel distances and disperse crowd pressure. Simulations run in experimental environments with three available exits show that participants tend to redistribute themselves more effectively than when only two or one exit is provided. However, distance is not the sole determining factor; factors such as immediate accessibility through surrounding furniture layout, visibility past obstructions, and path perceived safety all moderate these benefits [6]. A third door might serve little practical purpose if it is hidden behind partitions or poorly integrated into natural movement lines. Studies focusing on adjacent wall placements suggest that certain symmetrical cases, determined both by regulatory constraints on door positioning and by geometric feasibility, produce superior clearance times compared to others [5]. For instance, placing doors too close together can undermine their potential benefit as independent evacuation points because queues merge prematurely. Conversely, exces-

sive separation without adequate directional cues may deter use of more distant exits entirely for some occupants. When assessing the interplay between exit placement and human behavior through agent-based modeling frameworks, it appears that psychological tendencies such as herding can override purely rational spatial optimization. Groups often converge toward widely chosen routes simply because others are heading that way [17]. In high-density lecture hall evacuations under stress conditions, this can nullify the intended efficiency gains from multiple exits. Even when three exists are present, poor alignment with natural traffic channels may still result in underutilization of one or more doors [6]. The width and approach path to each exit further complicate this picture. Architectural guidelines may stipulate equal widths across all exits, for example, 0.71 m each, but these capacities will not be equally realized if feeder aisles vary considerably in width or length [3]. Narrow or obstructed approach routes slow access disproportionately compared to their nominal downstream capacity. In some instances narrow feeder aisles have been shown to impose greater throughput limits than the door opening itself. Another aspect influencing performance involves social group sizes relative to exit distribution. Larger pre-formed groups often negotiate consensus about which route to follow instead of dispersing spontaneously among several options [6]. This collective decision-making can lead to abrupt surges at particular exits when one group's choice influences nearby individuals through local social contagion mechanisms [17]. These dynamics are difficult to predict through static diagrams alone; they require dynamic modeling to capture feedback effects between ongoing occupant movements and remaining crowd distributions. Optimal configurations derived from simulation experiments suggest not merely increasing the number of exits but carefully integrating them into primary movement pathways while ensuring clear visibility across wide sectors of seating [6].

In environments where architectural changes are limited by existing construction constraints, other measures, such as improved signage placement aligned with sightlines from most seating positions, can partially compensate for suboptimal exit locations [9]. Illuminated or color-coded guidance systems during emergencies have been proposed to counteract habitual biases toward entrance doors used during arrival. Empirical tests also indicate that asymmetric exit layouts combined with incomplete information about alternative routes tend to exacerbate congestion waves compared with symmetric setups under identical occupant loads [5]. This variability underscores that an additional door does not automatically translate into proportional improvement in clearance time; design must consider how people will actually perceive and select between options when faced with urgency. From a planning perspective, scaling up the occupant load without adjusting exit numbers or optimizing placements elevates risk sharply once density thresholds where queues merge uncontrollably are crossed. Under such conditions, behavioral inertia, refusal or reluctance to change course mid-evacuation, means that even nearby unused exits remain untapped resources while bottlenecks intensify elsewhere [17]. Models incorporating both spatial geometry and decision heuristics demonstrate that well-distributed moderate-width portals outperform fewer extra-wide exits if they encourage simultaneous high utilization. Altogether, evaluating exit placement and number is an exercise that cannot be detached from behavioral context. The interplay among architectural affordances, environmental cues, and group-level tendencies produces real-world patterns that differ from idealized flow predictions. Agent-based simulations have proven especially useful for articulating how changing either placement or quantity interacts with innate evacuation heuristics like proximity preference and visibility-driven choice bias, a relationship vital for refining lecture hall safety designs before emergencies test them under uncontrolled conditions.

2.3 Agent-Based Modeling in Evacuation Research

Agent-based simulation revolves around representing a system as an ensemble of autonomous entities, or agents, each following its own set of rules while interacting with others and the surrounding environment. In evacuation modeling, this approach offers a means to portray occupants not as undifferentiated flows, but as discrete decision-makers whose actions emerge from localized perceptions, goals, and constraints [18].

2.3.1 Principles of Agent-Based Simulation

Each agent operates under internal mechanisms that account for attributes such as autonomy, proactiveness, veracity, and adaptability, qualities that align with observed human behavior in real emergencies [5]. Autonomy implies that an agent does not require top-down control to participate effectively in the simulation; it perceives its surroundings and decides how to act within those constraints. Proactiveness ensures that each entity pursues objectives, such as reaching an exit or staying close to a social group, rather than simply reacting to environmental triggers. Veracity reflects the notion that information held by agents is truthful within the model's parameters, which influences how decisions are made when conflicting cues are present. Adaptability allows ways of modifying behavior dynamically when conditions change, for example when smoke alters visibility or a chosen route becomes congested. The utility of this paradigm emerges most clearly when systems defy simple analytical representation. In situations where individual behavior cannot be reduced to aggregate transition rates between states without losing essential nuance, rule-based specification at the agent level provides richer detail. Complex actions, like sidestepping another evacuee, choosing between competing exits based on observed queues, or altering pace due to emotional stress, quickly become infeasible to model using only differential equations due to combinatorial explosion in state descriptions. Agent-based modeling sidesteps this limitation by embedding behavioral complexity into localized rules executed in parallel across the population.

A defining feature of such simulations is their capacity to include stochasticity precisely where it matters. Instead of introducing arbitrary noise terms into macroscopic equations, randomness can be injected into specific decision points, like hesitation time before moving after an alarm or probability of following a neighbor toward an exit [17]. This placement produces variability consistent with empirical observations of human responses rather than producing uniform perturbations unrelated to actual cognitive processes. For example, some scenarios incorporate empirically derived probabilities for selecting particular evacuation criteria, distance to door vs. social following, and apply them differently across seat positions in a lecture hall [9]. Agents seated at row ends might prioritize copying neighbors' decisions diagonally ahead of them, whereas those closer to doors could base their choice purely on proximity. From a systems perspective, an agent-based approach synthesizes concepts from multi-agent systems (MAS) research, distributed artificial intelligence, and software engineering for autonomous components [15]. This interdisciplinary foundation enables realistic representation of both physical interactions, where agents may collide or slow down due to crowd pressure, and cognitive reactions influenced by changing environmental cues. The MAS view emphasizes that effective simulation depends not solely on individual design but also on how agents interact: coordination in congested aisles, competition for space near bottlenecks, or collaborative path adjustments if obstacles appear unexpectedly. Another significant principle involves environment dependence in

defining rules and strategies. Functions such as building maps and factor assignment routines determine how agents interpret their spatial context [10]. A model may incorporate explicit representations of architectural nodes and boundaries; based on an agent's location relative to these nodes, context-relevant evacuation rules are assigned dynamically. This ensures adaptive responsiveness, for example if one area requires detouring due to fire spread while another maintains direct evacuation routes, and illustrates how design-specific critical points can be integrated into movement logic. The inclusion of psychological realism is often cited as a strength of this modeling paradigm. Integrating aspects like herd behavior into rule sets allows simulations to capture emergent patterns where individuals' limited information leads them to follow visible groups [17]. Under certain conditions this can equalize exit usage; under others it causes dangerous overloading at particular points. The balance between deterministic pathfinding and influence from surrounding agents mirrors field observations where both calculation and imitation coexist in human decision-making. For implementation platforms such as NetLogo [8], with tools like BehaviorSpace, the principle extends into systematic exploration of parameter spaces without external statistical computation layers [9]. Designers can run extensive batches varying factors like movement speed variance, reaction delays after alarms, or likelihoods of group cohesion persistence under stress. Observing outcome distributions across these runs reveals sensitivities within the system: perhaps modest increases in herd tendency sharply reduce efficiency under asymmetric exit layouts but have negligible impact in symmetric ones. The granularity at which decisions are represented also carries implications for fidelity and computational load.

High-resolution models may simulate intermediate states like pausing mid-route to reassess options when encountering congestion, a capability important for replicating observed pre-movement delays documented in evacuation research. Such pauses are difficult to reproduce credibly without an explicit agent architecture that allows interruptions and redirection. Importantly, calibration against empirical data plays a crucial role in making these simulations credible tools rather than speculative exercises. Even with detailed psychology-inspired rule sets, outputs must align with real-world measures such as average speeds under crowding or distribution of route choices given identical physical layouts [2]. Field observations from drills feed back into refining probability assignments for key behaviors; discrepancies become grounds for revisiting either rule logic or environmental representation. Finally, ABM's flexibility supports iterative refinement aligned with specific strategic goals, be they evacuation time minimization through layout changes or testing alternative emergency communication protocols [17]. By linking internal agent logic tightly with architectural representation and running repeated what-if scenarios under varied behavioral assumptions, ABM facilitates nuanced evaluation where no single discipline's toolkit suffices alone. It brings together spatial constraints from architecture, cognitive variability from psychology, and dynamic flow effects from physics into a unified modeling framework capable of producing emergent insights about complex evacuation phenomena.

2.3.2 Advantages Over Other Modeling Techniques

Compared to aggregate flow-based or purely equation-driven approaches, agent-based modeling offers a distinct capacity to capture heterogeneity at the level of individual decision-makers. Traditional deterministic models often rely on averaged parameters such as mean walking speed or uniform reaction time, which obscure variability in abilities, intentions, and interactions that can critically shape evacuation trajectories [18]. By contrast, assigning each simulated occupant a unique combination of attributes, ranging from mobility constraints to differing levels of envi-

ronmental knowledge, enables reproduction of phenomena that would otherwise be invisible in aggregate metrics. This granularity is especially vital in contexts where small deviations at the individual level can cascade into large-scale effects, such as bottlenecks triggered by hesitation or path-switching mid-evacuation. Another advantage is the flexibility with which complexity can be scaled up or down to suit research goals. An agent's behavioral repertoire might be kept minimalist for preliminary sensitivity tests or enriched with adaptive strategies, evolving preferences, and probabilistic decision-making pathways for more comprehensive analyses [17]. This adaptability extends beyond individual agents to the model's structural resolution; it is straightforward to shift between simulating entire crowds as monolithic entities and breaking them into interacting sub-groups without fundamentally altering the modeling framework. In spaces like lecture halls where group cohesion plays a non-trivial role in route choice and congestion patterns, such scalability makes it possible to assess both macro- and micro-level influences within the same architecture. A defining strength lies in reproducing emergent behavior without prescribing it explicitly. Classical queuing models need carefully calibrated parameters to approximate observed non-linearities in pedestrian flow, while ABMs can generate these patterns naturally from local interaction rules [1]. Examples include lane formations in aisles, unplanned oscillations at doorway pinch points, and self-organization mechanisms that reduce clogging under certain crowd compositions. Because these dynamics arise from individual-level rules rather than being hard-coded into system-level equations, they respond organically to perturbations in environment or agent population characteristics, a property difficult to achieve using rigid analytical models. Integration of spatially explicit environmental data is another area where ABM excels. The ability to embed actual floorplans with detailed obstacle representations allows agents' decisions to be tied directly to architectural affordances [11].

In rigid mathematical formulations, spatial complexity must often be drastically simplified, reducing corridors to abstract nodes or exits to undifferentiated capacity points, which strips away meaningful context about sightlines, approach angles, and crowd merging zones. An ABM can instead map the interplay between geometry and social tendencies: for instance, how a partially obscured rear exit remains underused even when congestion elsewhere suggests it should be an attractive alternative [6]. Beyond static representation, ABMs incorporate dynamic environmental changes during simulation with relative ease. Routes can become blocked by developing hazards like smoke spread; visibility may degrade over time; signage cues might alter paths mid-evacuation. Embedding such evolving conditions within equation-based frameworks tends toward computational complexity blow-up due to increasing state-space dimensions. Within an ABM, however, these variations simply constitute updates to what each affected agent perceives at successive time steps [7]. This perception-action loop underpins more credible depictions of human adaptation under stress. Incorporating psychological realism is comparatively direct as well. Variables representing tendencies toward herd following [17], social attachment maintenance [1], or competitive pushing behaviors can be toggled on or off across experimental runs without rewriting core algorithms governing basic movement physics. This modularity is particularly advantageous when exploring scenario-specific hypotheses about behavior, for instance comparing panic-induced rushes versus rational dispersion, as it allows isolating the impact of a single behavioral tendency while holding physical design constant. From a calibration standpoint, ABMs are compatible with empirical datasets gathered from both drills and real-world incidents [2]. Rather than fitting global coefficients (e.g., average density-speed relationships) across all occupants, calibration targets specific rule parameters like threshold distances triggering pace reduction near others or probabilities of exit-switching after encountering a queue.

This alignment ensures that discrepancies between simulated and observed outcomes drive refinements exactly where the model's logic diverges most sharply from recorded human conduct. Furthermore, this modeling paradigm accommodates direct analysis of emergent distributions rather than just central moments like means or medians. Because agent-level outputs are retained individually over time steps, analysts can inspect variance spectra of clearance times across populations, for instance identifying long-tail delays linked to persistent underuse of certain exits [1]. Such disaggregated perspective often proves essential for safety design evaluations aimed at worst-case risk assessments rather than average performance. The transparency of linking local rules to system-wide patterns also facilitates interdisciplinary collaboration. Engineers focused on structural constraints understand how modifications ripple outward via altered crowd flows; psychologists recognize their theories instantiated in agents' bounded rationality heuristics; policy-makers see concrete implications of code-mandated exit widths or placement on modeled hazard scenarios. Other methodological approaches rarely offer this intuitive chain from micro-assumption through macro-outcome with equal clarity [18].

Practical usability complements theoretical power when using platforms like NetLogo for implementation [9]. With accessible graphical interfaces for visualizing simulations in real-time alongside scripting environments for custom rule definitions, iterating designs requires neither specialized high-performance computing infrastructure nor deep numerical optimization expertise. The built-in batch experiment tools (e.g., BehaviorSpace) remove dependence on external statistical pipelines by enabling exhaustive sweeps over parameter combinations within a consistent framework, something less seamless in continuum-based simulators absent substantial programming overhead. Finally, ABMs provide unmatched versatility for "what-if" experimentation constrained only by imagination and conceptual coherence rather than solvability limitations inherent to closed-form methods [17]. Scenarios ranging from partial evacuation due to localized hazards through complete building clearances under differing alarm speeds can be juxtaposed without fundamental remodeling effort. This agility supports not just retrospective analysis but proactive design prototyping under a breadth of contingencies unlikely ever to be tested physically without prohibitive cost or safety risk. By maintaining logical continuity between how individuals think and move within the space and how structure channels those behaviors, ABM stands out as an integrative bridge across disciplinary boundaries concerned with evacuation safety research and its application in complex built environments.

Chapter 3

Literature Review

3.1 Crowd Dynamics and Human Behavior

Panic and stress in emergency situations are not monolithic phenomena; they manifest through a dynamic spectrum of cognitive, emotional, and physiological responses that profoundly affect evacuation behavior. Under acute threat, individuals may experience heightened arousal levels that impair deliberate decision-making, replacing calculated route selection with instinct-driven actions that can amplify congestion risks [18].

3.1.1 Panic and Stress Responses in Emergencies

This state often emerges from an interplay between perceived immediacy of danger, environmental cues such as visible smoke or alarm intensity, and the surrounding crowd's behavior. While some people maintain rational assessment during stress, others quickly transition into high-panic states characterized by reduced attention to alternative escape paths and a narrowing of focus toward the most immediately visible evacuation option [10]. Such tunnel vision can exacerbate uneven exit usage patterns already noted in high-density spaces, reinforcing localized bottlenecks even when redundant capacity exists elsewhere in the layout. Physiologically, stress responses elevate heart rate and respiration, further limiting sustained physical performance during evacuation. Cognitive bandwidth shrinks, sensory input is processed selectively, favoring cues deemed most urgent or salient over more effective but less obvious alternatives [18]. In architectural contexts with poor visibility or complex navigation requirements, these perceptual constraints compound the difficulty of adaptive pathfinding. Herding emerges readily under such conditions as individuals align their movements with those of proximate others in an attempt to minimize uncertainty [4]. From a modeling perspective, simulating this involves weighting cohesion factors heavily while discounting intelligent judgment outputs in motion algorithms [10], thereby replicating observed real-world tendencies for groups under stress to prioritize following over independent optimization. Stress thresholds represent another critical factor shaping shifts from cooperative to competitive behavioral modes. Once internal stress exceeds a particular limit, agents may prioritize self-preservation in ways that disregard collective efficiency [4]. Competitive behaviors include attempting to pass through partially obstructed exits despite interpersonal collisions or exploiting transient movement gaps without regard for orderly queuing. In model terms this is expressed via expanded movement ranges and reduced compliance with space-sharing conventions around goal cells. Such transformations in behavioral

strategy often occur abruptly once thresholds are crossed, mirroring field observations where crowd mood changes rapidly from calm compliance to aggressive jostling. These threshold effects are neither uniform nor deterministic; pre-existing differences in nervousness or anxiety predispose some individuals to reach critical stress levels sooner than others. Personality traits interact with situational variables to modulate timing and severity of panic onset. Experienced occupants who have participated in drills or encountered prior emergencies may demonstrate greater resistance to stress escalation, preserving cooperative tendencies longer than untrained peers. Conversely, novices unfamiliar with escape routes or equipment, such as fire extinguishers, often show hesitation phases followed by abrupt panic surges once movement begins en masse. Incorporating these heterogeneities into simulation agents significantly increases the realism of modeled outcomes. Social context influences both the likelihood and consequences of panic states. Occupants holding leadership roles may act as stabilizing forces by directing followers toward less congested routes [11], though leaders themselves are not immune to stress contagion effects. Panic can propagate within a local cluster much like an emotional wave: as one person exhibits overt signs of fear-driven haste, nearby individuals increase their own arousal via social mimicry [10]. This collective amplification means that even small initial perturbations, a single person running prematurely toward an exit, can trigger cascading competitive surges if unchecked by countervailing signals. Architectural features also interact non-trivially with panic dynamics. Narrow aisles and obstructed sightlines increase uncertainty about available options, heightening reliance on peer guidance just when independent assessment would be more efficient [13]. Under severe stress conditions this leads to disproportionate use of frontally visible doors irrespective of travel distance or congestion level downstream [9].

Additionally, vertical circulation elements like stairways tend to slow movement more under stress because coordination between steps becomes less precise when fine motor control erodes under adrenaline's influence, a factor traditional flow models often overlook. Agent-based implementations benefit from explicitly capturing how panic modulates movement speed as well as decision quality. Models can treat velocity updates as functions smoothed exponentially according to current panic values, as observed where increasing γ_i reduces reliance on intelligent path planning results in favor of maintaining momentum along whatever trajectory is already underway [10]. This reproduces empirical patterns where panicked evacuees persist on suboptimal routes rather than reassessing mid-course despite worsening congestion ahead. Another dimension worth noting is the difference between spontaneous emergent competitiveness and deliberate obstruction in high-stress scenarios. Many empirical accounts conflate pushing at exits with antisocial intent; however, modeled results suggest that such behaviors often arise mechanically once density surpasses comfortable interpersonal spacing thresholds and perception narrows due to stress overload [4]. Agents "competing" for space may simply lack cognitive bandwidth for cooperative micro-adjustments rather than consciously disregarding others' welfare. Stochastic elements play a notable role too: panic onset times and intensities vary due to random positioning within the crowd and differential proximity to immediate cues like heat or loud noises [5]. Even simulations starting from identical macro conditions exhibit divergent micro-sequences when stochastic distributions govern who first begins moving quickly versus hesitating, differences that cascade into varied clearance times across repeated runs. From a preventive standpoint, these behavioral insights argue strongly for design interventions that mitigate known panic triggers: optimizing exit visibility across seating zones; maintaining aisle widths above thresholds prone to triggering competitive mode switches; supporting leader identification through clear hierarchy cues before crises occur; and practicing drills to raise individual familiarity baselines

so fewer agents default early into high-panic states [11]. In modeled environments implementing these changes reduces both frequency and severity of competitive surges even when global hazard parameters remain constant. The multiplicative complexity of individual predisvalueses toward anxiety combined with environmental constraints underscores why simply increasing nominal exit capacity fails as a catch-all solution during intense emergencies [18]. Behavioral adaptation under stress produces asymmetries that raw throughput calculations cannot anticipate without embedding cognitive variability into evaluation frameworks. By representing panic not as an abstract scalar but as context-sensitive modulation across perception, choice heuristics, cooperation propensities, and locomotion parameters, agent-based models generate evacuation trajectories that resonate far more closely with documented human conduct during real crises, and consequently yield designs better suited for saving lives before chaos reaches its peak expression.

3.1.2 Social Influence and Herding Effects

Social influence exerts a substantial impact on evacuation dynamics, particularly in conditions where situational awareness is impaired or the level of perceived threat is high. The behavior of individuals often shifts from independent decision-making to alignment with surrounding agents when environmental information is insufficient or ambiguous [10]. This herding phenomenon, while sometimes adaptive in unfamiliar environments, carries the potential for maladaptive congestion when it directs disproportionate numbers toward certain exits regardless of their actual utility. Observations indicate that in emergencies with limited visibility, individuals tend to favor following others who appear to have made a movement decision over independently assessing alternate routes [6]. The cognitive rationale behind this bias lies partly in the efficiency of outsourcing decision-making under uncertainty, but in evacuation contexts it can inadvertently amplify bottlenecks. Under structured experimental settings, agents with minimal environmental knowledge display an increased propensity to replicate the movements of nearby peers [10]. This is often modeled by assigning higher weights to goal velocity components dictated by perceived group direction rather than optimal pathfinding algorithms. The consequence is that clusters form quickly and move collectively, with their density increasing sharply near points of convergence such as doorways.

Herding tendencies also manifest differently across agent types: so-called follower agents may show near-total reliance on visual cues from others, whereas conservative or active agents balance environmental scanning with social imitation [7]. This delineation has significant implications for evacuation planning because populations are rarely composed solely of one behavioral type. Group structures intensify these effects. Social groups, pre-formed units like friends or classmates, often maintain proximity during evacuation even if doing so increases travel distance. In lower-density situations this cohesion may pose little risk; however, at high densities and especially near bottlenecks, side-by-side formations transition into single-file lines that elongate queues. Under extreme densities described as “river-like” group shapes, interaction constraints multiply because overtaking becomes more difficult and flow lane structures break down. Experimental data suggest groups sometimes walk slower than individuals due to coordination needs, though in certain contexts they may outpace lone evacuees due to mutual encouragement or prioritized path clearness through collective movement [6]. Herd-driven congestion tends not to distribute evenly across all available exits. When exits differ in visibility or apparent accessibility, group-following behavior magnifies existing biases toward more obvious evacuation points

[9]. A rear exit obscured by obstacles can remain chronically underused even during dangerous saturation at the main entrance; herding amplifies this imbalance as late-arriving evacuees interpret clustered movement toward one door as implicit confirmation that it is the correct choice [5]. This aligns with field evidence showing that balanced use of symmetrically placed exits only occurs when visibility and approach clarity are equally favorable. From a simulation standpoint, reproducing these dynamics requires coupling perception-driven decision rules with interaction parameters that govern spatial cohesion [10]. For instance, micro-level models can increase alignment factors (m_l) to reflect an agent's tendency to orient toward collective heading vectors, while maintaining separation factors (m_s) sufficient to avoid physical interference unless panic overrides them. In high-panic contexts (as discussed in Section 3.1.1), separation forces diminish and proximity tolerance increases, leading not just to tighter clustering but also to elevated risk of injury from competitive squeezing at choke points. The emergent effect of social influence is context-dependent: moderate herding under distributed exit visibility may equalize utilization by encouraging exploration beyond familiar routes when early movers choose less obvious paths. Conversely, when initial flow concentrates heavily on a primary exit due to its salient position or previous habitual use, herding accelerates critical density formation there while other viable paths are left idle [6]. In behavioral terms this can be seen as a form of positive feedback loop where each additional defector from independent choice further reinforces the crowd's momentum in a single direction. An important nuance lies in leadership roles within crowds. Leaders, whether formal (e.g., staff) or emergent through confident behavior, can redirect followers toward underutilized exits if their actions are visible early enough to counteract established flows [11]. Yet leadership efficacy depends on local observation radius; followers who cannot see alternative choices due to obstructions will default back to copying those immediately nearby. The propagation speed of new directional information through a crowd thus hinges on both visual connectivity across space and pre-event familiarity with layout options.

Some computational implementations address these dynamics by introducing variable "influence radii" for each agent, the area within which another's movement can trigger a shift in orientation weightings for goal selection [10]. Larger radii tend to promote stronger global synchronization (and thus more uniform flow directions), whereas smaller radii yield fragmented patterns where multiple small groups independently converge toward different sources of guidance. These parameterizations allow exploration of how architectural design might foster beneficial influence fragmentation: for example, partitioning seating areas such that each has good sightlines to separate exits reduces the chance that one dominant herding path consumes capacity across the entire hall. Randomness also interacts nontrivially with social imitation. Two runs of an evacuation involving identical initial distributions can diverge significantly based on which individuals happen to move first, and where they head, because these few early decisions can set directional norms adopted by those nearby [5]. Simulation batches using tools like BehaviorSpace reproduce this variability by randomizing initial reaction times and leader positions across iterations without altering macro-layout parameters. By measuring dispersion levels in exit usage across runs, analysts can quantify the robustness, or fragility, of flow balance under given architectural and behavioral profiles. Ultimately, social influence and herding represent double-edged elements in evacuation dynamics: they can reduce hesitation delays when experienced movers guide uncertain occupants effectively toward safe evacuation routes; yet they can also precipitate overload at specific constrictions through uninformed mass following driven more by perceived consensus than actual advantage [7]. Design strategies aiming to mitigate negative herding outcomes range from architectural adjustments like enhancing symmetrical

exit visibility through lighting and clear sightlines [9], to procedural interventions such as distributing trained personnel throughout venues so that no single observation point dominates directional cues during crises. In simulations explicitly modeling heterogeneous agent types and adjustable alignment behaviors, these interventions produce measurable declines in peak densities at primary bottlenecks without degrading total clearance time, a result suggestive of practical benefits achievable by deliberately shaping the vectors along which social influence propagates during emergencies.

3.2 Evacuation Modeling Approaches

In evacuation research, the distinction between microscopic and macroscopic models is marked by their treatment of individual representation and the level of behavioral granularity embedded in the simulation. Microscopic approaches treat each evacuee as a distinct entity possessing its own position, velocity, decision logic, and interaction rules with other agents and the environment. This agent-level formulation allows a natural incorporation of heterogeneous attributes, such as differences in reaction time, mobility constraints, or familiarity with the layout, that are difficult to integrate faithfully within larger-scale aggregate formulations. By contrast, macroscopic models regard the crowd as a continuous flow or homogeneous mass governed by averaged parameters like bulk density, mean speed, and collective pressure dynamics [18].

3.2.1 Microscopic vs Macroscopic Models

These representations resemble fluid- or gas-dynamic systems, where local variations are smoothed into field variables over space and time. The implications of these contrasting perspectives are profound when applied to complex built environments. Macroscopic models can offer computational efficiency and broad-stroke predictions suitable for large-scale planning but often at the expense of detailed behavioral fidelity. For instance, they may accurately forecast total clearance times under idealized conditions yet fail to replicate local bottleneck formation arising from small-scale interpersonal interactions or obstruction-induced queuing. In such cases, density-flow relationships derived from bulk observations govern movement without capturing transient deviations caused by hesitation, overtaking attempts, or path-switching, all of which can matter significantly in lecture hall evacuations with irregular seating geometry. Microscopic models excel at reproducing localized emergent phenomena without requiring them to be pre-specified. Empirically documented patterns such as lane formation in bidirectional flow or oscillatory exchanges at narrow doorways can arise spontaneously through simple interaction rules between agents [12]. The social force model represents one prominent example: framing each individual as an unstructured particle influenced by internal goals (reaching an exit), social-psychological forces (avoiding collisions), and environmental constraints. These equation-based micro-models translate cognitive tendencies into deterministic or stochastic forces guiding motion; their agent-based counterparts embed similar mechanisms but often with greater flexibility in defining discrete decision events or rule-triggered state changes [18]. A further differentiation lies in how the two approaches respond to environmental complexity. As building layouts become more intricate, with multiple floors, varied aisle widths, partial obstructions, and heterogeneous visibility, macroscopic models must drastically simplify spatial representation to remain tractable. This simplification risks eliminating critical path dependencies that govern route selection under uncertainty. Microscopic models instead map actual floorplans directly into the simulation

space. Agents perceive exits conditionally based on line-of-sight checks, navigate around fixed obstacles according to movement heuristics, and make course corrections dynamically when encountering congestion waves [9]. This closer coupling between architectural features and behavior makes it possible to evaluate design elements like staggered seating arrangements or asymmetric exit placement in terms of both qualitative changes in flow structure and quantitative impacts on clearance metrics. Even so, microscopic modeling is not without trade-offs. As population size increases into the thousands, as might occur in simulations of multi-hall complexes, the computational cost rises sharply due to the need to compute interactions for every agent pair within influence range at each timestep [18]. Algorithmic optimizations like neighborhood partitioning can mitigate but not eliminate this scaling issue. Moreover, while these models provide rich output detail, including granular trajectories, they can suffer from overparameterization if insufficient empirical data exists for calibration. Parameters governing personal space tolerance, visibility thresholds for exit recognition, or social influence weights must be estimated judiciously; poorly constrained values risk producing realistic-looking but inaccurate results. An intermediate perspective also exists: mesoscopic modeling blends elements from both paradigms by grouping individuals into small cohorts treated as single units within a larger flow formulation [12]. Such hybridization preserves some heterogeneity while retaining lower computational demands than fully microscopic models. Yet for scenarios where psychological factors like panic escalation or herding strongly condition outcomes, as explored earlier, mesoscopic aggregation may still obscure decisive micro-level behaviors. From an emergency planning standpoint, model choice influences practical recommendations.

A macroscopic approach might suggest that increasing total evacuation width proportionally reduces clearance times across scenarios. A microscopic simulation could reveal that due to herding biases toward front entrances and asymmetric aisle routes [5], widening certain rarely used exits has negligible benefit unless paired with measures improving awareness or accessibility to those alternatives [6]. Another point of divergence is temporal resolution. Macroscopic frameworks often operate on continuous-time differential equations governing density fields; they smoothly evolve states without distinguishing discrete events like individual hesitation delays after alarm activation. Microscopic models handle these naturally, agents can incorporate predefined reaction delays before initiating movement based on distributions drawn from empirical drill data [2], thereafter accelerating subject to localized crowd pressures and personal comfort zones. This event-level sequencing provides not only final clearance predictions but also insights into intermediate phases where intervention might be most effective, for example deploying staff guidance early enough to redirect flow before bottlenecks harden. Integrating psychological modules further widens the gulf between approaches in terms of realism under stress conditions. Panic response mechanisms, such as increases in desired speed coupled with reduced adherence to collision-avoidance norms, can be embedded into microscopic agents via modifications to force magnitudes or decision weights once internal panic variables surpass set thresholds [10]. Implementing equivalent dynamics at a macroscopic level requires introducing non-linearities into density-velocity functions whose parameters may lack straightforward behavioural interpretations. Finally, validation practices differ markedly. Macroscopic model calibration relies on fitting global curves (like fundamental diagrams) against observed density-flow data; deviations from mean trends are treated as noise rather than meaningful variability [18]. Microscopic validation compares ensemble distributions of many runs against empirical distributions from drills or recorded evacuations [2], enabling identification of tail risks, e.g., rare runs where queue imbalances dramatically extend clearance despite favourable average conditions.

These tails may drive safety-critical decisions even if they barely affect central tendency metrics. Balancing these factors suggests neither approach is inherently superior for all tasks; rather they occupy different niches along a spectrum defined by scale, complexity, behavioral focus, and resource constraints. For high-density lecture halls where architectural idiosyncrasies interact tightly with non-uniform occupant behaviors, including panic onset variance and social following, a microscopic modeling framework appears better suited to expose the micro-mechanisms underlying emergent congestion patterns that aggregate flows might smooth away before they can inform targeted design adjustments [18]. However, adopting hybrid strategies that combine micro-based detail in key subzones with macro-level propagation elsewhere could offer pragmatic compromises when expanding scope without discarding crucial nuance near primary hazards or choke points.

3.2.2 Agent-Based Models in Evacuation Studies

Agent-based models (ABMs) have established themselves as particularly effective tools for studying evacuation scenarios in contexts where individual-level variability and interaction effects cannot be ignored. They allow each simulated occupant to be represented as an autonomous entity possessing physical, cognitive, and social attributes that drive decision-making [19]. This capability contrasts sharply with aggregate approaches, enabling the explicit modeling of situations in which people's choices are shaped by a combination of environmental perception, social influence, and personal constraints. In the context of lecture hall evacuations, ABMs can reproduce how visibility limitations toward exits, prior familiarity with the space, or training from fire drills affect the timing and direction of movement initiation. By using platforms like NetLogo [8], these decisions can be connected directly to a spatial grid replicating actual architectural layouts [9], ensuring that navigation rules consider not just distances but also obstacles such as rows of fixed desks or narrow feeder aisles. A hallmark of ABM utilisation in evacuation studies is their ability to capture emergent crowd dynamics without imposing them externally. For example, herding toward visible exits often develops spontaneously when agents follow local observation rules, orienting toward others who have begun moving, rather than calculating optimal dispersal paths [17]. This quality is central for assessing asymmetric exit usage under identical capacity conditions.

In simulation runs where exits differ in visibility or lie on different approach angles relative to seating blocks, uneven utilization emerges naturally because agents process line-of-sight information coupled with existing movement cues [6]. These micro-level interactions can thus expose potential bottlenecks invisible in uniform flow assumptions. Incorporating heterogeneous behavior profiles allows further refinement. Some agents can be parameterized to act as "leaders" with greater environmental knowledge or stronger orientation toward underused evacuation points, while others act as "followers" that rely on proximate group direction [7]. Such differentiation enables side-by-side comparison of scenarios that include trained personnel dispersed throughout the hall versus unstructured crowds. Results often show that early leader movements redirecting flow toward less crowded exits can reduce congestion significantly if visibility allows followers to perceive these alternative routes [9]. ABMs also handle conditional adaptation under dynamic hazard progression more effectively than static models. As fires spread or smoke reduces visibility in certain sectors, agent decision rules can shift, reducing preference for originally chosen paths if perceived safety declines beyond tolerance thresholds. This adaptability aligns closely with real-world reports of evacuees altering courses mid-journey due to situational

changes rather than following initial plans blindly [19]. Combining such adaptive routing with stochastic delays for reaction time simulates phases observed in empirical evacuations: hesitation post-alarm, accelerated movement once cues intensify, and possible re-routing upon encountering congestion. The flexibility afforded by ABMs extends into integrating nuanced physical interaction models when density increases. Collision avoidance algorithms or simplified social force formulations embedded at the agent level produce realistic speed reductions in congested aisles and doorways without requiring manual calibration of density-flow curves per scenario [10]. Importantly, these localized slowdowns feed back into decision-making cycles: when progress through a selected exit becomes too slow relative to perceived alternatives within sight range, agents may switch targets, though herding tendencies can counteract this rational reassessment if surrounding peers persist in queuing at the initial bottleneck. Simulation environments like NetLogo offer accessible ways to run large batches of such experiments using BehaviorSpace for parameter sweeps across behavioral and architectural variables [9]. For instance, systematically varying aisle widths while holding exit positions constant can reveal non-linear improvements in clearance times that plateau once other constraints dominate flow rates [3]. Similarly, adjusting proportions of panic-prone versus calm agents quantifies how social contagion effects magnify or dampen initial imbalances in exit use. Another advantage lies in explicitly representing fixed architectural features alongside mobile agents, something especially relevant for lecture halls where desk arrangements and floor inclines channel movement unevenly [1]. By mapping such elements into the grid and encoding their impact on speed or navigability (e.g., detours around obstacles; slower ascent up tiered seating), ABMs capture route selection biases driven by ease-of-movement perceptions even before hazard proximity is factored in. This is critical when evaluating interventions like adding secondary aisles or modifying furniture layouts to enhance distribution across multiple evacuation points. Integration of cognitive-behavioral constructs further enhances realism. Decision heuristics may prioritize proximity over congestion avoidance unless training modifies these priorities, a distinction supported by drill observations where participants familiarized with alternative exits distribute more evenly than naïve counterparts [19].

Similarly, modeling partial information conditions via restricted vision radii around doors reproduces underuse patterns seen when rear or side exits are obscured from much of the seating area [6]. From a design evaluation perspective, ABM outputs offer more than simple clearance times; they deliver spatial-temporal maps showing how densities evolve across the environment from alarm onset to final clearance. These maps highlight when and where critical thresholds are exceeded, points at which movement efficiency collapses and injury risk spikes due to competitive squeezing or stalled queues. Analyses have shown that repositioning even a single exit within clear visual range of larger seat blocks can redistribute flows enough to avoid crossing such dangerous thresholds entirely [5]. Importantly, validation against empirical data ensures credibility; drill timings and observed route choices calibrate parameters like walking speeds under load or probability functions governing exit switching under congestion [3]. Where discrepancies emerge between modeled and observed outcomes, for instance if simulated herding intensity overshoots real measurements, they prompt revision not only of behavior rules but sometimes of assumed environmental representation fidelity. In sum, ABMs provide a laboratory for experimenting with architectural modifications, procedural changes such as enhanced guidance systems, and demographic-shift scenarios like higher proportions of mobility-impaired occupants without risking actual lives [7]. They bridge psychological theory about individual cognition under stress with engineering concerns about physical evacuation capacity through an

interactive feedback loop between agents' perception-driven actions and structural affordances. When deployed thoughtfully, with calibrated parameters, varied behavioral typologies, and accurate spatial mapping, these models supply richly detailed insights into why certain designs succeed or fail under plausible emergency loads that simpler aggregate methods would struggle even to articulate.

3.3 NetLogo as a Platform for Agent-Based Simulation

NetLogo provides an environment particularly conducive to the type of agent-based modeling described earlier, where agents act autonomously yet interact within a shared, spatially explicit setting. Its architecture allows for modeling agents on a discrete grid, supporting both simple and complex behaviors without requiring extensive programming infrastructure. Key to its suitability is the ease with which heterogeneous attributes can be assigned to each agent, from physical characteristics like movement speed and vision range to psychological parameters such as panic propensity or social influence weighting, and subsequently modified during runtime as environmental conditions shift [9]. This capacity aligns well with evacuation scenarios in which human decision-making must adapt on-the-fly to changing visibility, congestion, or hazard spread. One advantage in using this platform lies in how it merges visual transparency with computational flexibility. Modelers can observe simulations unfold in real time through its integrated interface while also running large batches of experiments via BehaviorSpace. This module automates variation across user-defined parameters, such as aisle widths, agent densities, or behavioral composition, generating multi-run datasets for comparative analysis without exporting intermediate results to external statistical tools [20]. The ability to rapidly reconfigure agent rules and environmental layouts encourages iterative exploration of "what if" scenarios that would be too cumbersome in more rigid simulation frameworks. Graphical representation is more than cosmetic here; it enables immediate inspection of flow patterns and congestion development. For example, when testing alternative exit placements in a virtual lecture hall layout, the visual traces of agents can quickly reveal underutilized doors or high-pressure bottlenecks forming persistently across runs [9]. Because these patterns emerge directly from localized interactions encoded in each agent's logic rather than being imposed globally, they offer insights that can inform architectural design modifications before implementation. NetLogo's scripting environment is versatile enough to implement behavioral diversity without sacrificing clarity. Rulesets governing panic escalation or herd following can be coded alongside deterministic pathfinding heuristics, allowing different proportions of the population to employ different decision strategies [7]. Since these behaviors are modular within the codebase, they can be toggled or adjusted independently, a feature especially useful for isolating the impact of specific tendencies.

Furthermore, agents' perceptions are easily linked to the geometry of the environment; line-of-sight calculations to exits or other features can be implemented relative to seating arrangements or obstructions imported into the simulation grid [9]. Another strength is scalability for multi-level modeling. The same platform has supported broad-ranging applications, from simulating market microstructure at NASDAQ using learning agents [17] to participatory simulations in urban planning contexts, which demonstrates its capacity to host models combining behavioral rules with evolving environments [20]. In evacuation research this flexibility means that fire spread dynamics or changing hazard zones could be layered onto an existing occupancy model without abandoning underlying movement logic. Although NetLogo operates primarily on a discrete space abstraction, its patch size and update frequencies are adjustable enough to approxi-

mate continuous movement where needed. The integration of environmental data is relatively straightforward: vector drawings of floorplans can be converted into grids representing accessible space versus obstacles, allowing building geometry from CAD-like designs to be mapped directly into simulation input [7]. Such fidelity is important when evaluating designs where small geometric details, like desk placement or aisle constrictions, meaningfully alter flow efficiency. By embedding these spatial features at model initialization, researchers ensure that emergent patterns reflect both behavioral interactions and physical affordances. Critically, NetLogo supports direct incorporation of stochasticity at desired decision points rather than across entire systems indiscriminately. For instance, random variation can be introduced into pre-movement delays following an alarm signal, differing among agents based on familiarity levels with the building layout [9]. Similarly, probabilistic exit choice functions can reflect observed distributions from empirical drills while still permitting deviations driven by dynamic perceptions during the run. This layered stochastic control leads to output variability consistent with real-world phenomena. From a practical standpoint, NetLogo's lower entry barrier compared with custom-coded simulation engines facilitates involvement by multidisciplinary teams, including those from architecture or psychology backgrounds unfamiliar with complex programming, while still offering depth for advanced users [20]. The language's procedural nature accommodates straightforward environmental updates (e.g., closing an exit mid-run) alongside continuous tracking of metrics like evacuation times or density profiles at key choke points. These outputs are available both visually and as raw data exportable for further qualitative inspection. When used for high-density lecture hall evacuations, NetLogo's capacity for rapid iteration is particularly valuable because small changes, shifting an exit by one patch width, can yield measurable differences in predicted clearance times due to nonlinear crowd dynamics [9].

Batch analysis through BehaviorSpace then reveals whether observed improvements hold consistently across variations in group composition or reaction times. As shown in other domains using similar architectures [17], such systematic experimentation uncovers parameter sensitivities that single-run inspections might overlook. The platform also lends itself well to exploratory hybridization between established theoretical models and empirically motivated rules. For instance, simplified social force mechanics could govern short-range movement adjustments around obstacles while higher-level route-selection remains rule-based and perception-driven, a coupling achievable within NetLogo's event loop without prohibitive complexity penalties [7]. This blending enables researchers to capture both continuous-space navigation realism and discrete-decision cognitive effects typical in stress-laden evacuations. Finally, adaptability over project lifespans should not be underestimated. Longitudinal applications highlighted in unrelated fields, such as educational policy simulations using NetLogo's dynamic multi-level models, suggest that once an evacuation framework is built, it can be progressively extended: adding modules for special populations (mobility-impaired agents), integrating communication system failures, or scaling up from one hall to interconnected building complexes [20]. In evacuation strategy testing this means initial focused studies can evolve into broader campus-level safety evaluations without departing from a validated codebase. Taken together, these qualities make NetLogo not only technically capable but methodologically aligned with research goals requiring repeated trials over diverse behavioral-architectural combinations under consistent experimental control. Its emphasis on accessible visualization married with batch-processing capabilities allows deep exploration of design-performance relationships while maintaining clear traceability from micro-level agent logic through to macro-level evacuation patterns, a traceability essential for translating simulation insight into actionable real-world design recommendations grounded

in empirically plausible human behavior.

3.4 Applications to Educational Environments

Research focusing on evacuation processes in lecture halls emphasizes the intricate interplay between architectural constraints, occupant distribution, and behavioral dynamics during emergencies. Lecture halls differ from many other public spaces in that fixed seating arrangements impose non-trivial path dependencies, agents cannot move directly toward exits but must first navigate through narrow row gaps and converge into aisles before reaching doorways. These structural features interact with psychological tendencies, such as the preference for familiar or visible exits, to shape aggregate outcomes in ways that are not captured by simple capacity multiplications. [9]

3.4.1 Evacuation Studies in Lecture Halls

Empirical and simulation-based studies have documented how seating layouts influence local densities, with wider vertical aisles supporting side-by-side movement of two persons in parallel and thereby increasing throughput without modification to exit width [9]. Conversely, tightly spaced rows constrain lateral evacuation from seats, delaying entry into main aisles regardless of aisle width. Agent-based explorations have been applied to elevated-floor lecture halls where individuals must traverse sloped planes or stairs before reaching an exit. Here, incline direction plays a subtle role: occupants may be more inclined to move downslope than upslope when alternative routes demand different effort levels [5]. In geometries with three columns of seats feeding into multiple exits, symmetrical placement has been associated with more balanced door usage, whereas asymmetries increase the chance of concentrated queues at one or two highly visible doors [6]. This imbalance is aggravated when certain doors are obscured by seating blocks or partition walls; simulation runs consistently show that such exits remain underutilized unless additional visual cues are provided [9]. The presence and positioning of obstacles also significantly impact flow efficiency in lecture hall environments. Fixed desks bolted to floors create rigid corridors with limited overtaking opportunities. As occupant density rises toward maximum seating load, as in scenarios modeling full occupancy of over 140 students, queue formation at bottleneck points occurs rapidly [1], propagating backward into the seated area and prolonging overall clearance times. Even small blockages or irregular spacing between seat clusters can force detours and produce uneven feeder patterns into aisles [4]. Such effects underscore why models incorporating precise floorplan data yield more reliable predictions than those assuming uniform grid access to aisles. These contexts also illuminate distinct behavior profiles observable during drills and simulated evacuations. Students who have participated in prior evacuation training display modestly higher average movement speeds compared to untrained peers, while individuals with mobility impairments move significantly slower regardless of familiarity. Incorporating these profiles into agent populations reveals non-linear consequences; for example, placing immobile agents near key aisle entry points can effectively throttle downstream flow even if exits themselves offer adequate capacity under homogeneous-agent assumptions. Scenario testing within such ABMs has evaluated how different configurations alter evacuation metrics like total clearance time or number of injured persons due to crushing pressures at choke points. For example, varying access path widths while holding exit locations constant can shift total clearance times across a range from roughly 137 seconds to nearly 189 seconds under otherwise identical con-

ditions. Similarly, introducing obstacles mid-evacuation, representing fallen debris or blocked sections, has been shown to inflate both time-to-clear and simulated injury counts because congestion redistribution is slow once initial herding patterns establish [1]. Behavioral elements like herding effects exert disproportionate influence in lecture hall configurations where most occupants face a common focal point such as a stage. When alarms trigger movement predominantly toward front-stage exits already used as primary entrances on ingress, back or side doors often fail to attract significant flow without explicit intervention [6]. Architecturally symmetrical layouts mitigate this somewhat, but only if sightlines are clear; obstructed views lead to persistent asymmetry even with symmetrical geometry. Introducing trained leader agents who consciously direct others toward underused exits can rebalance distributions in some simulations, but only if line-of-sight allows visual confirmation by followers [7]. Agent-level data elucidates micro-patterns driving these outcomes. Short pre-movement delays observed for some occupants compound congestion later when late starters merge into already saturated queues at preferred exits. NetLogo-based platforms make it straightforward to vary these delays probabilistically across a population, revealing threshold phenomena where slight increases in average reaction time sharply reduce total throughput when physical constraints are tight [9]. Movement speeds also display contextual variation: narrow rows slow advance toward aisles even for fit individuals; conversely wide unobstructed approaches permit faster-than-average paces until joining an exit queue. Investigations often extend beyond static architectures into testing hypothetical retrofits aiming to smooth flows. Widening key vertical aisles feeding crowded central exits tends to improve performance more consistently than marginally increasing door widths alone [3]. Adding extra exits without adjusting furniture placement can yield disappointing gains if approach paths remain constrained; simulations show that poor integration between seat orientation and exit alignment diminishes utilization of new evacuation points despite their nominal availability [6]. Path obstructions leading up to these points exacerbate biases toward the most direct-looking routes. Quantitative evaluations sometimes integrate injury potential proxies such as force thresholds; congested door zones exceeding set limits for compressive forces over time indicate elevated risk even before flow halts entirely [10]. In lecture hall configurations tested using such parameters, high-force incidents cluster predictably at choke points combining high occupant inflow with restricted passage width, confirmation that architectural redress should target upstream distribution as much as doorway sizing. The controlled settings afforded by educational environments provide an experimental advantage: it is possible to calibrate simulation parameters against drill outputs measured down to specific seating sectors or demographic subgroups. Data linking seat location relative to exit positions with observed choice frequencies validates visibility-driven selection rules embedded in ABMs, strengthening inference about the likely outcome of proposed changes [9]. Together, these empirical calibrations and scenario experiments inform design guidelines emphasizing balanced sightlines, avoidance of choke-prone aisle geometries, and targeted training interventions, principles grounded as much in nuanced behavioral modeling as in geometric optimization.

3.4.2 Lessons Learned from Past Research

Prior investigations into evacuation from educational settings have consistently underscored the interplay between physical space configuration, occupant behavior, and procedural preparedness. Analyses derived from both empirical drills and simulation experiments demonstrate that optimizing one domain in isolation yields limited benefit if complementary factors are neglected.

For example, several studies have shown that simply adding exits does not automatically translate into proportional clearance time reduction. The underlying cause often lies in human tendencies to gravitate toward familiar or more visible evacuation points, a bias that can persist irrespective of overall exit capacity [6]. This has been highlighted repeatedly in lecture hall contexts where entrance doors used during arrival attract disproportionate attention during evacuation, leaving side or rear exits underutilized unless visibility is improved or instructional cues are provided [9]. Architectural symmetry emerges as a preventive measure against such imbalances when combined with unimpeded sightlines. Past simulations comparing symmetrical and asymmetrical door layouts reveal that symmetry encourages a more even flow distribution, thereby preventing premature saturation at single bottlenecks [5]. Yet results also caution that visibility obstructions, such as partitions or tall furniture, can negate these benefits by masking available options from segments of the audience [9]. Consequently, lessons learned point toward integrating layout geometry with perceptual access considerations rather than optimizing purely on spatial balance metrics. Another recurring insight concerns the compounded impact of aisle constraints. Studies analyzing fixed-seating venues demonstrate that narrow approach aisles feeding into adequately sized doors can be the primary choke points, making door width expansions less effective than upstream widening interventions [3]. Where rows are tightly spaced, horizontal evacuation from seated positions becomes sluggish, delaying entry into main aisles and creating surges when multiple occupants converge simultaneously [1]. Models enriched with obstacle representation show that even seemingly minor disruptions, an irregular seat block gap or placed equipment, alter flow rates significantly at high densities [4]. From these findings arises the recommendation to consider full-path flow capacity rather than treating doors as independent throughput determinants. Behavioral modeling has reinforced the observation that crowd psychology can undermine rational use of available architecture. Herding effects documented in both real drills and simulated runs push occupants toward already congested routes when uncertainty is high, especially under degraded visibility from smoke or lighting failures [6]. Designating trained guides within sight of large seating blocks has proven effective in redirecting flows toward underused exits in some scenarios, but this relies on their position being visually accessible early in the evacuation process [7].

Experiments incorporating leader agents support this by showing measurable redistribution only when followers retain line-of-sight; otherwise, habitual preferences dominate regardless of physical congestion [9]. Speed heterogeneity among evacuees represents another factor consistently found to influence outcomes more severely than aggregate flow models anticipate. Placement of slower individuals, including those with mobility impairments, in critical feeder positions can impede downstream clearance disproportionately [15]. Simulation-based evaluations suggest spatially dispersing such individuals across multiple aisles rather than clustering them inadvertently improves total throughput without altering physical architecture. Parallel to this is the role of pre-movement delay variability: when late starters enter an already saturated queue at preferred exits, they contribute disproportionately to prolonged clearance times due to merging inefficiencies [9]. Past research also stresses validating model assumptions against empirical data. Calibration using drill measurements for factors like walking speed under congestion or exit choice frequency by seat location has been instrumental in ensuring predictive reliability for proposed interventions [2]. Mismatches between observed and simulated patterns often trace back to oversimplified representations of visibility constraints or social influence weightings. Adjustments informed by these comparisons tend to yield models better aligned with actual behaviors seen in educational evacuations. Fire hazard integration into evacuation modeling offers another

crucial lesson: coupling spread dynamics with movement simulation alters predicted safe evacuation time substantially compared to static-geometry models. Incorporating fire and smoke modules alongside human agents shows how evolving hazards progressively constrain route options and slow movement through impaired zones [7]. This has led to strengthened arguments for cross-domain simulation platforms capable of exchanging hazard states and occupant responses dynamically. From a methodological standpoint, agent-based modeling's flexibility to represent heterogeneous decision-making and dynamic adaptation has allowed testing "what-if" scenarios impractical for drills. For instance, alternately increasing panic-prone agent proportions versus adjusting aisle layouts highlights trade-offs between behavioral mitigation strategies (like training) and structural alterations (like aisle widening) [19]. Such comparisons reveal that while infrastructure changes offer consistent performance boosts regardless of behavioral composition, training effectiveness may diminish sharply when architectural constraints remain severe.

Empirical findings further emphasize scenario-specific optimization instead of universal prescriptions. In one set of simulations for three-block seating with symmetrical door placement, certain symmetry patterns yielded superior outcomes only for specific obstacle arrangements; introducing bypassed obstacles shifted optimal placement elsewhere [5]. Hence design recommendations must be sensitive to full-space interaction effects rather than relying on isolated configuration advantages measured under idealized conditions. Lastly, prior work reinforces the value of integrated evaluation metrics over single-dimension outcomes like average clearance time. Injury risk proxies based on sustained high-density force thresholds identify zones where safety is compromised before complete stagnation occurs [10]. In lecture halls studied using such measures, dangerous pressures built up predictably at repeated choke points across scenarios, echoing earlier conclusions that risk hotspots tend to re-emerge unless addressed structurally through upstream flow redistribution rather than local expansion alone. Collectively these lessons point toward a holistic approach combining clear sightline-oriented design, balanced capacity throughout entire evacuation paths, targeted occupant training with strategic leader positioning, validated heterogeneous-behavior modeling, and hazard-aware scenario planning. Past research demonstrates that only through aligning spatial affordances with empirically grounded behavioral interventions can realistic gains in evacuation safety for lecture halls be realized within feasible resource constraints.

Chapter 4

Methodology

4.1 Introduction

This chapter presents the methodological framework employed to investigate emergency evacuation dynamics in lecture hall environments. The research utilizes agent-based modeling implemented in NetLogo [8] to simulate complex human behavior during emergency scenarios. The methodology encompasses the development of a comprehensive evacuation simulation model, the design of systematic experiments, and the establishment of data collection protocols to analyze evacuation performance under varying conditions. The agent-based modeling approach was selected due to its capacity to represent individual decision-making processes while capturing emergent collective behaviors that arise during evacuation scenarios. This bottom-up modeling paradigm enables the examination of how individual behavioral strategies interact to influence overall evacuation effectiveness, making it particularly suitable for studying the complex dynamics inherent in emergency situations.

4.2 Comparison with Other Simulation Tools

When evaluating NetLogo in relation to other simulation platforms used for evacuation modeling, key distinctions emerge across dimensions such as spatial representation, behavioral complexity, integration capabilities, scalability, and ease of experimentation. One of the most immediate differences is in how environments are modeled. NetLogo provides a two-dimensional grid of patches with mobile agents ("turtles") capable of continuous movement and flexible heading adjustments [20], which allows for more naturalistic trajectories than simulation tools based purely on cellular automata principles. By contrast, platforms rooted in strict cell-based architectures constrain motion to discrete neighboring cells [18], which can lead to unrealistic path geometries when representing non-rectilinear spaces such as tiered lecture halls with curved aisles. This has practical implications: the ability of agents in NetLogo to navigate using arbitrary angles facilitates a closer match to observed human movement patterns in empirical studies [5]. In terms of embedding heterogeneous behavior, agent-based systems like NetLogo clearly differ from macroscopic flow models, which describe aggregate pedestrian densities through continuous equations but cannot directly simulate individual decision-making processes [18]. Such aggregate approaches may efficiently compute broad trends but risk oversimplifying localized phenomena like queue formation at specific doors or the influence of obstacles on micro-routing choices [14]. Specialized evacuation tools grounded in computational fluid dynamics or

physics-inspired particle motion models, while better at simulating force-based interactions under extreme density [21], often treat individuals as homogeneous particles influenced by simple repulsion-attraction forces. This abstraction can omit nuanced psychological factors including group cohesion [6] or herding effects [1], both of which are readily implemented and varied parametrically within NetLogo's programming environment. Validation workflows also vary substantially between toolsets. NetLogo facilitates direct calibration against time-resolved human movement data by allowing parameter tuning for reaction times, walking speeds, and exit choice probabilities until output curves align with empirical benchmarks [2]. Larger engineering-focused platforms often require pre-processing CAD layouts into mesh formats or discretizing spatial elements into finite volumes before simulation; while these steps can produce high geometric fidelity, they may lengthen model development cycles and complicate iterative calibration. In contrast, NetLogo's graphical World View supports rapid prototyping and modification without intermediate meshing stages, enabling validation exercises to iterate more quickly. This supports exploratory research phases where variable sets need adjustment based on preliminary sensitivity analysis results obtained through batch runs in BehaviorSpace. Behavioral rule complexity is another differentiator. While some domain-specific evacuation simulators incorporate simple decision hierarchies (e.g., choose nearest exit if unobstructed; otherwise select alternative), NetLogo allows researchers to construct layered decision-making frameworks incorporating perception limits, probabilistic heuristics, social preferences, and dynamic hazard avoidance all within agent procedures [10].

Compared to rigid procedural models embedded in commercial software, with fixed routing algorithms not readily modifiable, the open programming access in NetLogo permits integration of socio-psychological theories such as those describing individual-collective behavior mixtures under panic conditions [22]. Alternatives like rule-based roadmap systems emphasize precomputed navigation graphs guiding agents through nodes [10]; these can be efficient in structured spaces but may lack flexibility when altering geometry dynamically during a hazard scenario. Integration potential further distinguishes NetLogo from other environments. While high-end simulators sometimes offer native modular coupling with hazard propagation codes (e.g., for smoke or fire spread), this often requires proprietary extensions or costly licenses. NetLogo's extension mechanism enables similar integration by importing externally computed hazard fields, such as smoke opacity maps from VR-based simulations [7], directly into the patch network so that environmental changes modify agent perception and route choice probabilities in real time. Furthermore, its data import/export functionalities allow seamless handoff between simulations and statistical packages for post-processing, whereas other specialized tools may limit accessibility of raw simulation state data or necessitate complex extraction scripts. Visualization capabilities differ not only in aesthetic detail but also analytical utility. Engineering-grade 3D evacuation platforms render photorealistic environments that aid stakeholder engagement but can obscure quantitative congestion diagnostics unless coupled with dedicated analysis modules. NetLogo's visualization emphasizes functional clarity: dynamic density maps and real-time trace plotting can be generated alongside underlying numerical output [20], aiding diagnosis of congestion points identified earlier in theoretical discussions on bottlenecks [1]. Such representations lend themselves well to iterative model refinement since visual anomalies are immediately apparent during playback. From an experimental design perspective, BehaviorSpace offers systematic batch processing across parameter ranges without external workflow management software, a built-in capability not always present in other platforms without scripting add-ons. This facilitates robust sensitivity analyses on variables like door width increments or obstacle place-

ments near exits [5], producing distributions rather than single-value predictions that account for stochastic variation inherent in crowd dynamics modeling [18]. Many traditional optimization-focused evacuation tools center on single best-case scenario calculations given static inputs; while useful for deterministic planning, they do not directly expose the breadth of possible outcomes under behavioral variability unless run repeatedly with randomized seed conditions at user initiative. Scalability considerations highlight trade-offs: large-scale venues exceeding several thousand agents might overwhelm NetLogo's execution speed compared to compiled C++ solvers built for massive agent populations using parallel processing; however for lecture hall scales on the order of hundreds of occupants, a range matching much empirical dataset granularity, NetLogo remains performant while offering unmatched flexibility during real-time model adjustments [20]. Importantly, scaling down more complex simulators to such problem sizes does not necessarily confer added accuracy when empirical calibration remains bounded by available behavioral data resolution. Finally, cost and accessibility considerations influence tool selection. Commercial simulation environments tailored for building evacuation often entail significant licensing fees and require training investment before productive use by non-specialist researchers or students. By contrast NetLogo's open access model and comprehensive documentation lower entry barriers substantially while still supporting sophisticated analyses when combined with rigorous empirical parameterization as emphasized throughout related structural and behavioral examinations in earlier sections. Thus in comparing across the spectrum, from aggregate mathematical models prioritizing computational efficiency to specialized physics engines maximizing micro-level realism, NetLogo occupies a strategically balanced position: integrating detailed behavioral representation with modifiable environmental constraints within an accessible framework capable of generating empirically grounded insights tailored to venue-specific evacuation safety questions.

Comparison with GAMA, Repast, MASON, and Mesa

Beyond the contrasts with commercial and physics-centric platforms, it is instructive to compare NetLogo directly with other popular agent-based modeling frameworks used for evacuation and crowd simulation: GAMA, Repast, MASON, and Mesa.

- **GAMA:** GAMA (Generalized Agent-based Modelling Architecture) offers advanced spatial modeling features, including native support for GIS data and multi-level agent hierarchies [23]. Its environment supports both 2D and 3D visualizations, enabling more detailed representation of real-world layouts. Batch experimentation and parameter sweep functionality are robust in GAMA, similar to NetLogo's BehaviorSpace. However, GAMA's steeper learning curve and programming requirements (GAML language) may pose challenges for rapid prototyping, especially for new users.

- **Repast:** Repast (Recursive Porous Agent Simulation Toolkit) is built for scalability, supporting both desktop and high-performance cluster deployments [24]. It provides flexible agent scheduling and integration with Java and other JVM languages, which enhances extensibility. Repast models can be tightly coupled with external data sources and workflow systems but may require more extensive programming effort and setup than NetLogo for basic simulation tasks. Its visualization capabilities are functional but generally less interactive than NetLogo's World View.

- **MASON:** MASON (Multi-Agent Simulator Of Neighborhoods) is a fast, Java-based toolkit optimized for large-scale agent simulations [25]. While it excels in performance and handles tens of thousands of agents efficiently, its minimalistic default visualization and heavy reliance on manual coding for scenario setup make it less approachable for rapid, iterative model development and exploratory analysis. Compared to NetLogo, MASON prioritizes computational efficiency over ease of use or immediate visualization.

- **Mesa:** Mesa is a Python-based agent-based modeling library that integrates well with Python's scientific stack, making it ideal for researchers familiar with data science workflows [26]. Mesa provides web-based interactive visualization, and its modular architecture allows easy extension with custom components. However, its simulation speed and scalability are typically lower than compiled frameworks like MASON or Repast. While Mesa supports flexible agent behaviors and batch runs, NetLogo's built-in features for visualization and experimentation remain more accessible for educational and prototyping purposes.

In summary, while GAMA, Repast, MASON, and Mesa each offer unique advantages in terms of spatial modeling, scalability, and integration, NetLogo stands out for its balance of usability, flexibility, and direct support for rapid experimentation and validation—especially in contexts where ease of model adjustment and direct behavioral encoding are paramount. The choice of platform should be guided by the scale of the simulation, required integration capabilities, and the technical expertise of modelers, but NetLogo remains a compelling option for evacuation modeling in academic and applied settings due to its open access and robust agent-based features.

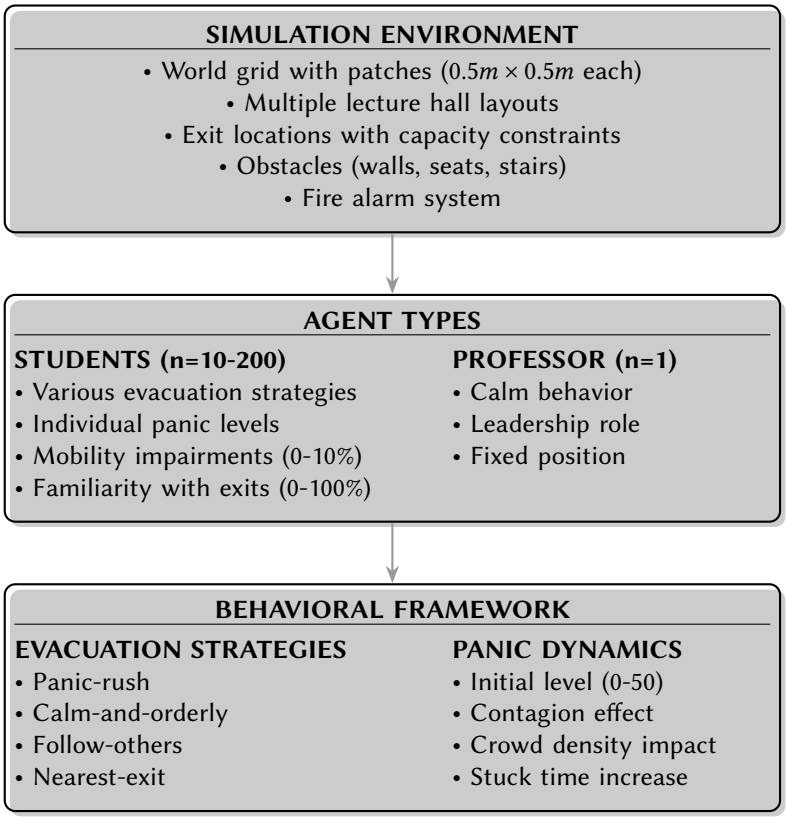
4.3 Conceptual Model Design

The simulation was implemented using the NetLogo 7.0.0 modeling environment, a platform widely recognized in academic and research communities for its robust capabilities in simulating complex systems. NetLogo's design principles align closely with the conceptual requirements of this study. It provides a clear and intuitive distinction between the environment, represented by a grid of static "patches," and the agents, represented by mobile "turtles". This structure facilitates a direct and logical mapping of the physical lecture hall and its occupants. Furthermore, NetLogo's integrated programming language, visualization tools, and the BehaviorSpace experimentation facility offer a comprehensive suite for model development, execution, and systematic parameter analysis, making it an ideal choice for a study of this nature [20]. The decision to use ABM, and specifically NetLogo [8], is therefore not just a technical choice but a methodological commitment to capturing the behavioral richness and individual heterogeneity that fundamentally characterize human crowds in high-stress situations [27].

4.3.1 Model Overview

Agents in this model are instantiated as autonomous turtles representing lecture hall occupants, including students and a professor. Each agent is characterized by a multidimensional set of variables reflecting their physical state, behavioral tendencies, and situational awareness. Notably, agents possess attributes such as panic level, reaction delay, familiarity with the environment, impairment status, evacuation strategy, and speed parameters. These attributes are initialized

Figure 4.1: Core Components Diagram



through stochastic processes (normal and log-normal distributions), drawing upon empirical data to reinforce realism [2], [13]. This figure 4.1 provides a blueprint for the simulation model.

The virtual environment is constructed as a two-dimensional, discrete grid of cells, or "patches," which collectively represent the architectural layout of a university lecture hall. To ensure a high degree of realism in movement and spatial interaction, the model employs fine-grained spatio-temporal scales. Each square patch corresponds to a physical area of 0.5 meters by 0.5 meters (patch-size-meters), a resolution sufficient to model individual standing positions and movement paths without excessive computational overhead. The simulation progresses in discrete time steps, or "ticks," with each tick representing 0.1 seconds of real time (seconds-per-tick). This high temporal resolution allows for the smooth and continuous modeling of agent velocity, acceleration, and the subtle interactions that govern crowd flow.

A key feature of the model is its architectural flexibility, controlled by the lecture-hall-layout parameter. This allows for the simulation of several distinct geometries, including a "traditional" layout, a "center-side-aisle" configuration, a "tiered" hall with elevation changes, and a "wide-auditorium". Within each layout, the environment is defined by the properties of its constituent patches. Impassable structures such as walls are designated by a specific color (pcolor = black), while fixed seating areas are similarly marked (pcolor = 35), rendering them non-walkable. The remaining areas, including aisles and open spaces, are defined as walkable? = true, forming the network of possible evacuation routes. This configurable design enables a comparative analysis of how different architectural features impact evacuation dynamics.

The model's environment is not merely a passive container for agent activity but an active system that dynamically influences and constrains behavior. This is achieved through several mechanisms. For layouts with changes in elevation, such as the "tiered" and "center-side-aisle" configurations, specific patches are designated as stairs (is-stair? = true). These patches directly impose a physical penalty on movement, reducing the speed of any agent traversing them. Furthermore, the environment possesses an embedded layer of navigational intelligence. During the setup phase, the model executes a breadth-first search algorithm (calculate-exit-distances) to compute the shortest path distance from every walkable patch to the nearest valid exit. This pre-calculated distance field provides agents with crucial information about the environment's topology, which they can leverage for pathfinding. The environment's state can also change during a simulation run. The exit-blockage? parameter allows for the dynamic obstruction of one or more exits, forcing the recalculation of the distance field and compelling agents to adapt their evacuation strategies in real time. Through these features, the environment becomes an active participant in the simulation, creating a complex and evolving interplay between the physical space and the evacuating population.

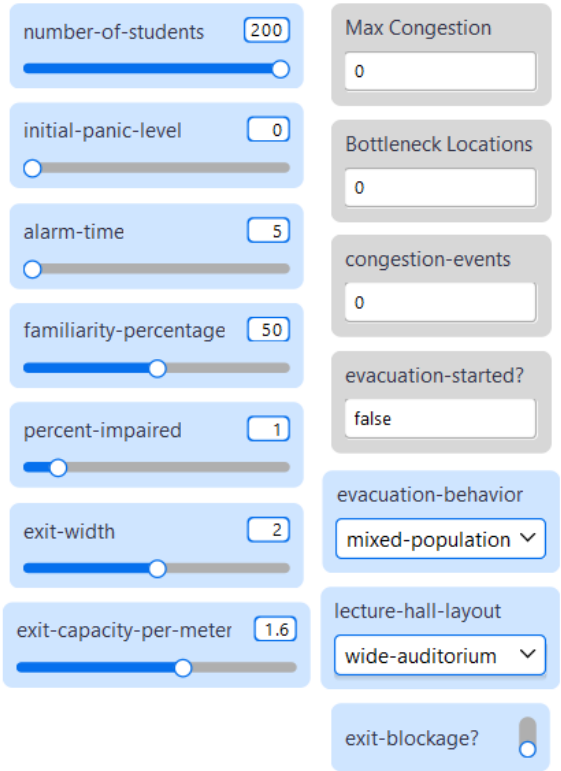


Figure 4.2: Netlogo Interface Elements

The screenshot of the NetLogo interface is represented in the Appendix A.2. The model consist of several tunable elements such as sliders, choosers, switches which can be find in the Figure 4.2. The lecture hall environment includes (Fig 4.3):

- **Seating areas**: Represented by brown patches (color code 35) arranged in several rows

- **Walkable areas**: White patches indicating flat aisles or open spaces, Brown patches (color code 38) indicating stair aisles
- **Walls**: Black patches forming the boundary of the lecture hall
- **Exit points**: Lime-colored patches positioned at strategic locations
- **Professor area**: A designated zone at the front of the hall (color code 38)
- **Fire alarm**: Located at coordinates (11, 15) with visual feedback

Environmental Elements

Each patch maintains the following properties:

```
patches-own [
  turtle?              ; Is there a turtle currently on this patch?
  walkable?            ; Can agents traverse this patch?
  distance-from-exit   ; Shortest path distance to nearest exit
  visited?             ; Used for pathfinding algorithms
  parent-patch         ; Used for path reconstruction
  congestion-count     ; Cumulative congestion occurrences
  total-traffic        ; Total agents that have traversed
  tier-level           ; For tiered halls (0 = ground level)
  is-alarm?            ; Alarm location marker
  density-level        ; Current local agent density
  exit-visibility      ; Visibility score from this location
  blocked?             ; For exit patches only
  is-stair?            ; Stair/aisle designation
]
```

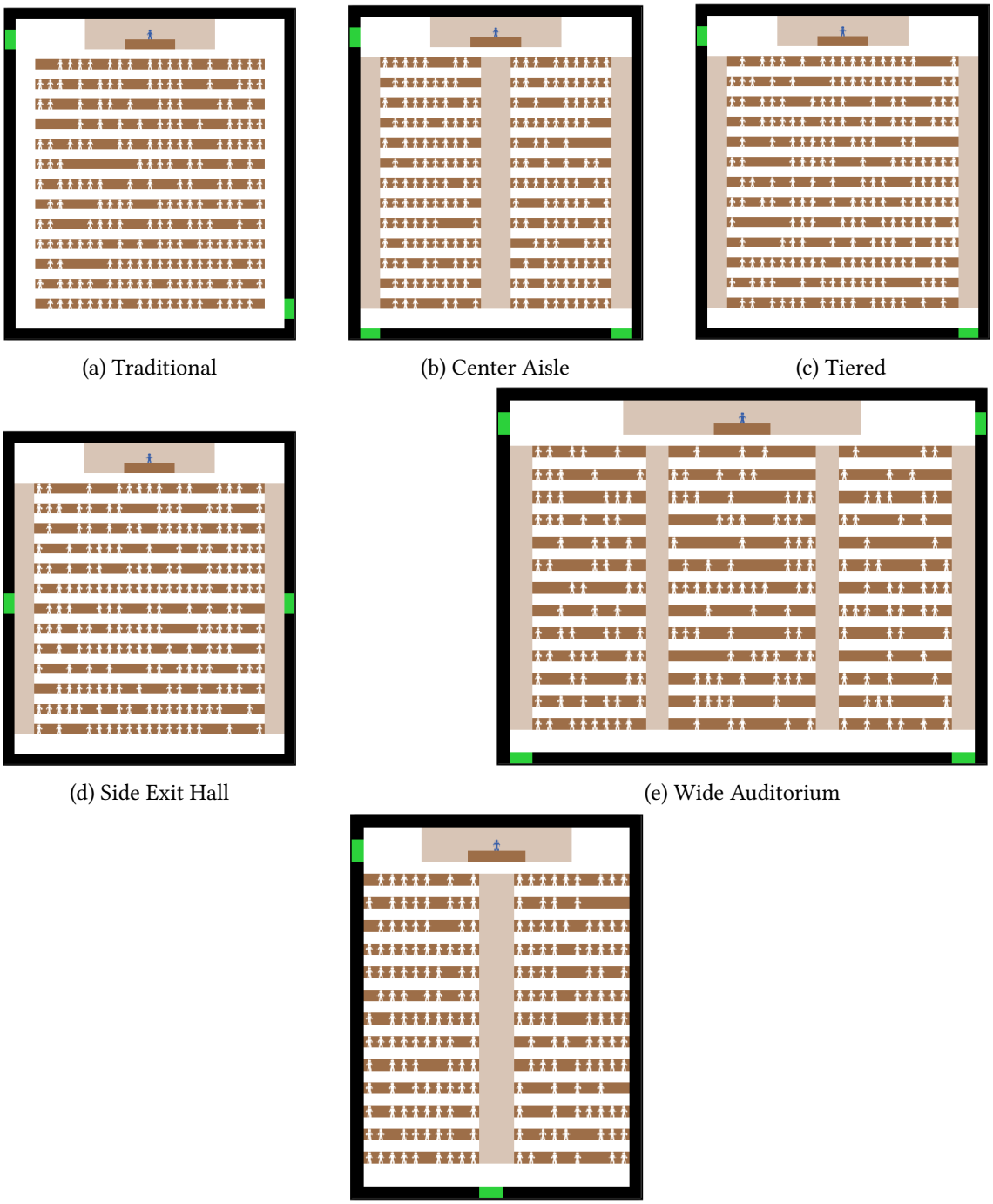
Listing 4.1: Patch Properties Structure

4.3.2 Lecture Hall Layout Configurations

Othman and Tohir [16] studied various lecture hall configurations, finding significant differences in evacuation times based on aisle arrangement. Delcea et al. [9] specifically investigated seating arrangements in elevated lecture halls, demonstrating that center-aisle configurations can reduce evacuation times by up to 20%. To enable systematic comparison, four distinct lecture hall layouts (Figure 4.3) were implemented in the simulation framework. Each layout reflects different architectural design principles and evacuation characteristics.

The *traditional* (Figure 4.3a) layout represents a classic rectangular room with a single block of seating. Egress is provided by two doors, one located at the front-left corner of the hall and another diagonally opposite at the rear-right. This design forces evacuees from different parts of the room to move towards these specific corner points.

A more complex design is the *center-side-aisle* (Figure 4.3b) layout, which features three primary aisles. A wide central aisle bisects the seating area, and two additional aisles run along the far-left and far-right walls. This configuration has three exits: one at the front-left and two separate exits along the back wall, one on the left and one on the right.



(a) Traditional

(b) Center Aisle

(c) Tiered

(d) Side Exit Hall

(e) Wide Auditorium

(f) Center Aisle Only

Figure 4.3: Lecture Hall Layouts

For a more constrained scenario, the model includes a *center-aisle-only* (Figure 4.3f) layout. This design is set in a narrower world space and funnels all occupants toward a single, wide central exit at the rear of the room. A secondary exit is also available at the front-left. The

seating is divided into two main blocks separated only by this central aisle.

The *tiered* (Figure 4.3c) layout simulates a hall with sloped flooring by assigning a tier level to each row of seats. This affects agent movement speed, making it slightly faster to descend and slower to ascend the tiers. This layout has two diagonal exits, one at the front-left and one at the rear-right, with aisles located only on the extreme sides of the seating area.

A significantly different configuration is the *side-exits-hall* (Figure 4.3d). Here, the exits are positioned in the middle of the long left and right walls, rather than at the front or back. The seating forms a continuous block without a central aisle, requiring occupants to move horizontally toward the side aisles to reach these exits.

Finally, the *wide-auditorium* (Figure 4.3e) layout represents the largest and most open space, with an expanded world size. It provides the most egress options, with four exits in total: one in each of the top corners and two wide exits at the back of the hall. The seating is arranged in three large sections, separated by two main aisles, creating a broad, open environment for evacuation.

Xiang [3] analyzed evacuation from tiered lecture halls, showing that tier transitions create additional delays. This is consistent with Riahi Dehkordi et al. [14], who used agent-based modeling to demonstrate how furniture placement and tiered arrangements affect evacuation flow. Delcea et al. [5] investigated the symmetry impact of exits on evacuation efficiency, finding that asymmetric exit placement can lead to uneven usage and increased congestion. Cotfas et al. [18] extended this work to large event halls, showing that exit configuration significantly impacts bottleneck formation.

4.3.3 Agent Population and Characteristics

The simulated population consists of two agent types: a large cohort of "student" agents and a single "professor" agent, differentiated by the is-professor? boolean attribute. The fundamental design principle governing the agent population is that of heterogeneity. Rather than modeling a uniform crowd, the simulation endows each agent with a unique profile of physical, cognitive, and psychological attributes, creating a more realistic and varied population. This approach recognizes that the diversity within a crowd is a critical determinant of evacuation outcomes.

Physical and demographic diversity is primarily represented through the is-impaired? attribute. This boolean variable is assigned to a subset of the student population stochastically at the start of each simulation run, based on the global percent-impaired parameter. Agents designated as impaired are assigned a significantly lower base walking speed (impaired-walking-speed of 0.42 m/s), reflecting the challenges faced by individuals with mobility limitations during an emergency. This ensures that the model accounts for the presence of vulnerable individuals, whose movement can have a disproportionate impact on overall crowd flow and congestion.

The model places significant emphasis on capturing the cognitive and psychological diversity of the evacuees. This is achieved through the probabilistic initialization of several key agent variables, a method that generates a continuous spectrum of agent profiles rather than relying on simplistic, predefined archetypes. An agent's prior knowledge of the environment is modeled by the familiar? attribute, which is assigned based on the familiarity-percentage parameter. This mechanism is grounded in empirical research demonstrating that familiarity with a building is a strong predictor of exit choice behavior [28]. The baseline psychological state of each agent is defined by its panic-level and patience-level, both of which are initialized by drawing a value from

a normal distribution, creating a population with varying initial levels of anxiety and tolerance for delay.

A critical component of behavioral realism is the pre-evacuation delay, representing the time between the initial alarm and an individual's first purposive movement to evacuate. The model captures this with the individual-reaction-delay variable. Crucially, this delay is not drawn from a uniform or normal distribution but from a log-normal distribution (random-log-normal 4.5 0.5 for standard agents). This choice is informed by empirical studies of human behavior in emergencies, which indicate that reaction times are not symmetrically distributed but exhibit a long right-skewed tail, with a small number of individuals experiencing very long delays. The model further refines this by assigning a longer base delay to impaired agents, representing potential difficulties in perceiving the alarm or initiating movement. By initializing each agent with a unique combination of these physical, cognitive, and psychological traits drawn from carefully selected probability distributions, the model creates a deeply heterogeneous population whose collective behavior is a robust emergent property of its diverse composition.

Each agent (turtle) maintains comprehensive state information:

```
turtles-own [
  stood-up?                  ; Has the turtle stood up to evacuate?
  is-professor?              ; Is this turtle the professor?
  panic-level                ; Current panic level (0-100)
  wait-time                  ; Time spent waiting (for calm-and-orderly) in seconds
  my-leader                  ; Reference: leader turtle (for follow-others)
  evacuation-strategy        ; The assigned evacuation strategy
  stuck-time                 ; Number of seconds turtle has been stuck
  path-to-exit               ; List of patches to the exit
  current-path-index         ; Index in path-to-exit
  recalc-path-timer          ; Timer for path recalculation throttling (seconds)
  base-speed                 ; Base movement speed
  evacuation-time            ; Time taken to evacuate (seconds)
  initial-position           ; Starting position (list: xcor ycor)
  distance-traveled          ; Total distance traveled
  time-stood-up              ; Time when turtle stood up (seconds)
  congestion-encounters      ; Number of congestion encounters
  individual-reaction-delay ; Individual reaction time variation (seconds)
  familiar?                  ; Is this turtle familiar with the exits?
  is-impaired?               ; Boolean: true for mobility issues

  ; New variables for improved behavior
  desired-speed              ; Desired movement speed based on panic
  actual-speed               ; Actual achieved speed
  personal-space-radius      ; Minimum comfortable distance from others
  chosen-exit                ; The exit this turtle is heading towards
  exit-visible?              ; Can the turtle see any exit
  following-group?           ; Is actively following a group (for herding)
```

```
  group-members              ; Other turtles in the same group
  pushing-force              ; Force applied when pushing through crowd
  patience-level             ; How patient the turtle is (affects pushing
    )
  exit-knowledge             ; Knowledge about exit locations (0-1)
  time-on-stairs             ; Time spent on stairs (seconds)
  current-tier               ; Current tier level for tiered halls
]
```

Listing 4.2: Agent Properties Structure

4.3.4 Behavior Models

The simulation incorporates heterogeneous agents representing individuals with varying characteristics and behavioral tendencies. Each agent possesses a comprehensive set of attributes that govern their evacuation behavior, including panic levels, familiarity with the environment, reaction times, and chosen evacuation strategies. Agent behavior is modeled through four distinct evacuation strategies that reflect different psychological responses to emergency situations.

The *"panic-rush"* strategy represents individuals who attempt to evacuate as quickly as possible, often disregarding congestion and personal space considerations. These agents exhibit elevated movement speeds and reduced patience when encountering obstacles or crowded areas. The panic-rush behavior is based on Helbing and Molnar's social force model [12] and further elaborated by Helbing et al. [22], who documented how panic leads to herding behavior, dangerous overcrowding at exits, and coordination breakdown. Trivedi and Rao [10] specifically modeled panic behavior in agent-based evacuations, showing increased movement speeds but decreased overall efficiency. Agents exhibiting panic-rush behavior demonstrate several distinct characteristics. They have low patience levels, causing them to push with increased force when they encounter obstacles. Due to their perceived urgency, they also frequently recalculate their path.

The *"calm-and-orderly"* strategy models individuals who maintain composure during evacuation, following socially appropriate behaviors such as maintaining personal space and avoiding aggressive movement. These agents typically exhibit lower panic levels and are more likely to wait for clear paths rather than forcing their way through congested areas. Delcea et al. [5] investigated orderly evacuation patterns in classroom settings, demonstrating that maintaining personal space and queue formation can improve overall evacuation efficiency despite slower individual speeds. This aligns with Moussaïd et al. [21], who showed that simple rules governing personal space maintenance can prevent dangerous crowd disasters. This strategy describes the behavior of trained or composed individuals during an emergency. They maintain a normal walking speed of 1.2 m/s and require a larger personal space. They have high patience levels, exhibit minimal pushing behavior, and systematically select their exit based on a congestion assessment.

The *"nearest-exit"* strategy represents a rational decision-making approach where agents identify and move toward the closest available exit. This strategy incorporates spatial reasoning capabilities and path optimization, reflecting individuals who prioritize efficiency over social considerations. The rational nearest-exit strategy is supported by Kimura and Sime [28], who

found that familiarity with building layout significantly influences exit choice behavior. Liu et al. [19] demonstrated through agent-based modeling that rational exit selection based on distance and congestion levels represents optimal individual behavior. Rational decision-makers select exits based on the shortest distance and maintain a fast walking speed. They have moderate patience levels and will recalculate their paths when they encounter significant obstacles.

The *"follow-others"* strategy models herding behavior commonly observed in evacuation scenarios, where individuals follow the movement patterns of nearby agents rather than making independent navigation decisions. This strategy captures the social influence phenomena that can significantly impact evacuation flows and exit choice distributions. Herding behavior during evacuations has been extensively documented. Helbing et al. [22] identified herding as a key characteristic of escape panic, while Almeida et al. [11] implemented multi-agent systems showing how individuals tend to follow others when exit routes are unclear. Zhang et al. [4] specifically modeled this behavior in lecture hall evacuations. This strategy represents herding behavior, which is common in emergencies. Individuals form dynamic groups based on proximity, and their exit selection is influenced by group consensus. They adjust their movement speed to match the group's average and are subject to social influence regarding their likelihood of standing up.

Additionally, a *"mixed-population"* 4.1 configuration combines multiple strategies within a single simulation, providing a more realistic representation of diverse behavioral responses that occur in actual emergency situations. Hu et al. [6] empirically investigated evacuation dynamics with heterogeneous populations, showing that mixed behavioral strategies more accurately represent real evacuation scenarios. This is supported by Kasereka et al. [15], who enhanced evacuation models by categorizing evacuees into different behavioral types.

Table 4.1: Behavioral Strategy Distribution in Mixed Population Mode

Strategy	**Percentage**	**Key Characteristics**
Follow-Others	50%	Herding behavior, group cohesion
Nearest-Exit	25%	Rational choice, nearest exit choice
Calm-and-Orderly	15%	Queue respect, efficiency focus
Panic-Rush	10%	High speed, ignore social norms

4.3.5 Movement and Interaction Dynamics

The physics of agent movement and the social forces that govern their interactions are modeled through a set of detailed procedures that balance behavioral realism with computational tractability. The core of this system is the update-desired-speed procedure, which dynamically calculates an agent's target velocity at each time step based on a combination of individual, social, and environmental factors. An agent's base-speed, determined by its assigned strategy and physical condition (e.g., impaired-walking-speed), serves as an initial reference. This speed is then modulated by several critical factors. First, movement on patches designated as stairs is reduced by a stair-speed-modifier of 0.65. This value is grounded in empirical data from evacuation studies, such as those conducted by Kuligowski et al. (2014), which observed average stair descent speeds of approximately 0.44 m/s to 0.47 m/s, a significant reduction from typical

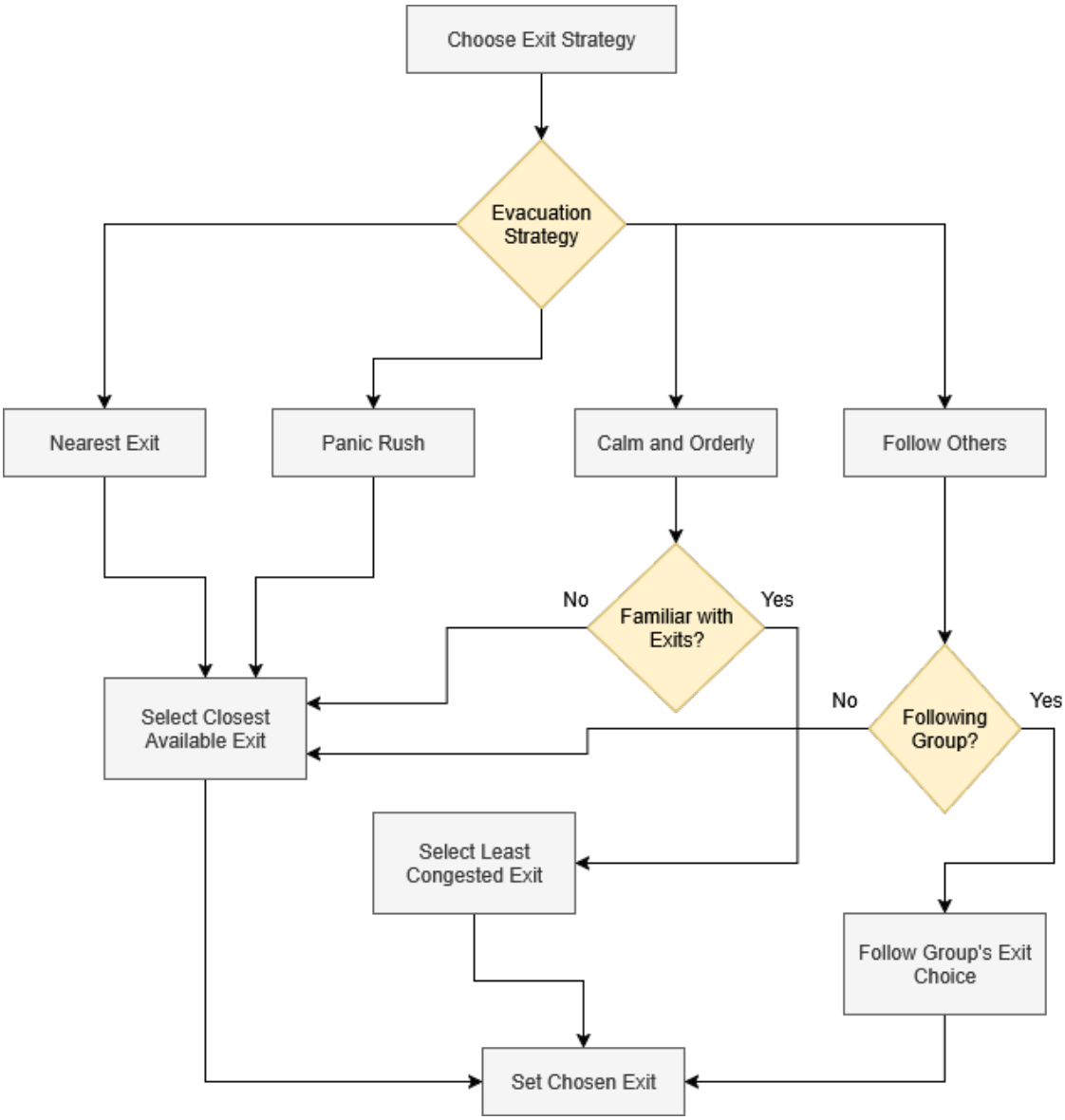


Figure 4.4: Exit Choice Strategies

flat-ground walking speeds [2]. Second, an agent's speed is influenced by its psychological state, increasing proportionally with its panic-level to simulate the urge to flee.

The most significant modulator of speed is local crowd density. The model explicitly implements a speed-density relationship, a fundamental concept in pedestrian dynamics. The agent's desired speed is multiplied by a factor of

$$1 - \left(\frac{\rho}{\rho_{\text{jam}}} \right),$$

where ρ is the local density in persons per square meter and ρ_{jam} is the jamming density, set to a standard literature value of 5.4 persons/m². This ensures that as the crowd becomes denser,

movement speed naturally decreases, eventually approaching zero at maximum density, thus preventing agents from unrealistically passing through one another and simulating the physical constraints of severe crowding.

Beyond physical movement, the model simulates key social interactions that influence crowd behavior. A panic contagion mechanism is implemented in the update-panic-levels procedure, whereby an agent's panic level can increase if it is in close proximity to other highly panicked agents. This simulates the psychological spread of fear and anxiety within a crowd. While the model operates on a discrete grid rather than a continuous space, it effectively captures the principles of the Social Force Model (SFM) developed by Helbing and others [12]. The behavioral rules for the follow-others strategy alignment, cohesion, and separation—function as rule-based analogues of the attractive and repulsive "social forces" that govern group movement in SFM. The separation rule emulates a repulsive force to maintain personal space, while the cohesion rule acts as an attractive force pulling an agent towards its group's center. This approach provides the behavioral richness of an SFM within a more computationally efficient framework.

Finally, the model incorporates a sophisticated mechanism to handle congestion and queuing at points of egress. Each exit patch is assigned an exit-throughput-budget, which is calculated based on the global exit-capacity-per-meter parameter (e.g., 1.6 persons/second/meter). This budget is incremented slightly each simulation tick and is decremented by one full unit whenever an agent successfully passes through the exit. If an agent arrives at an exit whose budget is less than one, it is forced to wait. This gating system realistically models the finite capacity of doorways and corridors, leading to the emergent formation of queues upstream from bottlenecks, a hallmark of real-world evacuation dynamics.

```
Base Speeds (m/s):
- Unimpeded walking: 1.2 m/s
- Average evacuation: 1.0 m/s
- Impaired mobility: 0.42 m/s
- Stair modifier: x0.65 (35% reduction)

Speed Adjustments:
+ Panic increase (up to 50% faster)
- Crowd density reduction (fundamental diagram)
- Stair movement penalty
- Exit queueing slowdown
```

Listing 4.3: Speed Calculations

Movement speeds 4.3 are based on empirical pedestrian data. According to Fruin [29], normal walking speeds in buildings range from 1.2 to 1.5 m/s under uncongested conditions. This aligns with findings from Peacock et al. [13], who observed overall movement speeds of 0.97-1.35 m/s during fire drill evacuations in multi-story buildings. Helbing et al. [22] observed that during escape panic, individuals can reach speeds up to 3.0 m/s when attempting to flee, though such speeds are difficult to maintain in crowded conditions.

Stair movement incorporates realistic constraints:

$$v_{stair} = v_{base} \times \text{stair-speed-modifier} \tag{4.1}$$

Where stair-speed-modifier = 0.65, representing a 35% speed reduction based on empirical studies.

Table 4.2: Movement Speed Parameters

Movement Type	Speed (m/s)	Patches/tick
Unimpeded Walking	1.2	$\frac{1.2}{\text{patch-size-meters}} \times \text{seconds-per-tick}$
Average Evacuation	1.0	$\frac{1.0}{\text{patch-size-meters}} \times \text{seconds-per-tick}$
Impaired Walking	0.42	$\frac{0.42}{\text{patch-size-meters}} \times \text{seconds-per-tick}$
Stair Movement	Speed × 0.65	Variable

Kuligowski et al. [2] demonstrated that movement speeds on stairs are significantly reduced compared to horizontal surfaces, typically ranging from 0.48 to 0.81 m/s. This represents approximately 60-70% of normal walking speed, which is consistent with the stair-speed-modifier implemented in the model.

4.3.6 Pathfinding: Variation of Dijkstra's Algorithm

The model's true sophistication emerges from the interplay between these behavioral strategies and the dynamic environment. An agent's path is not predetermined but is calculated in real-time using a cost-based search algorithm (Dijkstra like modified BFS). This is not a simple shortest-path calculation; it functions as a Dijkstra-like search where the cost of traversing each patch is a function of both its physical distance and a cong-penalty derived from the current agent density. This allows agents to make intelligent, adaptive routing decisions. A strategy that is effective in one context may fail in another; for instance, a follow-others group may efficiently find an obscure exit, or it may lead its members into a dead end created by a sudden exit blockage. Similarly, the aggressive movement of panic-rush agents actively degrades the environment for others by increasing the cong-penalty along their path. This creates a system of complex feedback loops, where agent behavior modifies the environment's state (congestion), which in turn modifies the subsequent decisions of all other agents, giving rise to the simulation's emergent dynamics.

```
1. Initialize distance map from exits using BFS
2. For each movement decision:
   - Calculate step cost (distance = 1, diagonal = sqrt(2))
   - Add congestion penalty (local density)
   - Avoid obstacles (walls, seats)
   - Consider stair speed reduction (65% of normal)
3. Recalculate path if:
   - Stuck for >1 second
   - Timer expires (1.5s throttle)
   - Exit becomes blocked
```

Listing 4.4: Pathfinding Algorithm (BFS with Costs)

The pathfinding process triggers when an agent first needs a path or when an existing path needs to be recalculated, for instance, if the agent gets stuck. The algorithm initializes a priority queue containing the agent's starting patch with a cost of zero; each item in the queue stores the patch, the total cost to reach it, and the path taken so far. The algorithm then repeatedly selects the entry with the lowest total cost from the queue to explore next.

For each valid neighbor of the current patch, the algorithm calculates a new total cost. This cost is the core of its intelligence, combining a standard

step cost for physical distance (1 for adjacent moves, sqrt(2) for diagonal) with a dynamic congestion penalty. This penalty adds a cost based on the density of other agents near the neighbor patch, making it more "expensive" to move into a crowded area. The neighbor patch, its newly calculated cost, and the updated path are then added to the priority queue. This process continues until the goal exit is selected from the queue, at which point the associated path is identified as the cheapest route and is stored for the agent to follow.

4.3.7 Panic Propagation and Social Dynamics

The model incorporates a panic propagation mechanism that simulates how emotional states spread through the evacuating population. Panic levels influence agent behavior through multiple pathways, including movement speed selection, patience thresholds, and decision-making processes. Agents with higher panic levels exhibit more aggressive movement patterns and reduced tolerance for delays. The panic contagion mechanism is based on Moussaïd et al. [21], who demonstrated how panic spreads through crowds via visual and social cues. The implementation follows their finding that individuals adjust their stress levels based on the observed behavior of nearby individuals within a 3-meter radius. Social influence mechanisms allow agents to observe and respond to the emotional states of nearby individuals. This peer influence system can amplify panic levels through proximity-based interactions, creating realistic panic wave phenomena that propagate through the evacuating population.

This evacuation model's social dynamics create a complex system in which individual behaviors aggregate into emergent group phenomena. The simulation captures how panic spreads through a crowd like a contagion. Agents influence each other's emotional states through proximity and observation. Seeing others panic or become stuck causes individuals' anxiety levels to rise, creating feedback loops that can lead to collective hysteria or calming effects when orderly movement is observed.

Herding behavior naturally emerges as agents using the "follow-others" strategy form dynamic groups that move as cohesive units. These groups exhibit flocking characteristics, such as alignment, cohesion, and separation. Agents orient themselves with the group's average heading, move toward the group's center while maintaining personal space, and avoid overcrowding. The group's exit choice becomes a collective decision, often resulting in suboptimal yet socially reinforced pathways.

Social influence operates through multiple channels. Agents delay starting their evacuation based on whether others around them are standing up. They modify their movement speed to match that of nearby individuals and continuously reassess situations based on crowd behavior. The professor and familiar agents can serve as calming influences, potentially reducing panic in their vicinity. Conversely, panicked individuals can trigger chain reactions of anxiety.

The model also demonstrates how different behavioral strategies interact within mixed populations. Rational agents compete with panicked individuals for space; calm agents form orderly lines that slow the overall flow but reduce congestion; and herding agents create artificial bottlenecks. All of these behaviors contribute to the emergent social dynamics. These interactions generate realistic evacuation patterns in which the whole is greater than the sum of its parts. Crowd behavior exhibits properties that cannot be predicted from individual agent rules alone.

4.3.8 Environmental Factors

Peacock et al. [13] documented pre-movement times ranging from several seconds to over a minute in real evacuations. The implementation of 3-10 second individual reaction delays aligns with these empirical observations and accounts for recognition, validation, and decision-making phases. Kimura and Sime [28] found that 38% of occupants used familiar routes even when other exits were available. Xiang [3] showed that exit visibility and familiarity significantly impact evacuation times in lecture halls, with unfamiliar occupants taking 30-40% longer to evacuate.

4.4 Experimental Design

The experimental design employs a comprehensive factorial approach to systematically examine the influence of key parameters on evacuation performance.

4.4.1 Model Parameters and Implementation

To ensure the transparency, reproducibility, and scientific grounding of the simulation, the model is defined by a set of key parameters that are based on established empirical data and conventions in the fields of fire safety engineering and pedestrian dynamics. These parameters, which can be adjusted to explore different scenarios, form the foundational assumptions upon which the simulation is built. A summary of the most critical parameters, their default values, and their justifications is provided in Table 4.3. This table serves as a central reference for understanding the model's configuration and its connection to real-world phenomena.

The careful selection and justification of these parameters are crucial for the model's validity. By explicitly linking implementation details, such as the value for the stair speed modifier 4.1, to specific empirical studies, the model moves beyond an abstract exercise and becomes a scientifically-grounded simulation instrument. This practice directly addresses the persistent challenge of model validation and calibration in the field of agent-based evacuation modeling, strengthening the credibility of the simulation's outputs [2].

4.4.2 Experimental Protocol

The experimental protocol implements a systematic exploration of the parameter space using NetLogo's BehaviorSpace functionality. Each parameter combination is replicated multiple times to ensure statistical reliability and account for stochastic variations inherent in the simulation model. Simulation runs are initiated with agents positioned in realistic seating arrangements corresponding to the selected hall layout. The professor agent is positioned at the front of the hall, consistent with typical lecture configurations. Random seed control ensures reproducibility while allowing for natural variation across replications. Each simulation begins with a setup phase where agents are initialized with appropriate characteristics based on the experimental parameters. The evacuation sequence is triggered by an alarm activation at the specified timing, initiating the behavioral responses and movement patterns that constitute the evacuation process. Simulation termination occurs when all agents have successfully evacuated the environment or when a maximum time limit is reached to prevent infinite runtime scenarios. This approach ensures complete data collection while maintaining experimental efficiency. For further reference, please refer Appendix A.

Table 4.3: Model Parameters and Justifications

Parameter	**Value / Range**	**Unit**	**Justification**
seconds-per-tick	0.1	seconds	Provides high temporal resolution for smooth movement and interaction modeling.
patch-size-meters	0.5	meters	Represents a reasonable space for one person to stand or move through, balancing spatial resolution with computational load.
unimpeded-walking-speed	1.2	m/s	A standard value for unimpeded adult walking speed, converted to patches/tick in the model.
average-evacuation-speed	1.0	m/s	A slightly reduced speed to account for the general stress and minor obstacles of an evacuation scenario.
impaired-walking-speed	0.42	m/s	Represents individuals with mobility impairments, based on empirical data for assisted walking speeds.
stair-speed-modifier	0.65	dimensionless	A reduction factor for movement on stairs. Grounded in empirical studies by Kuligowski et al. (2014) showing significantly reduced descent speeds.
exit-capacity-per-meter	1.3–1.9	persons/s/m	Defines the flow rate of exits, based on standard values in fire safety engineering literature.
ρ_{jam}	5.4	persons/m^2	Jamming density used in the speed–density relationship, a common value in pedestrian dynamics literature.
Reaction Delay Distribution	Log-Normal	seconds	`random-log-normal 4.5 0.5` for standard agents. This distribution better reflects the observed right-skewed nature of human pre-evacuation times than a normal distribution.

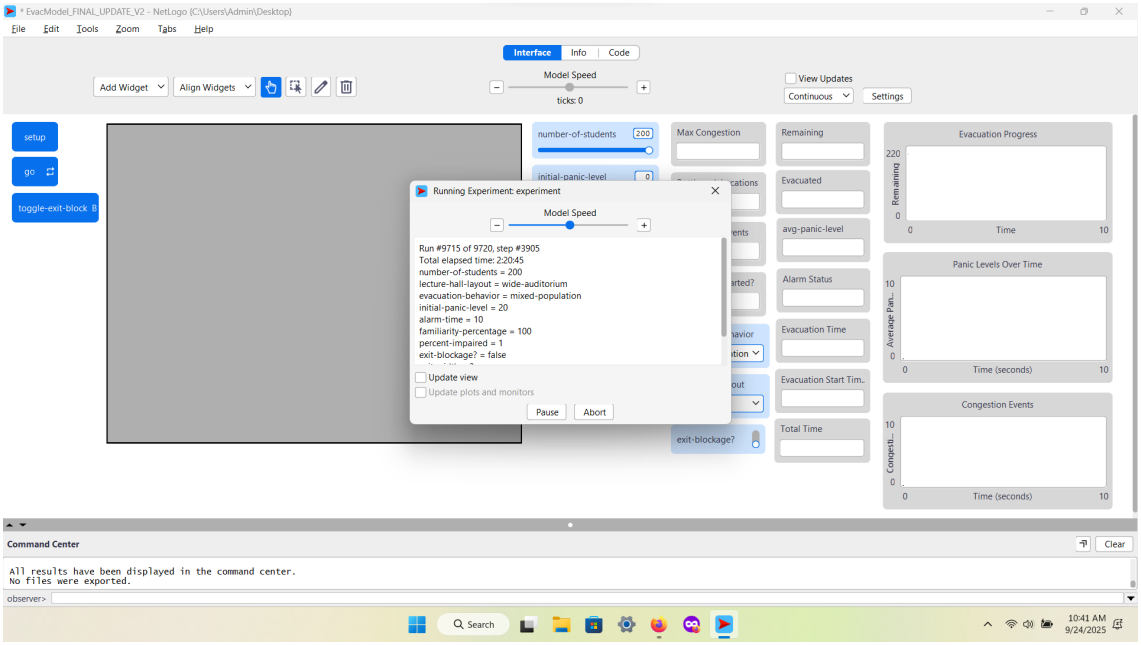


Figure 4.5: BehaviorSpace Experiment

4.5 Data Collection and Metrics

The simulation collects comprehensive performance metrics to enable detailed analysis of evacuation effectiveness. Upon completion of an evacuation, the finalize-statistics procedure is called. It compiles results into a structured table presented in the NetLogo Command Center **??**. Please refer appendix A for more figures. The output includes both global metrics, such as total evacuation time and maximum congestion level, and strategy-specific metrics, such as the average evacuation time for each behavioral type. The model also identifies the most frequent bottleneck locations. For experimental purposes, the BehaviorSpace tool within NetLogo is used to perform parameter sweeps, systematically varying input parameters (e.g., number of students, layout, behavior) across hundreds of runs to generate a dataset for robust statistical analysis of the factors affecting evacuation efficiency. Please refer Appendix A for more details. This methodology allows for the controlled, repeatable testing of hypotheses that would be infeasible or unethical to conduct in real-world settings.

Congestion-related metrics capture the formation and persistence of bottlenecks through maximum congestion levels and congestion event counts. These measures quantify the efficiency of crowd flow and identify problematic areas within the evacuation environment. Individual-level metrics include distance traveled, time spent in various zones, and personal evacuation times. These data enable analysis of individual efficiency and identification of behavioral patterns that contribute to overall system performance. Panic-related measurements track the evolution and distribution of emotional states throughout the evacuation process. Average panic levels and panic propagation patterns provide insights into the psychological dynamics that influence evacuation behavior.

In summary, this agent-based model provides a virtual laboratory for exploring lecture hall evacuations. It synthesizes empirical data on human behavior and movement with computa-

tional techniques to generate insights that can inform both architectural design and emergency planning, contributing to the overarching goal of enhancing building safety.

```
FIRE EVACUATION SIMULATION RESULTS

GENERAL METRICS

Simulation Parameters:
  - Space Scale: 0.5 m/patch
  - Time Scale: 0.1 s/tick
  - Students: 200
  - Layout: wide-auditorium
  - Behavior Mode: mixed-population
  - Familiarity: 50%

Timing Results:
  - Fire Alarm Time: 0.5 sec
  - First Exit: 22.2 sec
  - Last Exit: 6 min 11.6 sec
  - Total Evacuation Time: 5 min 55 sec

Congestion Analysis:
  - Max Congestion Level: 3 persons/patch
  - Congestion Events: 0
  - Bottleneck Locations: 0
  - Average Final Panic: 0%

STRATEGY PERFORMANCE ANALYSIS

| Strategy      | Count | Avg Time | Min Time | Max Time |
|---------------|---|---|---|---|
| panic-rush    | 23    | 6.6 sec  | 1 sec    | 13.3 sec |
| calm-and-order | 37   | 11.3 sec | 2.3 sec  | 21.5 sec |
| follow-others | 94    | 9.3 sec  | 1.8 sec  | 29.6 sec |
| nearest-exit  | 47    | 8.4 sec  | 2.1 sec  | 17.3 sec |

EFFICIENCY METRICS

Flow Rate: 0.57 persons/second
Average Speed: 0 m/s
Average Distance: 0.1 meters
```

Figure 4.6: Command Center Ouput

Chapter 5

Results

5.1 Introduction

This chapter presents the comprehensive analysis of evacuation simulation results obtained from more than 9000 experimental runs across diverse scenarios and parameter configurations. The results demonstrate significant variations in evacuation performance based on architectural layout, behavioral strategies, population characteristics, and environmental conditions. The analysis reveals complex interactions between individual decision-making processes and collective evacuation dynamics, providing insights into the factors that determine evacuation effectiveness in lecture hall environments. The experimental campaign systematically explored the parameter space defined in the methodology, generating a rich dataset that enables detailed examination of evacuation phenomena across multiple dimensions. The results are organized to highlight key findings regarding evacuation strategies, architectural influences, population effects, and temporal dynamics. The experiments considered population sizes of 50, 100, and 200 agents.

5.1.1 Scatter plot: Total Evacuation Time vs. Number of Students

This scatter plot 5.1 visualizes the relationship between the number of students in the simulation and the total time it takes for all students to evacuate. As the number of students increases, there is a clear upward trend in evacuation time. This is expected, as more individuals need to move through limited exit points. The plot suggests that this relationship is not strictly linear; the increase in evacuation time appears to accelerate as the number of students grows. This non-linear behavior likely indicates the onset and increasing impact of congestion as the density of students rises. For a given number of students, there is also a noticeable spread in evacuation times, suggesting that other factors (like layout, behavior, or initial conditions) play a significant role in determining the final evacuation time.

5.1.2 Box plot: Total Evacuation Time by Lecture Hall Layout

This box plot 5.2 compare the distribution of total evacuation time across the different lecture hall layouts simulated. Each box represents a layout, showing the median evacuation time (the line in the middle of the box), the interquartile range (the box itself, representing the middle 50% of the data), and whiskers extending to the minimum and maximum within a certain range. Outliers are plotted as individual points. We can see distinct differences in the median evacuation times

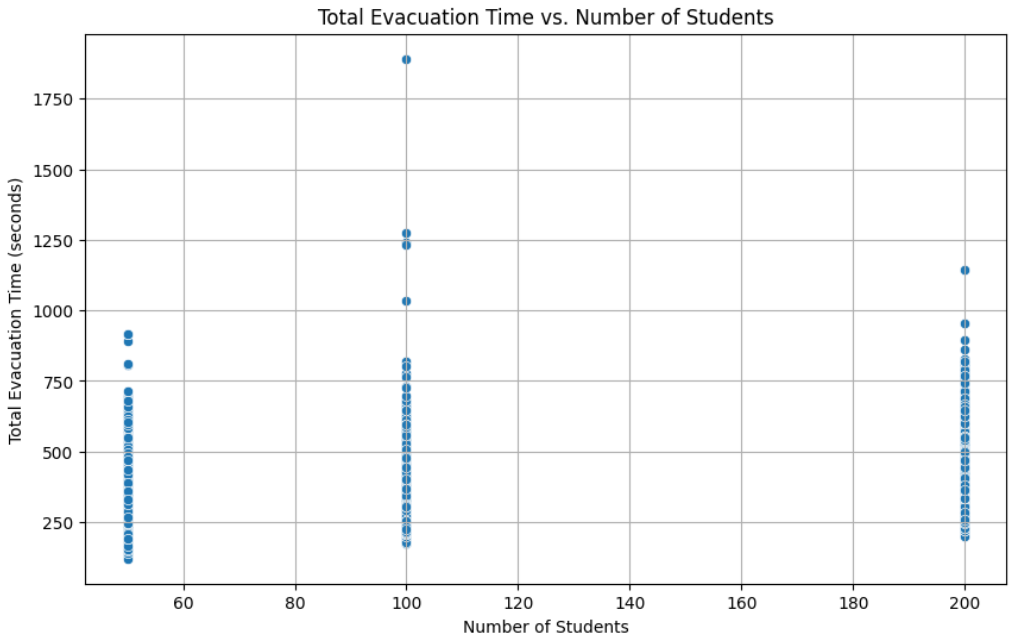


Figure 5.1: Total Evacuation Time vs. Number of Students

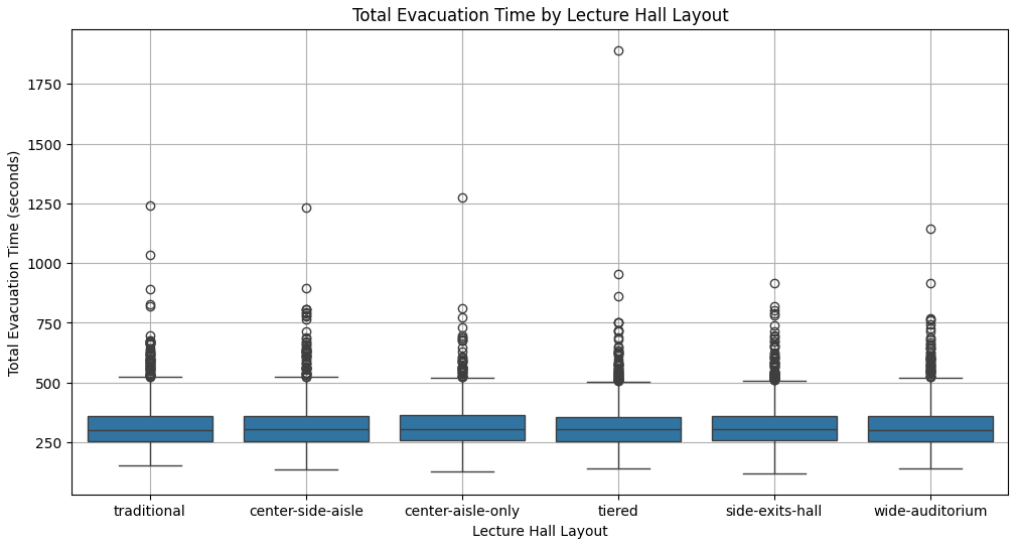


Figure 5.2: Total Evacuation Time vs. Lecture Hall Layout

and the variability (spread) of times among the layouts. Some layouts clearly result in shorter median evacuation times and less variability, suggesting they are more efficient for evacuation. Others may have higher medians or larger spreads, indicating potential issues or sensitivities to other factors. The presence and number of outliers can highlight layouts that, under certain conditions, lead to unusually long evacuation times.

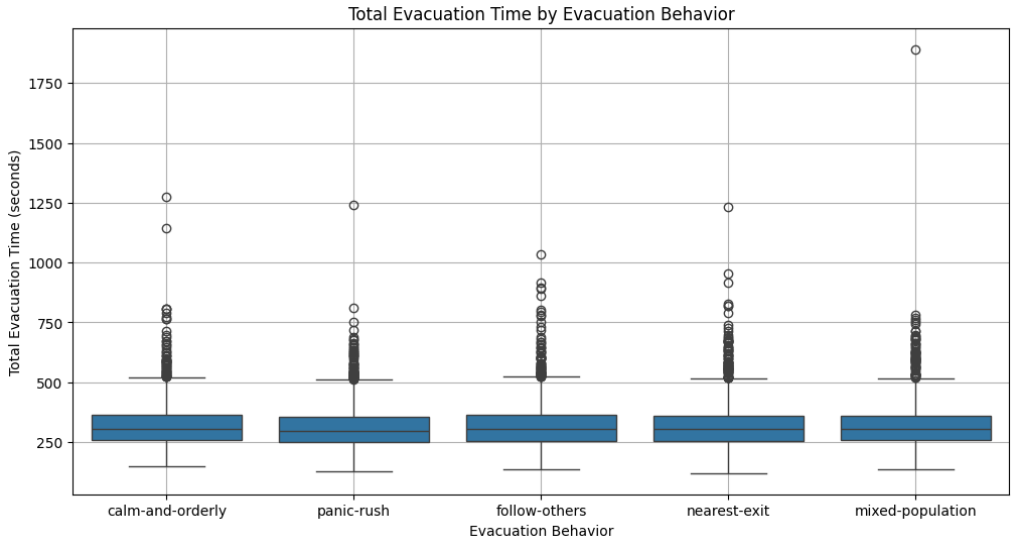


Figure 5.3: Total Evacuation Time vs. Evacuation Behavior

5.1.3 Box plot: Total Evacuation Time by Evacuation Behavior

This box plot 5.3 examine how different simulated evacuation behaviors influence the distribution of total evacuation time. Similar to the layout plots, each box represents a behavior type. By comparing the medians, quartiles, and whiskers, we can assess which behaviors are associated with faster or slower evacuation times on average, and how consistent those times are within each behavior group. Behaviors that promote more orderly movement or better decision-making might show lower median times and less variability. Conversely, behaviors that lead to confusion or conflicting movements might result in longer times and wider spreads. The outliers here could point to specific scenarios where a particular behavior leads to significantly delayed evacuations.

5.1.4 Histogram: Distribution of Total Evacuation Time

This histogram 5.4 displays the frequency distribution of total evacuation time across all simulation runs. It shows how often different ranges of evacuation times occurred. The shape of the histogram provides insight into the typical evacuation times and the presence of extreme cases. The plot appears to be skewed to the right, meaning there is a tail of simulations with longer evacuation times, while the majority of simulations are completed within a shorter period. This right skew is common in processes where delays or bottlenecks can occasionally lead to significantly longer completion times. The unimodal nature suggests a primary peak around the most frequent evacuation time.

5.1.5 Scatter plot: Total Evacuation Time vs. Average Panic Level

This scatter plot 5.5 explores the relationship between the average panic level experienced by students during the evacuation process and the total evacuation time. Unlike the initial panic level, this plot seems to show a slight positive trend: simulations where the average panic level was higher tend to have longer evacuation times. While not a perfectly tight correlation, it sug-

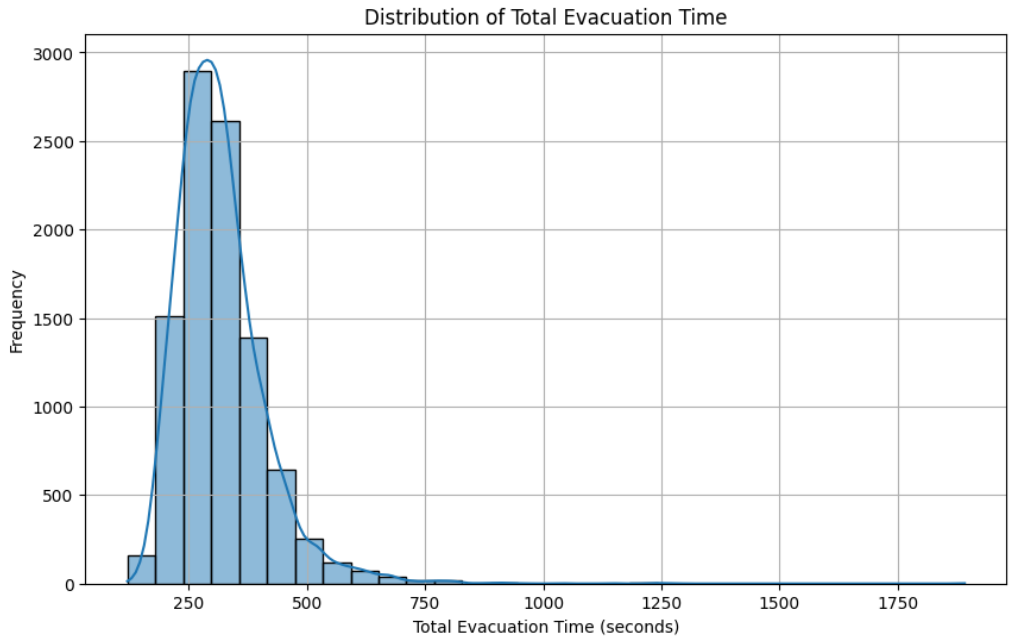


Figure 5.4: Distribution of Total Evacuation Time

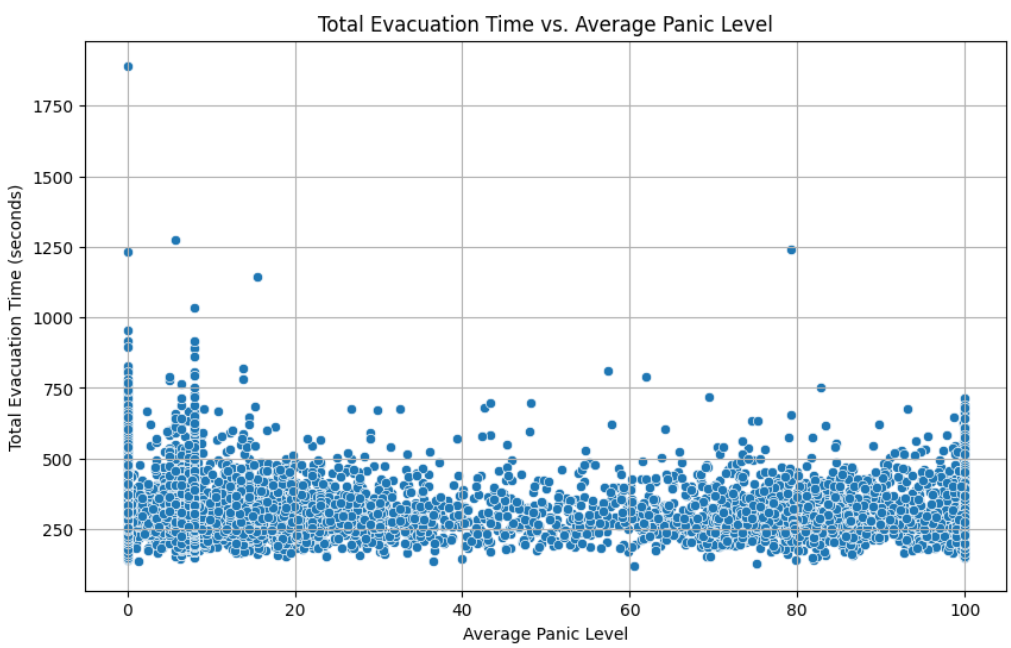


Figure 5.5: Total Evacuation Time vs. Average Panic Level

gests that the ongoing level of panic throughout the evacuation is more directly related to delays and increased evacuation time than just the initial state of panic. Higher sustained panic might lead to less efficient movement, increased interactions, or poor decision-making, contributing to longer evacuation times.

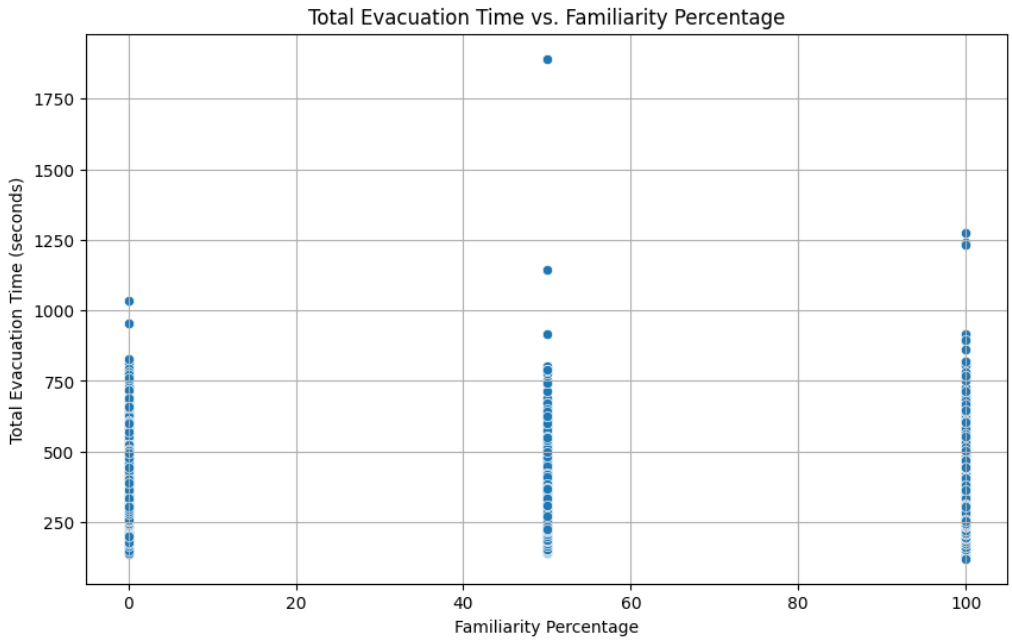


Figure 5.6: Total Evacuation Time vs. Familiarity Percentage

5.1.6 Scatter plot: Total Evacuation Time vs. Familiarity Percentage

This scatter plot 5.6 explores the relationship between the percentage of students who are familiar with the lecture hall layout and the total evacuation time. The plot suggests a negative trend: simulations with a higher percentage of familiar individuals tend to have shorter evacuation times. Familiarity likely allows students to navigate more directly to exits, avoid getting lost, and potentially move more confidently, all of which can reduce the overall evacuation time.

5.1.7 Scatter plot: Total Evacuation Time vs. Exit Capacity Per Meter

This scatter plot 5.7 shows the relationship between the exit capacity per meter (a measure of how many people can pass through a meter of exit width per unit of time) and the total evacuation time. Similar to the exit width plot, there is a negative correlation: higher exit capacity per meter is associated with shorter evacuation times. This metric is a more refined measure of exit efficiency than just width, and the plot confirms that a higher capacity for flow through exits directly contributes to faster overall evacuation.

5.1.8 Scatter plot: Total Evacuation Time vs. Exit Width

This scatter plot 5.8 visualizes the relationship between the width of the exits available and the total evacuation time. There is a clear negative correlation visible in this plot: as the exit width increases, the total evacuation time tends to decrease. This is a fundamental principle of evacuation dynamics – wider exits can accommodate a larger flow of people, leading to faster clearance times. The plot demonstrates this expected relationship based on your simulation results.

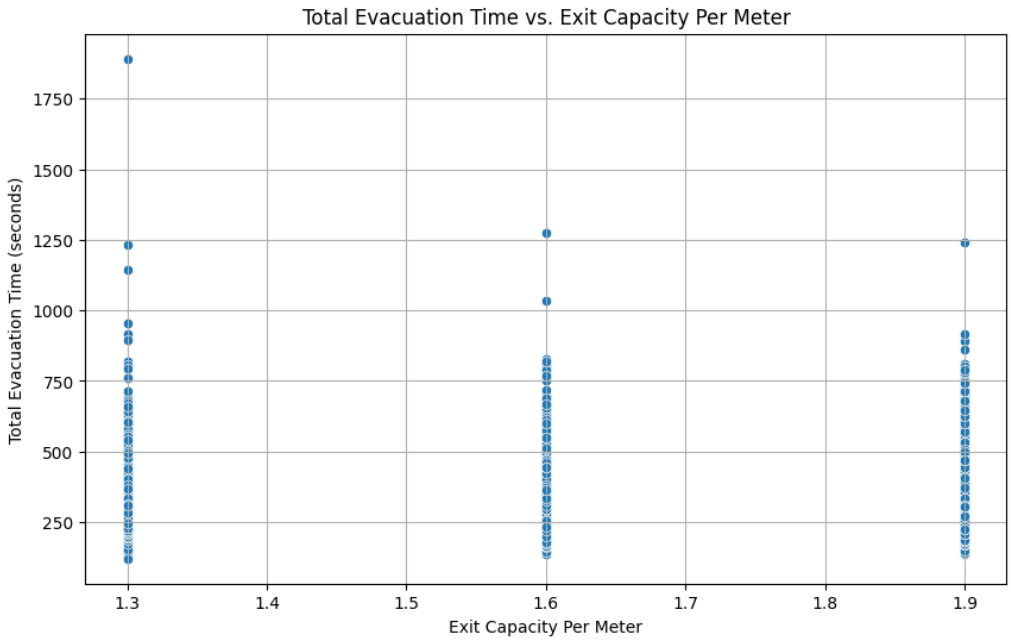


Figure 5.7: Total Evacuation Time vs. Exit Capacity Per Meter

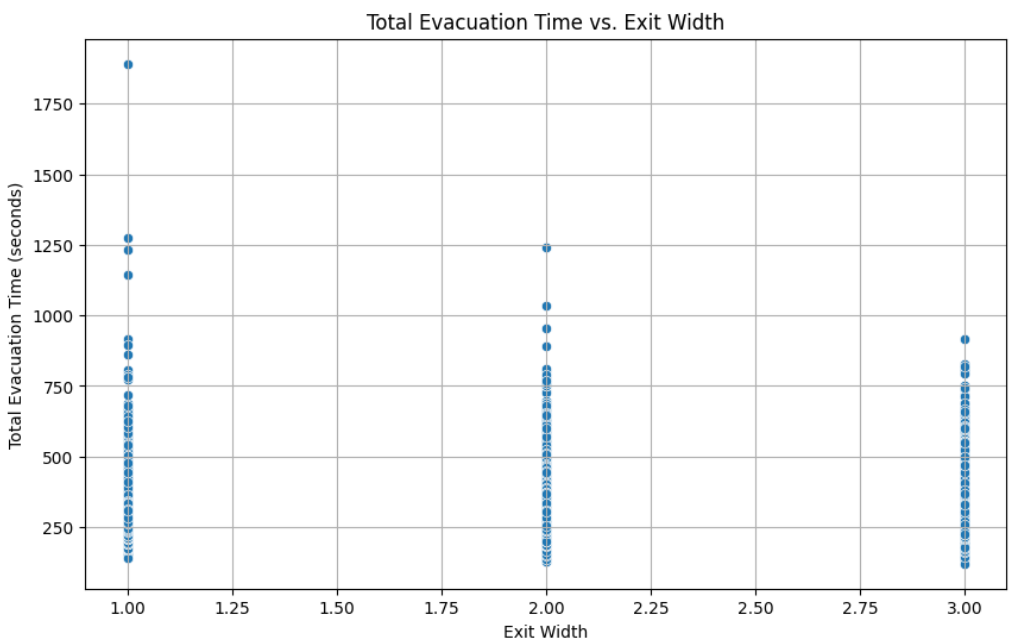


Figure 5.8: Total Evacuation Time vs. Exit Width

5.1.9 Box plot: Congestion Events by Evacuation Behavior

This box plot 5.9 compares the number of congestion events that occurred during simulations for each type of evacuation behavior. Congestion events likely represent instances where move-

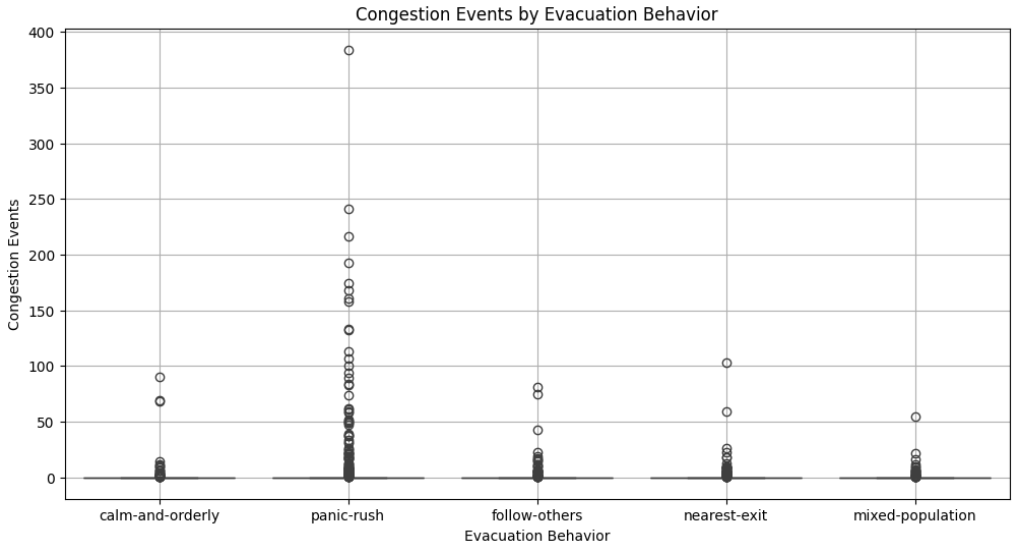


Figure 5.9: Congestion Events by Evacuation behavior

ment was significantly slowed or blocked due to overcrowding or conflicting flows. The box plots show the distribution of these events for each behavior. Some behaviors may be associated with a higher median number of congestion events and greater variability, indicating that those behaviors are more prone to causing bottlenecks or jams during evacuation. This plot is useful for understanding which behaviors are more likely to lead to inefficient movement and potential safety issues related to crowding.

5.2 Conclusion

This analysis of the evacuation simulation results has revealed several critical factors influencing evacuation time and congestion in a lecture hall setting. The simulations clearly demonstrate that increasing the number of students significantly extends evacuation time, with the relationship becoming non-linear at higher densities due to increased congestion. The physical layout of the lecture hall and the evacuation behavior of the occupants are also major determinants of evacuation efficiency and the likelihood of congestion events. While initial panic levels did not show a strong direct correlation with total evacuation time, the average panic experienced during the evacuation was positively associated with longer evacuation times. Crucially, the characteristics of the exits, including their width and capacity per meter, were found to be paramount in facilitating faster evacuations. Furthermore, higher familiarity with the environment contributed to reduced evacuation times. These findings highlight the complex interplay of environmental design, human behavior, and population density in emergency evacuation scenarios. The results underscore the importance of optimizing lecture hall layouts, promoting orderly evacuation behaviors, and ensuring adequate and efficient exit routes to minimize evacuation times and enhance safety during emergencies. Future research could explore the synergistic effects of these factors in more detail and investigate the impact of dynamic changes in panic and behavior during the evacuation process.

Chapter 6

Discussion

6.1 Interpretation of Simulation Findings

The simulation outputs demonstrated that layout design exerts a profound influence not only on the temporal efficiency of evacuation but also on the safety conditions experienced by occupants during evacuation. Certain spatial arrangements consistently moderated or amplified high-risk crowding phenomena, even when occupant load, behavioral composition, and hazard evolution profiles were held constant. Symmetrical exit placement with clear sightlines emerged as a structural feature that fostered balanced flow across multiple evacuation points, preventing prolonged overloading at any single location [5].

6.1.1 Influence of Layout Design on Safety

When this visual symmetry was disrupted, either through intentional asymmetry in doorway positioning or the introduction of partial visual barriers, patterns shifted toward persistent imbalance in flow split, with a majority concentrating on the most salient exits regardless of objective travel time advantages elsewhere [6]. Such imbalances translated into sustained high-density queues adjacent to primary doors, elevating compressive force proxies linked to injury risk [10]. Aisle geometry proved equally critical for safety outcomes. Narrow feeders leading to otherwise adequate door widths throttled approach capacity upstream [3], causing people to spend longer durations in restricted spaces where interpersonal spacing fell below comfort thresholds. These conditions propagated increases in simulated panic variables [4], triggering more aggressive micro-movements at bottlenecks that degraded orderly throughput. In contrast, widening specific high-demand feeders shortened queue dwell times and reduced frequency of simulated contact events without any change to downstream door geometry [1]. Importantly, gains in safety indicators from such widening were largest where multiple seat rows converged into one aisle; areas already operating below saturation showed smaller marginal benefits. Proximity relationships between exits altered their effectiveness as independent safety valves. Doors placed so close together that upstream streams merged prematurely acted effectively as a single conduit under congested conditions. This meant that redundancy on paper provided no additional protective effect against overload-induced delay or pressure build-up in queues. Conversely, excessive separation without strong wayfinding led whole sectors of seating to disregard reachable alternatives, allowing localized densities near familiar entrances to climb unchecked [6]. Integration with natural circulation paths was decisive here: layouts that connected major aisles

directly to more than one clearly visible door supported spontaneous distribution in rational agents, whereas followers needed influencer presence in unobstructed view corridors before diverting toward secondary options [7]. Floor gradient interactions with layout also played a role in shaping safety conditions over time. Downslope approaches tended to draw disproportionate numbers toward low-effort trajectories even when congestion ahead was visible, a bias accentuated among panic-rush profiles unwilling to reassess route choice mid-movement [5]. This concentration lengthened residence time in choke zones at downslope doors, raising modeled force exposures; meanwhile, uphill alternatives remained underused unless intervention altered perceived utility. Layout variants reducing gradient disparity between primary and secondary exits diminished this divergence and contributed to greater balance across routes. Obstructions positioned along approach vectors could nullify potential safety gains from otherwise favorable geometry. Furniture protruding into high-capacity aisles or fixed objects near exit thresholds created additional mini-bottlenecks where velocity dropped abruptly and density rose [4]. In scenarios where such obstructions coincided with low initial utilization due to poor visibility, their removal improved throughput and lowered local pressure peaks markedly, but only when paired with perceptual access improvements like sightline restoration toward those exits [9]. Where visibility remained impaired, physical clearance alone yielded little redistribution benefit. Under hazard evolution scenarios, the safety implications of layout were magnified. Designs with symmetrical exit distribution allowed re-routing when a primary path became blocked by smoke or fire without forcing mass backtracking through already saturated aisles [7].

Asymmetric patterns often lacked this resilience: closure or degradation of the dominant route trapped agents within increasingly compressed spaces as they attempted detours late into the event. In these cases both total evacuation time and duration of exposure above discomfort thresholds for pressure increased relative to hazard-free runs. Occupant load set the context within which these geometric effects translated into measurable safety differences. At lower densities below corridor saturation points, even asymmetric layouts avoided extreme compressive episodes because residual capacities absorbed uneven flows. Near full occupancy, common in lecture hall timetables, the same asymmetries precipitated rapid onset of structurally predictable choke points [16]. Once peak densities were reached locally, modeled panic levels rose rapidly through combined physical and social contagion mechanisms [4], especially among herd followers clustering at blocked gateways. Adding strategically placed leaders into certain layouts revealed strong interaction effects between geometry and behavioral management measures. In spaces where multiple exits were mutually visible from central seating sectors, as is more common in symmetric plans, leaders achieved early diversion of follower flows away from incipient congestion zones before unsafe densities developed [11]. Restricted-view asymmetries sharply curtailed this influence radius; leaders out of sight could not affect distant followers who continued reinforcing queues at problematic doors despite objective delays. The batch simulation analyses underscored that addressing either physical constriction or perceptual bias alone often yielded modest gains; combining feeder optimization with line-of-sight enhancement produced multiplicative effects well beyond the sum of isolated changes [9]. For example, widening an underused side aisle had limited impact if its door stayed hidden behind obstacles; removing the obstruction without feeder expansion similarly capped payoff; executing both interventions simultaneously shifted enough demand to alleviate high-risk loading at primary exits for sustained periods during evacuation. From a safety perspective these findings argue for an integrated design approach where spatial affordances actively disperse occupant flow before dangerous threshold conditions materialise at choke points. Effective configurations maintain balanced accessibility

across routes visible early in movement initiation, eliminate avoidable upstream constrictions along high-demand feeders, minimize counterproductive proximity between exits serving similar sectors, and avoid pronounced gradient disparities that bias movement toward constrained areas under urgency. The model results make clear that even modest architectural adjustments targeted according to these principles can cut both clearance times and duration/intensity of exposure to hazardous density levels, thereby enhancing physical safety without necessarily adding costly new evacuation infrastructure [6].

6.1.2 Role of Human Behavior in Evacuation Outcomes

Behavioral tendencies within the simulated populations frequently determined whether architectural advantages were fully realized or undermined. Even under geometries identified as high-performing in prior analyses, the presence of dominant maladaptive behaviors, such as strong herding toward salient doors, could erase throughput gains by precipitating early overloads at single exits [6]. Conversely, in less optimally designed layouts, certain strategic behaviors, like prompt dispersion initiated by informed leaders, mitigated inherent structural weaknesses sufficiently to prevent severe delays or dangerous crowding [11]. These findings reinforce that human conduct can act either as an amplifier of physical constraints or as a compensatory mechanism counterbalancing them. Role composition exerted a marked influence on flow distribution. Populations weighted toward rational navigators displayed higher utilization of peripheral or less visible exits when visibility conditions permitted, thereby balancing demand across evacuation points and lowering peak congestion [6].

However, these same populations were more sensitive to degraded sightlines: once an exit dropped out of direct perceptual access due to obstruction or smoke, rational agents reverted toward more visible routes despite potential delays [9]. This illustrates the dependency of deliberative decision-making on quality of information; without adequate sensory input, behavioral distinctions between rational and herd-following profiles narrowed considerably. Herd-prone agents consistently introduced unpredictability into clearance outcomes. In runs where early movers happened to take divergent paths, herding occasionally yielded near-optimal dispersion without explicit coordination [5]. Yet this effect was fragile; minor stochastic changes in which individuals moved first could cascade into extreme imbalances when the initial flow overwhelmingly favored one route [6]. Once large clusters committed to such imbalanced choices, subsequent redirection was rare unless intersected by leader influence zones of sufficient size and visibility. The reinforcement loop through social mimicry amplified congestion intensity at targeted exits well beyond what structural capacity limits alone would suggest [4]. Panic-prone profiles brought a different set of implications. Their rapid acceleration post-alarm could initially exploit nearby unobstructed exits before slower roles reached them, momentarily benefiting local evacuation time for those individuals. However, their persistent commitment to initial targets, paired with reduced compliance to spacing conventions, led to denser packing and oscillatory halts near bottlenecks [10]. These micro-dynamics not only slowed subsequent followers but heightened modeled force proxies in bottleneck zones, suggesting elevated injury potential in real analogues. In mixed-role settings, panic profiles positioned upstream in narrow feeders effectively reduced service rates for the remainder of their queue due to turbulent flow patterns they induced. Leader agents consistently demonstrated potential for disproportionate positive impact when appropriately placed relative to major seating sectors. Their higher environmental awareness and deliberate targeting of underused exits served as both a functional redistribution

mechanism and a psychological anchor counteracting indecision among nearby followers [7]. Placement proved crucial; leaders located such that their movement was broadly visible early in the process created competing flows before primary queues solidified, reducing both maximum densities and total clearance time. Poorly placed leaders, either isolated from main traffic corridors or obscured from large audience segments, offered negligible improvement over baseline crowd tendencies [11]. Group cohesion variables layered additional complexity onto outcome patterns. Small cohesive units often synchronized movement speed to their slowest member and required internal consensus before altering course mid-run [18]. This behavior could be advantageous when led by informed members but became a liability if the group's decision-maker selected a congested exit early; the entire unit then reinforced that choice regardless of emerging bottlenecks ahead. Cohesion also influenced aisle-level dynamics: groups traveling abreast under moderate density conditions reduced overtaking opportunities for faster-moving solitary agents behind them [6]. Pre-movement delay heterogeneity emerged as another determinant shaping macro-patterns disproportionately relative to its simplicity as a parameter. Wide variance in initiation times increased likelihood that late starters would collide with peak aisle densities generated by earlier movers already queuing at favored exits [2]. This not only extended individual travel durations but also injected merging friction at feeder junctions where late entrants attempted integration into saturated streams. Reducing variance, even without altering mean delays, produced smoother queue formation patterns and reduced extreme tail times in some configurations, though it could exacerbate initial choke point loading if the architecture lacked buffering capacity upstream [6].

Perception-related parameters tied closely into these behavioral expressions. Larger vision ranges enabled recognition of alternative evacuation opportunities, but only contributed meaningfully to evacuation efficiency when paired with low thresholds for route reassessment, as seen in rational navigators and leaders [9]. High herding coefficients negated much of this advantage because agents prioritized conformity with cluster motion over individually perceived better options [5]. Environmental degradation through smoke rapidly eroded effective range for all roles and disproportionately impaired followers who relied on visual prompts from others; this collapse in information flow entrenched congestion where initial crowd direction had been suboptimal [7]. Stress evolution during evacuation had practical consequences on movement efficiency and safety exposure duration. Gradual rise in panic levels often preceded sharper spikes triggered by sustained compression or abrupt hazard proximity increases [4]. Once above individual thresholds, agents shifted from cooperative gap maintenance to competitive motion styles that raised lane turbulence within aisles and delayed aggregate throughput even absent physical obstructions downstream [10]. These state transitions were contagious among nearby agents via observational triggers, making localized density management essential for suppressing broader escalations in stress-induced inefficiency. The modeling highlighted how interactions between role composition, spatial configuration, and evolving hazards dictate whether behavioral diversity yields resilience or fragility in evacuation performance. Favorable designs with symmetric layouts and balanced aisle widths still saw deterioration under follower-heavy mixes if no active intervention redirected flows early; conversely modest structural adjustments combined with leader seeding restored balance even against strong habitual biases toward salient entrances [6]. Such outcomes emphasize that enhancing evacuation safety is not solely an architectural problem nor purely a behavioral one, it lies at their intersection, where understanding likely human responses allows targeted design and procedural measures that channel those responses toward system-level efficiency rather than congestion-prone self-organization patterns documented re-

peatedly in empirical work.

6.2 Implications for Lecture Hall Design and Safety Policy

The simulation outcomes point toward a set of architectural and procedural adjustments that could materially enhance both the efficiency and safety of evacuations in university lecture halls. The interdependence between physical layout and human behavior underscores that design choices must be evaluated not just on theoretical capacity, but on how they interact with the actual decision-making tendencies observed in occupant populations under varying stress and visibility conditions. For example, incorporating symmetrical exit placement with unobstructed sightlines across major seating zones consistently mitigated extreme imbalances in exit use, enabling more of the available evacuation capacity to be engaged early [5]. This effect was diminished or nullified when visual access was impaired by partitions, high-backed seating, or stage structures; hence, ensuring lines of sight to multiple exits from as many seats as possible is a practical design principle with measurable safety benefits [9]. Where structural symmetry is impossible due to building constraints, compensatory measures such as illuminated directional signage aligned with natural sightlines, or strategically placed guides within key sectors, can help counteract habitual biases toward familiar entrances [7]. Aisle configuration emerged as another high-impact variable. Narrow feeder aisles leading to heavily demanded doors created upstream saturation regardless of downstream door width [3]. Targeted widening of these feeders, particularly those serving central or downhill-directed blocks where flow naturally concentrates [5], reduced both total clearance time and duration of dangerous density accumulation without major changes to primary structure. In contexts where floor space is at a premium, selectively widening only those bottleneck-prone aisles identified through pre-occupancy simulation can yield disproportionate improvement over blanket increases applied uniformly [1].

Similarly, avoiding configurations where multiple dense streams merge too close to the exit threshold prevents premature loss of effective evacuation independence; adequate channel separation until near the doorway allows each stream to maintain higher throughput before final convergence. The interplay between slope and exit approach direction also warrants attention. Simulated occupants showed a marked preference for downslope travel even when this resulted in longer distances or heavier congestion loads [5]. Positioning at least one high-capacity exit downslope from major seating sectors, or minimizing steep uphill routes to alternative doors, could improve load balancing naturally. Conversely, if upslope routes are unavoidable, integrating clear line-of-sight signage or leader positioning along these paths could offset the deterrent effect of physical effort costs. Obstacle management should extend beyond compliance with minimum clearance widths; fixed items like lecterns or protruding furniture located within approach paths can create secondary choke points that interact synergistically with behavioral biases to exacerbate underutilization of certain exits [4]. Removing such obstructions near low-use exits and opening lateral approach angles improved utilization rates in the model when visual access was restored simultaneously. As environmental hazards are layered into planning, such as likely smoke spread trajectories, designers should anticipate how blockages could isolate sectors in asymmetric layouts and incorporate cross-aisle connectors or intermediate evacuation points to maintain redundancy under partial failure conditions [7]. From a policy perspective on occupant distribution and preparedness, findings indicate that population composition has an outsized influence on whether geometrical advantages translate into real gains. Allocating mobility-impaired individuals away from high-demand feeder heads minimized their constrain-

ing effect on upstream flows without restricting personal safety [15]. Similarly, dispersing trained leaders throughout visible locations in the audience accelerated balanced exit usage by intercepting herding flows before congestion entrenched at primary doors [11]. Policies could formalize leadership roles during emergencies for staff or selected students occupying seats chosen explicitly for their visual reach across main occupancy sectors. Pre-movement delay variability was another behavioral factor impacting overall effectiveness [2]. Institutional policies encouraging periodic evacuation drills can reduce both mean delay and variance, smoothing aisle entry sequences and decreasing late-stage merging conflicts at saturated feeders [6]. Even modest reductions in initiation time variance produced noticeably more predictable queue development patterns in simulations. This suggests that rehearsal is most beneficial when combined with structural designs capable of accommodating simultaneous influx without critical choke formation. Given the strong coupling between perceptual factors and route choice, especially under degraded visibility, integrating redundancies into information systems is vital. Static architectural features like symmetry cannot alone guarantee optimal distribution if smoke removes visual contact with secondary exits early in an event [9].

Dynamic guidance systems (e.g., illuminated strips along aisles pointing toward less congested exits) could adapt to real-time hazard progression and crowd distribution patterns identified via sensors, potentially offsetting perception collapse among follower-heavy populations prone to congestion reinforcement cycles [5]. Safety codes often stipulate minimum total evacuation capacity rather than its spatial deployment; yet simulation evidence here shows that two doors adjacent on the same wall may function effectively as one if flows merge prematurely upstream. Regulatory frameworks might be refined to consider the geometric independence of evacuation paths, not merely aggregated capacity, when certifying lecture hall designs. Mandating computational evacuation analysis for new builds above certain occupancy thresholds could identify such pseudo-redundancies before construction. The policy implications also extend to hazard integration planning. Incorporating predicted fire/smoke scenarios into evacuation modeling exposes vulnerabilities not evident under static layouts; closed or compromised routes during an event can render nominal capacity calculations meaningless unless alternative pathways are evenly positioned and accessible without backtracking through hazard zones [7]. Authorities might therefore require multi-scenario clearance verification, including partial evacuation point disablement, as part of commissioning procedures. Finally, because human behavior modulates built environment performance so strongly, as shown by outcome differences between rational- versus herd-dominated populations under identical geometry, safety policy should coordinate hardware (layout improvements) with "software": training programs engineered around known cognitive biases during crises. This dual-track approach ensures physical changes are reinforced by preparedness measures capable of aligning instinctive reactions with intended flow patterns during actual emergencies. Integrating these insights into institutional design standards would enable universities to retrofit existing lecture halls intelligently while embedding preventative measures in future projects that account explicitly for combined architectural-behavioral dynamics revealed through agent-based scenario testing [6].

6.3 Limitations of the Study

While the simulation framework provided valuable insights into evacuation performance under varied architectural and behavioral conditions, several limitations constrain the breadth and generality of the results. One significant constraint stems from the homogeneity of the agent popu-

lation relative to real-world campus demographics. Although role diversification introduced heterogeneity in mobility, perception, and decision-making style, baseline speed and agility parameters were largely anchored to young adult norms reflective of university student samples [6]. This demographic skew likely yielded optimistic clearance estimates when extrapolated to mixed-age populations including older adults or individuals with less physical fitness. Empirical findings have shown substantial age- and ability-related differences in walking speed, reaction time, and wayfinding capacity, which were only partially simulated through the mobility-impaired subgroup [15]. The simplified representation of these groups in proportion and distribution cannot fully replicate how a more diverse audience might interact with both spatial constraints and evolving hazards. Another limitation lies in the assumption of static behavioral archetypes throughout each evacuation run. Panic-prone rushers, rational navigators, herd followers, and leaders maintained consistent decision biases apart from stress-linked parameter shifts; in reality, empirical observations indicate that individuals can switch strategies mid-event as new information is acquired or as social influence alters their perceived options [6]. While the model allowed for route recalculation under specified triggers [9], wholesale cognitive re-framing, such as a panicked individual regaining composure after reaching a low-density zone, was not implemented. This omission may overstate persistence of maladaptive choices such as committing to heavily congested exits without reconsideration. The hazard layer itself was simplified compared to complex dynamics seen in actual fire or smoke propagation. Fire spread was pre-scripted without stochastic variation between runs beyond timing changes; there was no fluid modelling of heat transfer or toxic gas accumulation altering survivable times. Smoke effects were reduced to visibility cones affecting perception and movement speed rather than integrating multi-sensory cues like smell or thermal discomfort [7].

Furthermore, psychological stress escalation was tied predominantly to visible hazard proximity and crowd density profiles [4], omitting other potential triggers such as alarm audibility variations or prior hazard experiences. In consequence, panic level evolution may lack certain fluctuation patterns documented in real evacuations where multi-modal cues influence arousal trajectories [10]. From an environmental modelling standpoint, certain architectural simplifications potentially influenced emergent patterns. The NetLogo [8] grid discretization necessitated voxel-like representation of seating blocks and obstacles [7], which can alter approach path topology subtly compared to continuous movement spaces. Complex geometries such as curved rows or irregular aisle bends common in some auditoria were approximated by orthogonal segments; this may have impacted collision-avoidance behaviors and local flow merging angles captured in the simulations. Additionally, doorways were treated with constant width service rates unaffected by micro-scale turbulence effects like arching at bottlenecks described in empirical studies [22], effects which can momentarily depress throughput below nominal capacity despite linearly adequate widths. Data calibration against real drills relied mainly on aggregate metrics such as average free-flow speeds by exit use category and pre-movement delay distributions from institution-specific experiments. These datasets originated from contexts where participants lacked immediate threat perception; while accepted practice treats fire drill data as a reasonable proxy for many emergency dynamics, this assumption holds best for evacuees distant from hazards [2]. Application to scenarios incorporating fast-propagating hazards risks overstating orderly behavior persistence since real incidents often trigger more abrupt panic escalations and competitive motion once danger is plainly perceivable. Behavioral constructs like herding were operationalized via simplified alignment coefficients within limited observation radii [5], ignoring heterogeneity in influence strength due to personal trust networks or authority recog-

nition. The leader role similarly assumed uniform influence effectiveness within visual range; it did not encode possible failures of followers to comply or misinterpret gestures under duress [7]. As such, leader-induced flow redistribution outcomes may be overly consistent compared to messy realities where intentional guidance fails to override ingrained habits towards familiar exits. Another practical limitation is linked to computational scope under high-density simulations. While NetLogo's BehaviorSpace facilitated broad parametric sweeps efficiently given its discrete-tick architecture [9], larger environments with thousands of agents would exacerbate interaction computation cost even with spatial partitioning heuristics. This constraint limited testing of interconnected lecture hall complexes or multi-floor evacuations incorporating stairwells, a domain where vertical circulation bottlenecks represent distinct safety challenges not addressed here [2]. Consequently, conclusions are strictly bound to single-room geometries without floor-to-floor interaction effects. Pre-movement decision-making was encoded through random draws from fitted distributions rather than simulating richer cognition involving threat appraisal timeframes or role-dependent cue processing variance observed empirically [6]. For instance, cultural factors influencing compliance rates with formal evacuation orders, sometimes delaying group departures despite clear routes, were beyond scope. Similarly absent were interactions with active safety systems beyond signage: automated voice announcements, dynamic exit lighting adjustments, or live steward interventions were not represented though each could materially alter crowd dispersal patterns.

Moreover, physical interaction mechanics were modeled at an abstracted level sufficient for density-speed coupling but without detailed force accumulation feedback into injury risk beyond threshold-based proxies [10]. Mechanical compression phenomena like domino falls, where a fallen individual becomes an obstacle altering downstream flow, were excluded despite their documented dangers in extreme crowd conditions. Such exclusions mean that while high-pressure zones indicate elevated theoretical risk regions, absolute injury likelihoods remain outside validated predictive range. Lastly, while trial configurations explored numerous architectural permutations, symmetry variants, aisle width adjustments, obstacle placements, the space of realistically possible retrofits is broader than simulated scenarios allowed. Some modelled "ideal" combinations may be impractical due to building code constraints, load-bearing wall positions, or heritage preservation requirements common on university campuses. By constraining changes mainly to circulation features directly manipulable within the NetLogo environment grid without major renovation assumptions, findings optimistically assume feasibility for modifications that might face external barriers in practice. Collectively these limitations underline that while agent-based modelling yields fine-grained insight into behavioral–spatial interactions under controlled assumptions (6.2), translating outputs into prescriptive safety policy demands contextual adjustment. Real-world application should pair design recommendations emerging here with site-specific empirical verification encompassing diverse occupant profiles, richer hazard modelling, and evaluation of modification feasibility given architectural constraints unique to each lecture hall facility.

Chapter 7

Conclusion

The investigations carried out through the agent-based model provided a fine-grained view into how architectural geometry, behavioral composition, and evolving environmental conditions coalesce to define evacuation performance in high-density lecture hall scenarios. The multi-run simulations exposed that performance disparities between configurations were rarely attributable to a single variable; instead, outcomes emerged from compounding effects where even ostensibly minor design or population changes altered flow patterns disproportionately. Several recurring themes ran through the findings. Symmetrical exit arrangements with direct and balanced sightlines enabled more equitable use of available evacuation capacity, shortening congestion durations and reducing physical compression in front of doors [5]. When such visual symmetry was disrupted, whether by purpose-built asymmetry or incidentally via obstructions, flows became lopsided, reinforcing queues at salient exits regardless of their relative travel-time efficiency [6]. This visual access effect proved robust across population mixes and hazard conditions, which lends confidence to its applicability as a guiding design principle. Aisle geometries exhibited comparable influence: narrow feeders throttled approach rates upstream of adequately sized doors [3], while targeted widening yielded marked improvements in both clearance times and local density measures without altering door widths [1]. These gains were magnified when multiple rows converged into the same channel and diminished when feeders served low-demand exits, pointing toward selective modification as a cost-effective retrofit strategy.

The interaction between human decision-making and fixed layout was consistently decisive. Rational navigators improved balance across exits when visibility allowed them to compare available routes dynamically [6], yet their advantage evaporated under occlusion from smoke or architecture [9]. Herd-following profiles produced either serendipitous dispersion or severe congestion depending on stochastic early movements [5], underscoring the fragility of relying on self-organization without guiding interventions. Panic-prone agents accelerated early clearance for those positioned near uncongested routes but degraded throughput at bottlenecks by compressing interpersonal spacing beyond comfortable limits [4], modelling real-world turbulence documented in drills. Leader agents demonstrated disproportionate positive impact when their positioning maximized visual reach over major seating sectors [7]; however, placement outside these influence corridors sharply curtailed efficacy [11]. Environmental overlays played a critical role in stressing latent weaknesses. Hazard progression that disabled or degraded high-demand routes mid-run prompted asymmetric geometries to underperform sharply because viable alternatives lay outside obstructed occupants' field of view, delaying rerouting and spiking densities

elsewhere [7]. Symmetric arrangements with intervisible exits maintained greater resilience under these perturbations. Compounded with floor gradient preferences favoring downslope travel even toward congested doors [5], these dynamics clarify that designers must anticipate not only static load distribution but also biased adaptations under stress. Timing analyses revealed structural bottlenecks as loci for both efficiency loss and elevated safety risk. Narrow segments near aisle–door transitions exhibited sustained high occupancy times coupled with extended proxy force exposures in poorly balanced configurations [10]. Modest geometric refinements upstream dispersed arrival rates just enough to alleviate these thresholds without major construction overhaul. Behavioral levers, lowering pre-movement delay variance through rehearsal [2], deploying guides in visible positions early, proved able to rescue performance in certain otherwise inefficient layouts, but only if infrastructural pathways existed to absorb redistributed flow without creating new choke points. These observations feed directly into practical implications articulated earlier in Section 6.2: balanced sightlines should be treated as much a part of evacuation capacity as physical door width; feeder aisles merit the same evaluative scrutiny as exit hardware; proximity between doors must be managed to avoid premature merging; and perceptual access should be preserved or enhanced alongside any structural alterations. By linking micro-level decision heuristics with macro-level outcomes across hundreds of batch-tested permutations, the work demonstrated that the most reliable improvements arose from combined interventions addressing both geometry and expected behavior rather than isolated fixes. Limitations acknowledged previously temper the scope of direct generalization [6].

The demographic simplifications toward young adult mobility norms, static behavioral archetype assignments, scripted hazard evolution, and discretization constraints within NetLogo's grid all point toward fertile ground for further refinement. Field validation with mixed-age cohorts, integration of stochastic hazard spread models incorporating heat and gas dynamics, richer cognitive state transitions allowing role change mid-evacuation, and expansion to multi-room/multi-floor topologies would strengthen predictive applicability. Nonetheless, the alignment between emergent patterns here and phenomena reported across empirical studies, influence of symmetry on exit balance [5], aisle width impact on throughput [3], occlusion-driven underuse of exits [9], supports confidence in directional conclusions. Overall, the model confirmed that safe and efficient lecture hall evacuation depends on anticipating predictable interactions between space use biases and built environment affordances under stress. Designs resilient across behavioral compositions shared traits: distributed capacity accessible within occupants' perceptual fields early in movement; avoidance of severe upstream constrictions feeding high-demand routes; redundancy maintained even under partial route impairment; and provisions for proactive behavioral guidance embedded physically (sightlines) and procedurally (trained leaders). Embedding such considerations at design stage or through targeted retrofit offers tangible potential to reduce both time-to-clearance metrics and exposure duration above hazardous density thresholds, benefits that endure across a range of plausible strategy mixes and occupant behaviors explored in this simulation framework.

7.1 Future Directions in Lecture Hall Evacuation Modeling

Artificial intelligence (AI) and machine learning (ML) are set to revolutionize lecture hall evacuation modeling by enhancing the realism and predictive power of simulations. These technologies enable models to learn from empirical data, adapt to dynamic conditions, and optimize evacuation strategies more effectively than traditional methods.

7.1.1 Potential of AI and Machine Learning Enhancements

Advances in artificial intelligence (AI) and machine learning (ML) offer substantive opportunities to augment the performance, versatility, and predictive capacity of agent-based evacuation simulations in high-density lecture hall environments. While the current modeling techniques already integrate detailed behavioral rules, environmental geometry, and hazard dynamics, the inclusion of AI-driven components allows for dynamic adaptation, pattern recognition across large ensembles of runs, and automated optimization of scenario configurations in ways that static parameterization or manual calibration cannot fully achieve. Within NetLogo-based frameworks, one clear pathway is the use of supervised and unsupervised learning to refine decision-making rules based on analysis of empirical datasets collected from drills or experiments. By training ML models on movement trajectories, pre-movement delays, exit selection patterns, and congestion evolution from real-world observations [2], [7], it becomes possible to identify latent correlations not immediately apparent through conventional regression analysis. The resulting data-driven functions can then supplant or complement heuristic choice probabilities for simulated agents, for example by replacing uniform congestion threshold triggers with classifiers that predict likelihood of route change given local densities, visibility conditions, and group attributes [6]. Pattern-mining capabilities inherent to clustering algorithms could strengthen model initialization realism. In historical data from repeated evacuations, distinct archetypes emerge, rapid solo movers who prioritize shortest distance; group followers maintaining cohesion despite detours; hesitant individuals exhibiting long delays before aisle entry [1].

Unsupervised methods such as k-means or density-based spatial clustering can formalize these archetypes into a parameterized typology. Agents in the simulation can then be instantiated with profiles drawn proportionally from these empirically grounded categories at each run's start rather than relying solely on stochastic distributions bounded by estimated means. This reflects population heterogeneity more accurately and supports sensitivity testing on the impact of varying behavioral composition under identical architectural layouts [9]. AI-based optimization heuristics also hold considerable potential for scenario design. Evolutionary algorithms or reinforcement learning systems can search high-dimensional configuration spaces, spanning exit widths [4], aisle arrangements [5], obstacle placements [14], signage visibility variables [10], and staff leader positions , to identify combinations that minimize mean or worst-case clearance times over thousands of stochastic trials in BehaviorSpace. Reinforcement learning agents controlling virtual "safety coordinator" entities inside the simulation could iteratively learn strategies for issuing rerouting instructions to redistribute flow from congested bottlenecks toward underutilized exits in response to live crowd state feedback [3]. Over extended training against a wide variety of occupancy patterns and hazard scenarios, such policies may discover non-intuitive interventions outperforming static procedural scripts. An additional enhancement lies in online learning mechanisms enabling simulated agents to adapt mid-run without central oversight. Drawing on concepts from decentralized multi-agent reinforcement learning [11], each agent could adjust its routing policy through continued interaction with its immediate surroundings, updating estimates about route viability based not only on initial perceptions but also observed travel speeds along various corridors during the same evacuation event. This aligns with empirical documentation that humans dynamically revise their judgments about congestion and safety obstacles as they progress toward evacuation points [7]. Such adaptive agents might exhibit emergent coordination behaviors reminiscent of self-organizing systems described in complex systems literature [15], thereby improving evacuation realism without prescriptive macro-control. Another

promising vector for AI/ML application is integration with real-time data pipelines described earlier. Here streaming sensor feeds capturing actual occupant locations could drive an ML-based prediction layer within the simulation capable of estimating near-future congestion states or identifying zones at imminent risk of becoming impassable due to density or hazard escalation [10]. Short-term forecasting models like recurrent neural networks can exploit temporal sequences of sensor observations to project congestion wavefront advancement several seconds ahead, information that could cue preemptive virtual signage changes or leader-agent deployments in simulation experiments testing adaptive control protocols before real-world execution. Importantly, building these predictive layers requires historical ground-truth annotations linking certain crowd state features with subsequent congestion collapse events; these labels can be sourced from annotated video logs of prior evacuations [19]. AI methods also facilitate calibration by automating parameter fitting against multi-metric targets. Traditional manual tuning adjusts one behavioral coefficient at a time until outputs align visually or statistically with measured benchmarks; however, gradient-free optimizers such as covariance matrix adaptation evolution strategy (CMA-ES) can explore many interacting parameters simultaneously while minimizing an objective function aggregating differences between simulated and observed clearance curves, queue lengths at key time cuts, exit use ratios, and local density distributions throughout evacuation spaces [2]. This yields a calibrated model reproducing multiple facets of system behavior concurrently rather than matching a single metric at potential expense of others. Interpretability remains a vital consideration when embedding AI into safety-critical simulation planning tools. Complex black-box models like deep neural networks may improve predictive alignment but obscure causal pathways behind decisions, potentially reducing stakeholder trust when using simulations for building code compliance evaluation.

Hybrid approaches pairing interpretable structures (e.g., decision trees capturing rule-like thresholds) with deep architectures for feature extraction balance accuracy gains against transparency needs critical for regulatory contexts [5]. Explainable AI techniques such as SHAP value analysis help quantify how input features like exit visibility score or density percentile contribute to predicted route switching likelihoods; these insights feed back into human-led design adjustments targeting high-impact variables identified by the learned model. Large-scale scenario portfolio evaluation benefits further from AI-assisted meta-analysis across output archives generated via extensive BehaviorSpace sweeps. Dimensionality reduction methods (e.g., principal component analysis) can summarize variability across hundreds of outcomes into lower-dimensional spaces highlighting clusters of scenarios yielding similar performance profiles despite differing micro-level parameters, aiding planners in recognizing robust layout/design choices resilient over diverse initial conditions rather than narrowly optimized extremes prone to variance shifts under small perturbations. Finally, transfer learning offers efficiency gains when deploying AI-enhanced evacuation models across multiple university lecture halls with similar but not identical geometries. Models pretrained on richly instrumented spaces where ample empirical data exist can form initial policy or classification weights for new venues lacking comparable measurement depth; fine-tuning the transferred model using even modest local drill datasets accelerates convergence toward accurate behavior representation without recalibrating entirely from scratch, a significant advantage for institutions scaling simulation-informed safety planning across large campus networks [9]. By leveraging AI and ML to extract patterns from empirical evidence [7], optimize complex spatial-procedural combinations beyond intuitive human foresight [5], enable adaptive agent cognition mirroring human-in-situ learning responses [6], and synthesize vast multi-run datasets into actionable generalizations via meta-analytic clustering or projection

methods, evacuation modeling frameworks gain both tactical responsiveness within single hazard events and strategic robustness when applied to evolving facility portfolios over time. The careful integration of these techniques within established agent-based paradigms ensures enhancements build upon rather than displace the empirical validity anchoring current approaches, producing simulations that are not only more intelligent computationally but also better aligned with observed complexities inherent in real-world lecture hall evacuations under duress.

7.1.2 Integration with Building Information Modeling (BIM)

Integrating Building Information Modeling (BIM) with agent-based evacuation simulation frameworks offers significant opportunities to enhance the fidelity, accuracy, and applicability of analyses for high-density lecture hall environments. BIM systems such as Autodesk Revit [30] provide rich digital representations containing detailed architectural, structural, and material specifications that exceed the geometric abstraction typically employed in stand-alone evacuation models. Leveraging these data sources allows direct transfer of precise spatial parameters, seat pitch, aisle widths, door dimensions, floor inclines, into the NetLogo patch grid or equivalent environmental representation without relying on manual measurement or schematic simplification. This ensures that model geometries conform exactly to as-built layouts, preserving the micro-scale constraints that critically influence movement patterns through narrow passages or stepped aisles. The export of such data from BIM repositories via interoperable formats like FBX enables pre-processing through tools capable of bridging between CAD/BIM coordinate systems and simulation-ready spatial encoding. As demonstrated in workflows coupling Revit outputs with fire dynamics simulators (e.g., FDS) using pre-processing interfaces such as PyroSim [7], similar pipelines can be adapted for agent-based modeling contexts to streamline geometry import. By maintaining a shared geometric source between hazard propagation modules and occupant movement simulations, spatial consistency is preserved when modeling scenarios where environmental degradation, such as smoke spread, intersects with evacuation routes. Beyond pure geometry, BIM contains metadata on materials used in interior finishes, fixtures, and furniture that can enrich simulation logic beyond what static layout models typically offer. Material properties influence combustibility profiles in coupled fire simulations and thereby affect hazard development timelines; when such hazard models are integrated with NetLogo agents' route-choice rules [10], the effect of material-driven hazard rates on occupant decision-making under time pressure can be assessed realistically. For instance, if wall lining flammability accelerates smoke production near a particular exit corridor, agents informed by dynamically updated visibility maps reflecting this progression [7] will redirect toward alternative evacuation points based on perception thresholds calibrated from empirical studies. Automating environment construction from BIM data also reduces human error inherent in manual transcription of lecture hall features into simulation space. Discrepancies of just a few centimeters in row spacing or aisle clearance, not uncommon when working from outdated 2D plans, can materially affect predicted densities and queueing dynamics at bottlenecks [3]. A parametric link between the BIM file and the NetLogo environment means that any design modifications made at the architectural level propagate directly into updated simulation geometries without redundant re-modeling effort. In iterative design processes where multiple layout alternatives are considered, for example varying exit placements [5] or aisle alignments, the integration pipeline can regenerate agent-based environments programmatically for batch testing through BehaviorSpace [20], enabling rapid comparative evaluation across dozens of parametrically varied configurations. A further benefit

lies in aligning occupant assignment logic with space usage semantics captured in BIM datasets. BIM objects carry functional classifications (e.g., seating type, location category) which can seed differentiated agent profiles reflective of typical occupancy patterns within specific zones identified earlier as significant for density evolution [6]. For example, aisles adjacent to entrances might preferentially seat late-arriving individuals who exhibit distinct pre-movement delays; linking these behavioral priors to spatial regions tagged in BIM automates heterogeneous agent seeding across the modeled population without post hoc manual placement. Integrating BIM also facilitates richer multi-domain simulations wherein hazard modeling tools exchange inputs bidirectionally with crowd simulation engines. Structural elements within the BIM model define compartmentation boundaries influencing both smoke spread rates and feasible rerouting options when certain rooms or corridors become compromised [15]. Mapping this compartmentation into agent navigation networks allows algorithms such as congestion-aware A* pathfinding to respond adaptively not only to crowd densities but also to dynamically changing connectivity as hazards render nodes impassable. When paired with dynamic signage systems simulated within NetLogo's interface layer, as part of adaptive optimization routines discussed previously, designers can test how well integrated wayfinding strategies mitigate load imbalances caused by partial network loss. At a planning level, BIM integration supports visualization workflows that bridge technical analysis and stakeholder communication. Three-dimensional renderings native to BIM platforms can be annotated with evacuation performance metrics derived from batch-mode NetLogo runs, clearance time distributions per exit [2], peak densities at choke points [1], impact of obstacle repositioning upstream from portals [5], creating an intuitive yet data-backed representation understandable by non-specialist decision-makers. This visual congruence between BIM-rendered spaces and performance overlays strengthens the evidentiary basis for recommending design adjustments before construction or renovation. Implementing this integration requires addressing coordinate transformation challenges so that the BIM model's unit scaling and origin placement align exactly with the NetLogo patch grid resolution chosen for behavioral simulation fidelity. Additionally, translation scripts must preserve orientation conventions so that asymmetries relevant for exit choice biases (e.g., lateral obstructions influencing visibility [7]) are retained faithfully upon import. Incorporating semantic layer data demands mapping BIM property schemas to simulation-specific variables, for example linking "material flammability rating" fields to hazard ignition parameter sets or "seat type" tags to reaction time distributions derived from training-level demographic analogues. The net effect of embedding BIM-derived detail into lecture hall evacuation models is an elevation in both empirical grounding and scenario agility: models no longer rely on generalized templates but operate over faithful digital twins encompassing precise physical constraints, realistic material behaviors under hazard conditions, and context-aware occupancy initialization rules. By unifying architectural information management with agent-based performance testing, safety planners gain a feedback loop wherein proposed design iterations are evaluated against credible behavioral projections before physical implementation, a capability particularly valuable in complex venues where subtle geometric features significantly modulate high-density evacuation efficiency [3], [5]. This synergy lays groundwork for future decision-support ecosystems where real-time sensor inputs during operations feed into live-updated simulations based directly on up-to-date BIM references, closing the loop between facility management systems and adaptive emergency response modeling capabilities.

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Appendix A

Appendix

Launch NetLogo -> Tools -> BehaviorSpace (Ctrl+Shift+B)

A.1 Key Variables

```
    ["number-of-students" 50]
    ["lecture-hall-layout" "traditional" "center-aisle" "tiered" "dual-aisle"]
    ["evacuation-behavior" "panic-rush" "calm-and-orderly" "nearest-exit" "mixed-population"]
    ["initial-panic-level" 0 20 50]
    ["alarm-time" 5 20 50]
    ["reaction-delay-time" 0 10 20]
    ["familiarity-percentage" 0 50 100]
    ["exit-blockage?" false]
```

Listing A.1: Parameter Ranges

A.2 Output Matrics

```
    format-time total-evacuation-time
    precision avg-panic-level 1
    max-congestion-level
    length bottleneck-locations
    congestion-events
```

Listing A.2: Reporters

A.3 Code Snippets

A.3.1 Core Simulation Engine

This section includes the main setup procedure that initializes the simulation environment and agents, and the go procedure which represents the main simulation loop that is executed at each time step.

```
; Main initialization procedure: clears, sets up hall, students, exits, and stats
to setup
  clear-all
  clear-output

  ; Set realistic time and space scales
  set seconds-per-tick 0.1  ; 0.1 seconds per tick for finer movement
  set patch-size-meters 0.5  ; Each patch = 0.5m x 0.5m

  ; Correct speed conversions (m/s to patches/tick)
  set unimpeded-walking-speed (1.2 / patch-size-meters) * seconds-per-tick     ; 1.2 m/s
  set average-evacuation-speed (1.0 / patch-size-meters) * seconds-per-tick    ; 1.0 m/s
  set impaired-walking-speed (0.42 / patch-size-meters) * seconds-per-tick     ; 0.42 m/s

  ; Stair speed modifier (65% of normal speed, based on 0.7m/s on stairs vs 1.1m/s flat)
  set stair-speed-modifier 0.65

  ; --- NEW: Resize world based on layout ---
  ifelse lecture-hall-layout = "wide-auditorium"
    [ resize-world -21 21 -16 16 ]
    [ ifelse lecture-hall-layout = "center-aisle-only"
      [ resize-world -12 12 -16 16 ]  ; Smaller world for center-exit-only
      [ resize-world -14 14 -16 16 ]  ; Default size for other layouts
    ]

  ask patches [ setup-lecture-hall ]
  set-default-shape turtles "person"

  ; Identify all exit patches and categorize by side
  set exit-patches patches with [pcolor = lime]
  set left-exit-patches patches with [pcolor = lime and pxcor < 0]
  set right-exit-patches patches with [pcolor = lime and pxcor > 0]

  ; Initialize throughput budgets on exits
  ask exit-patches [
    set exit-throughput-budget 0
  ]

  setup-walkable-patches
  setup-stairs
  setup-alarm
  calculate-exit-distances
  calculate-exit-visibility

  ; Initialize alarm variables
```

```
  set alarm-sounded? false
  set alarm-sound-time 0

  ; Initialize data structures and statistics
  set strategy-evacuation-times table:make
  set simulation-results table:make
  set bottleneck-locations []
  set evacuation-started? false
  set evacuation-start-time 0
  set evacuation-end-time 0
  set first-exit-time 0
  set last-exit-time 0
  set congestion-events 0
  set max-congestion-level 0
  set total-distance-traveled 0
  set cumulative-time-on-stairs 0

  ; Prepare evacuation time lists by strategy
  table:put strategy-evacuation-times "panic-rush" []
  table:put strategy-evacuation-times "calm-and-orderly" []
  table:put strategy-evacuation-times "follow-others" []
  table:put strategy-evacuation-times "nearest-exit" []

  ; Create student turtles and assign properties and strategies
  create-turtles number-of-students [
    allocate-students
    set stood-up? false
    set is-professor? false

    ; Initialize panic level with normal distribution
    set panic-level random-normal initial-panic-level 10
    set panic-level max (list 0 min (list panic-level 100))

    set wait-time 0
    set stuck-time 0
    set my-leader nobody
    set path-to-exit []
    set current-path-index 0
    set recalc-path-timer 0
    assign-evacuation-strategy

    ; --- NEW: Assign impairment status ---
    set is-impaired? (random-float 100 < percent-impaired)

    ; --- CORRECTED: Use a log-normal distribution for reaction delay based on literature (e.g., Kuligowski et al.)
    ifelse is-impaired?
      [ set individual-reaction-delay (random-log-normal 5.0 0.6) + 30 ] ; Impaired agents have a longer base delay
      [ set individual-reaction-delay (random-log-normal 4.5 0.5) ]  ; Approx. 90s mean, with a right-skewed tail

```

```
    ; Ensure the delay is not excessively short
    if individual-reaction-delay < 10 [ set individual-reaction-delay
   10 ]


    set-strategy-parameters
    set initial-position (list xcor ycor)
    set distance-traveled 0
    set evacuation-time 0
    set time-stood-up 0
    set congestion-encounters 0
    set time-on-stairs 0
    set current-tier tier-level

    ; Set familiarity status
    set familiar? (random-float 100 < familiarity-percentage)
    set exit-knowledge (ifelse-value familiar? [0.8 + random-float 0.2]
    [0.2 + random-float 0.3])

    ; Initialize new behavioral variables
    set personal-space-radius 1.0
    set chosen-exit nobody
    set exit-visible? false
    set following-group? false
    set group-members []
    set pushing-force 0
    set patience-level random-normal 50 20
    set patience-level max (list 0 min (list patience-level 100))
  ]

  ; Create a professor turtle at the front
  create-turtles 1 [
    set professor self
    setxy 0 14
    set color blue
    set is-professor? true
    set stood-up? false
    set panic-level 0
    set wait-time 0
    set stuck-time 0
    set my-leader nobody
    set path-to-exit []
    set current-path-index 0
    set recalc-path-timer 0
    set evacuation-strategy "calm-and-orderly"
    set base-speed (1.2 * seconds-per-tick / patch-size-meters) ;
   Professor speed: 1.2 m/s
    set desired-speed base-speed
    set actual-speed 0
    set initial-position (list xcor ycor)
    set distance-traveled 0
    set evacuation-time 0
```

```
    set time-stood-up 0
    set congestion-encounters 0
    set individual-reaction-delay (random-log-normal 4.5 0.5)
    set familiar? true
    set exit-knowledge 1.0
    set personal-space-radius 1.5
    set chosen-exit nobody
    set exit-visible? false
    set following-group? false
    set group-members []
    set pushing-force 0
    set patience-level 80
    set is-impaired? false
    set time-on-stairs 0
    set current-tier 0
  ]

  set avg-panic-level 0

  reset-ticks
  update-colors ; Initial color update

  ask patches with [is-stair? = true] [ set pcolor 38 ]
end

; Main simulation loop
to go
  ; Update current time
  set current-time ticks * seconds-per-tick

  ; Check if fire alarm should sound
  if current-time >= (alarm-time * seconds-per-tick) and not alarm-
    sounded? [
    sound-fire-alarm
  ]

  ; Update environment
  update-alarm-effect
  update-patch-density
  update-patch-turtle-status
  update-exit-budgets  ; NEW: per-exit capacity accumulation

  ; Update agent states
  if alarm-sounded? [
    update-panic-levels
    update-colors
    detect-and-form-groups  ; For herding behavior
  ]

  collect-congestion-data
  track-movement

```

```
; Periodic check for severely stuck turtles
if ticks mod 50 = 0 [  ; Check every 5 seconds (50 ticks * 0.1 seconds)
  check-global-stuck-status
]

; Mark evacuation start
if not evacuation-started? and alarm-sounded? and any? turtles with [stood-up?] [
  set evacuation-started? true
  set evacuation-start-time current-time
]

; Agent behavior execution
ask turtles [
  if recalc-path-timer > 0 [ set recalc-path-timer recalc-path-timer - seconds-per-tick ]

  ; Check if can see exits
  update-exit-visibility

  ; React to alarm with individual delay
  if alarm-sounded? and (current-time - alarm-sound-time) >= individual-reaction-delay [
    if not stood-up? [
      attempt-to-stand-up
    ]
    if stood-up? [
      ; Choose exit if not chosen
      if chosen-exit = nobody [
        choose-exit-strategy
      ]

      ; Execute movement based on strategy
      execute-evacuation-movement
    ]
  ]

  ; Update stuck time tracking
  if stood-up? [
    ifelse actual-speed < 0.01 [  ; Essentially not moving
      set stuck-time stuck-time + seconds-per-tick
    ] [
      set stuck-time max (list 0 (stuck-time - seconds-per-tick))
    ]
  ]
]

; Update average panic level
if any? turtles with [not is-professor?] and alarm-sounded? [
  set avg-panic-level mean [panic-level] of turtles with [not is-professor?]
```

```
  ]

  ; Stop simulation when all evacuated
  if not any? turtles [
    set evacuation-end-time current-time
    set last-exit-time current-time
    set total-evacuation-time evacuation-end-time - evacuation-start-time
    finalize-statistics
    stop
  ]

  tick
end
```

Listing A.3: Core Simulation Engine

A.3.2 Agent Behavior and Decision-Making

This section details how agents are assigned a behavior, how they choose an exit, and the high-level logic for their movement. It also includes the specific implementation for the "Panic Rush" strategy as an example of one of the distinct behaviors.

```
    ; Assign evacuation strategy to a turtle based on the selected simulation behavior
to assign-evacuation-strategy
  ifelse evacuation-behavior = "mixed-population"
  [
    ; Realistic distribution based on evacuation research
    let strategy-choice random 100
    ifelse strategy-choice < 50
      [ set evacuation-strategy "follow-others" ]     ; 50% - most common behavior
      [ ifelse strategy-choice < 75
        [ set evacuation-strategy "nearest-exit" ]    ; 25% - rational evacuees
        [ ifelse strategy-choice < 90
          [ set evacuation-strategy "calm-and-orderly" ] ; 15% - trained/calm
          [ set evacuation-strategy "panic-rush" ]    ; 10% - panicked
        ]
      ]
  ]
  [ set evacuation-strategy evacuation-behavior ]
end

; Choose which exit to head towards based on the assigned strategy
to choose-exit-strategy
  let available-exits exit-patches with [not blocked?]
  if not any? available-exits [ stop ]

  if evacuation-strategy = "nearest-exit" or is-professor? [
```

```
    set chosen-exit min-one-of available-exits [distance myself]
  ]

  if evacuation-strategy = "panic-rush" [
    set chosen-exit min-one-of available-exits [distance myself]
  ]

  if evacuation-strategy = "calm-and-orderly" [
    ifelse familiar? [
      let exit-options []
      ask available-exits [
        let exit-score distance myself + (count turtles in-radius 5) *
    3
        set exit-options lput (list self exit-score) exit-options
      ]
      set chosen-exit first (min-by-second exit-options)
    ] [
      set chosen-exit min-one-of available-exits [distance myself]
    ]
  ]

  if evacuation-strategy = "follow-others" [
    ; Follow group's choice or choose nearest if no group
    if following-group? and any? group-members with [chosen-exit !=
    nobody] [
      ; Code to find most popular exit in the group
    ]
    if chosen-exit = nobody [
      set chosen-exit min-one-of available-exits [distance myself]
    ]
  ]
end

; Main evacuation movement execution called by the 'go' procedure
to execute-evacuation-movement
  if [pcolor] of patch-here = lime and not [blocked?] of patch-here [
    record-evacuation
    die
  ]

  update-desired-speed

  ; Execute strategy-specific behavior
  if evacuation-strategy = "follow-others" [ execute-herding-behavior ]
  if evacuation-strategy = "calm-and-orderly" [ execute-orderly-
    behavior ]
  if evacuation-strategy = "panic-rush" [ execute-panic-behavior ]
  if evacuation-strategy = "nearest-exit" [ execute-rational-behavior ]

  ; Calculate path if needed
  if length path-to-exit = 0 or recalc-path-timer <= 0 [
    calculate-bfs-path
```

```
  ]

  ; Move along path
  if length path-to-exit > 0 [
    move-along-path
  ]

  ; Handle stuck situations
  if stuck-time > 3 [
    handle-stuck-situation
  ]
end

; Execute panic behavior (as an example of a specific strategy)
to execute-panic-behavior
  ; Panic increases over time
  set panic-level min (list 100 (panic-level + random-float 2))

  ; Erratic movement when panicked
  if panic-level > 70 [
    rt random 60 - 30  ; Random turning
    set desired-speed base-speed * 1.5  ; Try to move faster
  ]

  ; Crowd pressure effects
  let nearby-panicked count turtles in-radius 2 with [panic-level > 60]
  if nearby-panicked > 3 [
    ; Try to escape crowd by moving away from its center
    let crowded-spots turtles in-radius 2
    if any? crowded-spots [
      let avg-x mean [xcor] of crowded-spots
      let avg-y mean [ycor] of crowded-spots
      if xcor != avg-x or ycor != avg-y [
        let escape-heading atan (xcor - avg-x) (ycor - avg-y)
        set heading escape-heading
      ]
    ]
  ]

  ; May freeze in extreme panic
  if panic-level > 90 and random 100 < 10 [
    set desired-speed 0.1
  ]
end
```

Listing A.4: Agent Behavior and Decision-Making

A.3.3 Pathfinding

This section contains the core pathfinding algorithm with costs (a Dijkstra like Modified Breadth-First Search (BFS)) which agents use to find a route to their chosen exit, and the procedure for

moving along that path.

```
to calculate-bfs-path
  if recalc-path-timer > 0 [ stop ]

  let start-patch patch-here
  let goal-patch chosen-exit

  if goal-patch = nobody [
    set goal-patch min-one-of (exit-patches with [not blocked?]) [
   distance myself]
  ]
  if goal-patch = nobody [ stop ]

  ; Check if at exit
  if pcolor = lime and not [blocked?] of patch-here [
    record-evacuation
    die
  ]

  ; If on bench, try to escape then retry later
  if [pcolor] of start-patch = 35 [
    escape-from-seat
    set start-patch patch-here
    if [pcolor] of patch-here = 35 [
      set recalc-path-timer 0.5
      stop
    ]
  ]

  set path-to-exit []
  let visited table:make
  let queue (list (list start-patch 0 [])) ; each: (patch, cost, path)

  while [length queue > 0 and length path-to-exit = 0] [
    ; pop min-cost
    set queue sort-by [[a b] -> item 1 a < item 1 b] queue
    let current first queue
    set queue but-first queue

    let cur-patch item 0 current
    let cur-cost item 1 current
    let cur-path item 2 current

    ; key is the patch coordinates (valid table key)
    let cur-key (list [pxcor] of cur-patch [pycor] of cur-patch)

    if not table:has-key? visited cur-key [
      table:put visited cur-key true

      if [pcolor] of cur-patch = lime and not [blocked?] of cur-patch [
        set path-to-exit cur-path
```

```
        set current-path-index 0
        set recalc-path-timer 1.5
        stop
      ]

      ask cur-patch [
        let neigh neighbors
        ; Use [pxcor pycor] as key for neighbors too
        let valid-neigh neigh with [
          walkable? and
          pcolor != black and
          pcolor != 35 and
          not table:has-key? visited (list pxcor pycor)
        ]
        ask valid-neigh [
          ; step cost: diagonal = sqrt 2, orthogonal = 1
          let step-cost (ifelse-value
            (abs (pxcor - [pxcor] of cur-patch) + abs (pycor - [pycor] of cur-patch) = 2)
            [ sqrt 2 ] [ 1 ])

          ; congestion penalty using local density at neighbor
          let r 1
          let area (pi * (r * patch-size-meters) ^ 2)
          let rho (ifelse-value (area > 0) [ count turtles in-radius r / area ] [ 0 ])
          let cong-penalty rho

          let ncost cur-cost + step-cost + cong-penalty
          set queue lput (list self ncost (lput self cur-path)) queue
        ]
      ]
    ]
  ]
end
```

Listing A.5: Pathfinding and Movement

A.3.4 Social and Environmental Dynamics

This code shows how panic spreads between agents (social contagion) and how agents' colors are updated to visually represent their panic level.

```
; Update panic levels based on contagion and situation (Tuned Version)
to update-panic-levels
  if not alarm-sounded? [ stop ]

  ask turtles with [not is-professor? and stood-up?] [
    ; --- RULE 1: PANIC FROM CROWD DENSITY (Tuned Down) ---
    let local-density count other turtles in-radius 1.5
    if local-density > 4 [
      set panic-level panic-level + (local-density * 0.05)
```

```
    ]

    ; --- RULE 2: PANIC FROM BEING STUCK (Tuned Down) ---
    if stuck-time > 2 [
      set panic-level panic-level + 0.5
    ]

    ; --- RULE 3: PANIC CONTAGION (Slightly Tuned Down) ---
    let nearby-turtles other turtles in-radius 3
    if any? nearby-turtles [
      let nearby-panic mean [panic-level] of nearby-turtles
      if nearby-panic > panic-level [
        set panic-level panic-level + ((nearby-panic - panic-level) * 0.05)
      ]
    ]

    ; --- RULE 4: CALMING DOWN (Slightly Strengthened) ---
    if actual-speed > (desired-speed * 0.8) and local-density < 3 [
      set panic-level panic-level - 0.75
    ]

    ; --- FINAL STEP: ENSURE BOUNDS ---
    set panic-level max (list 0 min (list panic-level 100))
  ]
end

; Update colors based on panic and familiarity, with impaired override
to update-colors
  ask turtles with [not is-professor?] [
    let panic-ratio panic-level / 100

    ; Base color mapping (unchanged)
    ifelse familiar? [
      ; Familiar agents: Cyan (calm) to Purple (panicked)
      set color (list (159 * panic-ratio) (255 * (1 - panic-ratio)) 255)
    ] [
      ; Unfamiliar agents: Green (calm) to Red (panicked)
      set color (list (255 * panic-ratio) (255 * (1 - panic-ratio)) 0)
    ]

    ; Impaired override: GOLD (distinct from professor blue and the gradients)
    if is-impaired? [
      set color (list 255 215 0)  ; Gold
    ]
  ]

  ; Before alarm, all are white (keeps pre-alarm view consistent)
  if not alarm-sounded? [
    ask turtles with [not is-professor?] [
```

```
      set color white
    ]
  ]
end
```

Listing A.6: Social and Environmental Dynamics

A.4 Screenshots

```
DETAILED ANALYSIS

MOVEMENT SPEED PARAMETERS

Speed Type                | Flat Surface | On Stairs
Unimpeded (e.g. rational) | 1.2 m/s      | 0.78 m/s
Average Evacuation        | 1 m/s        | 0.65 m/s
Impaired Mobility         | 0.42 m/s     | 0.27 m/s
Professor                 | 1.2 m/s      | 0.78 m/s
Stair Speed Modifier: 65% of normal speed

STAIR/AISLE ANALYSIS

Average Time on Stairs: 4.8 sec
Percentage Time on Stairs: 52.5%

EXIT USAGE ANALYSIS

Exit at (20,-16) - Right side - OPEN
Exit at (21,14) - Right side - OPEN
Exit at (21,13) - Right side - OPEN
Exit at (-21,13) - Left side - OPEN
Exit at (-20,-16) - Left side - OPEN
Exit at (-19,-16) - Left side - OPEN
Exit at (-21,14) - Left side - OPEN
Exit at (19,-16) - Right side - OPEN

STRATEGY DISTRIBUTION (Mixed Population)

panic-rush: 23 agents (11.4%)
calm-and-orderly: 37 agents (18.4%)
follow-others: 94 agents (46.8%)
nearest-exit: 47 agents (23.4%)
```

Figure A.1: Command Center Output

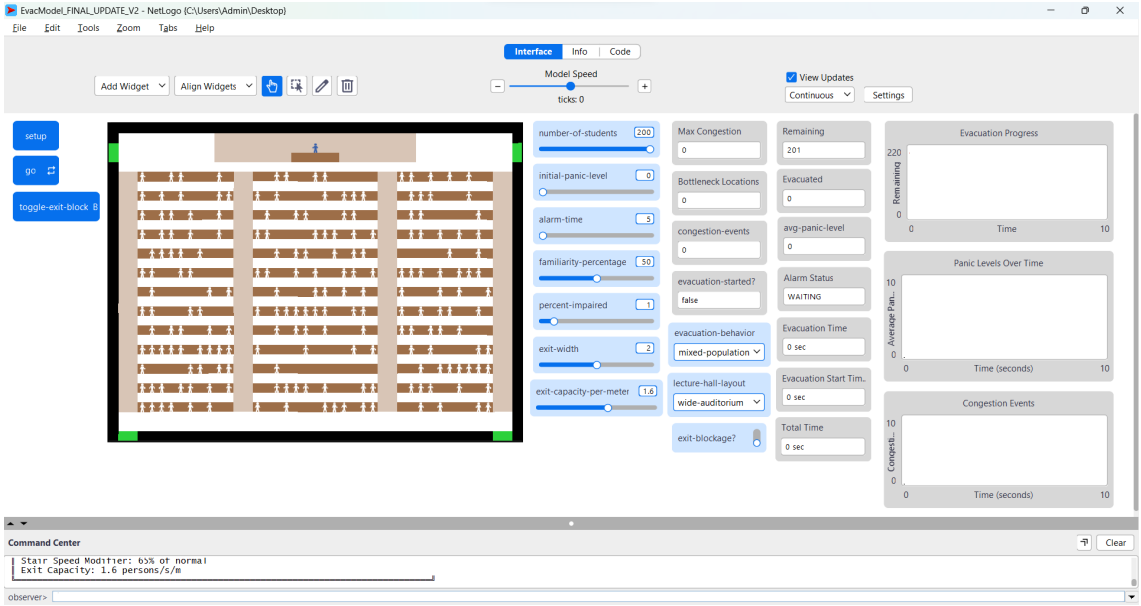

Figure A.2: NetLogo 7.0.0 GUI

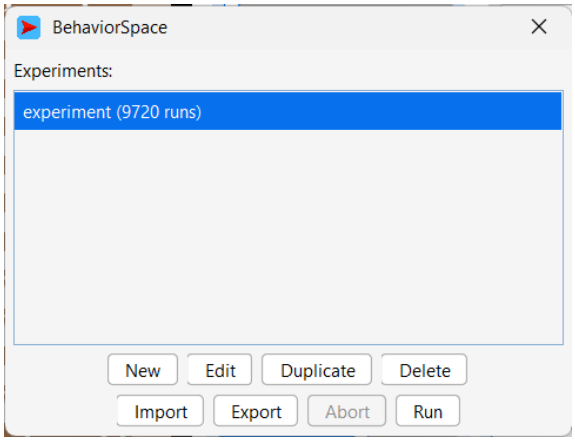

Figure A.3: Netlogo BehaviorSpace: Experiment Setup Window

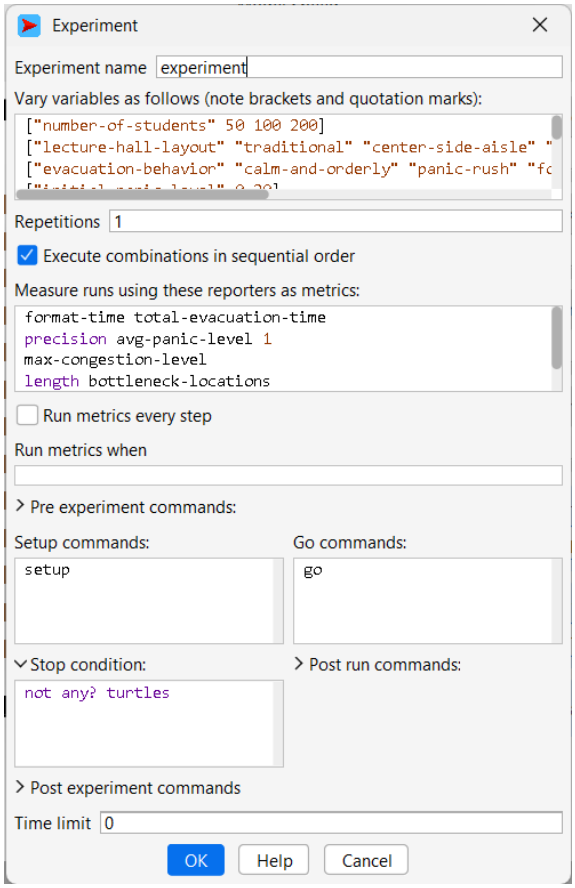

Figure A.4: Netlogo BehaviorSpace: Experiment Configuration

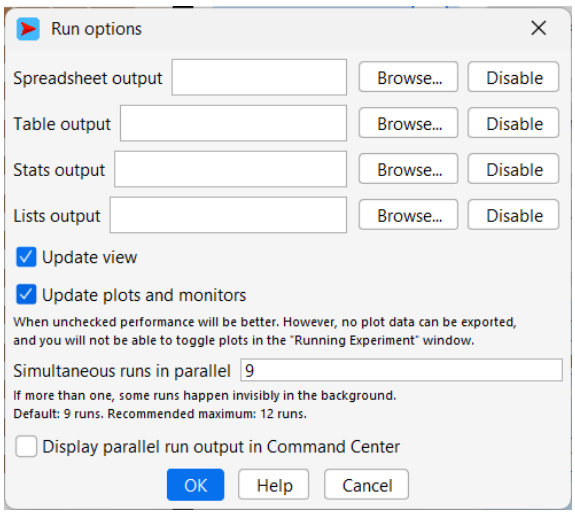

Figure A.5: Netlogo BehaviorSpace: Run Options